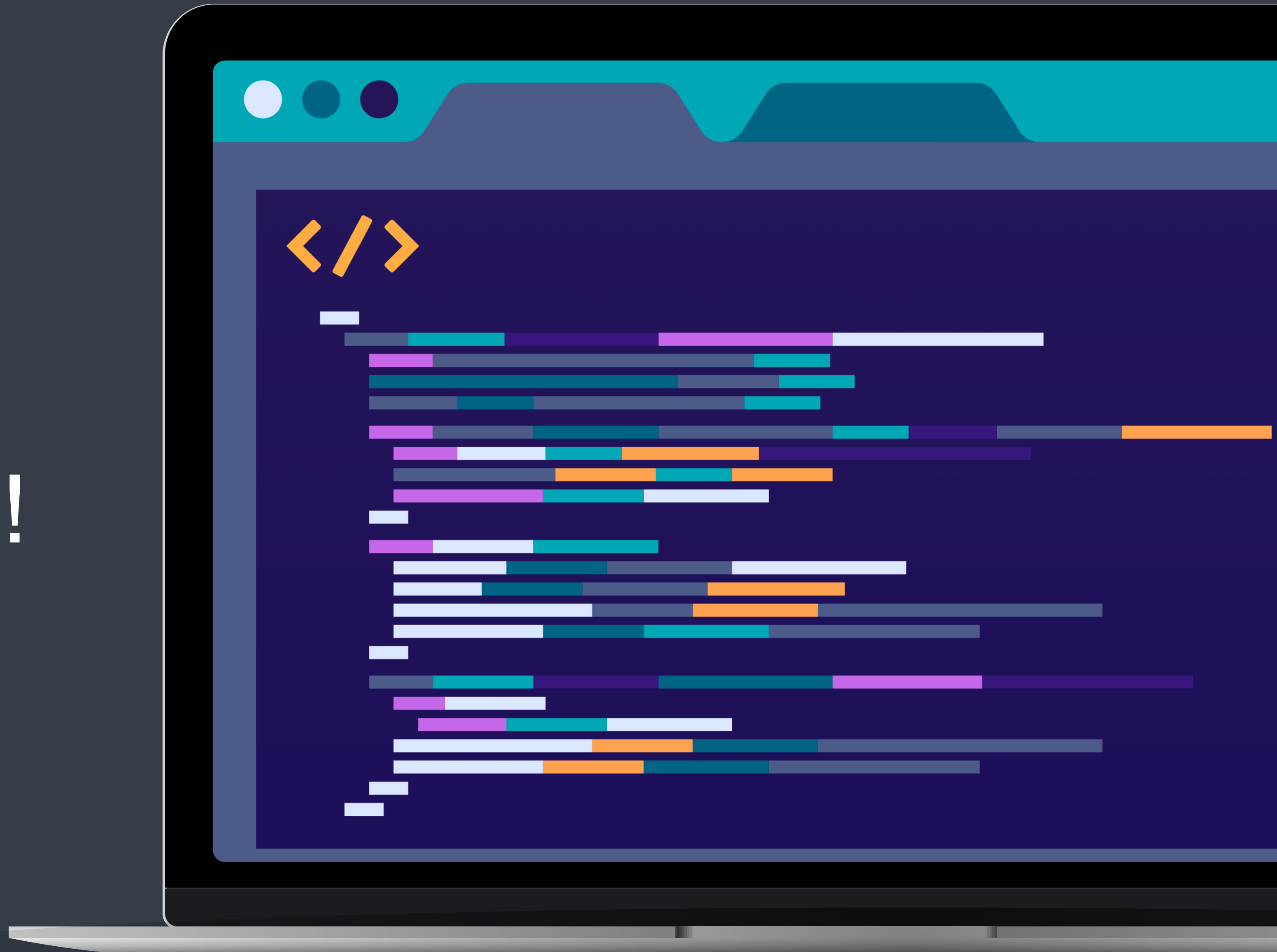


JUST BASH IT!

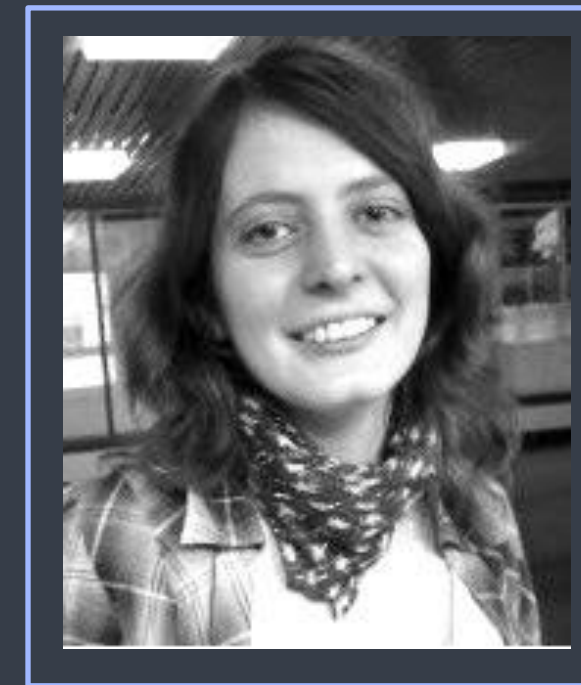
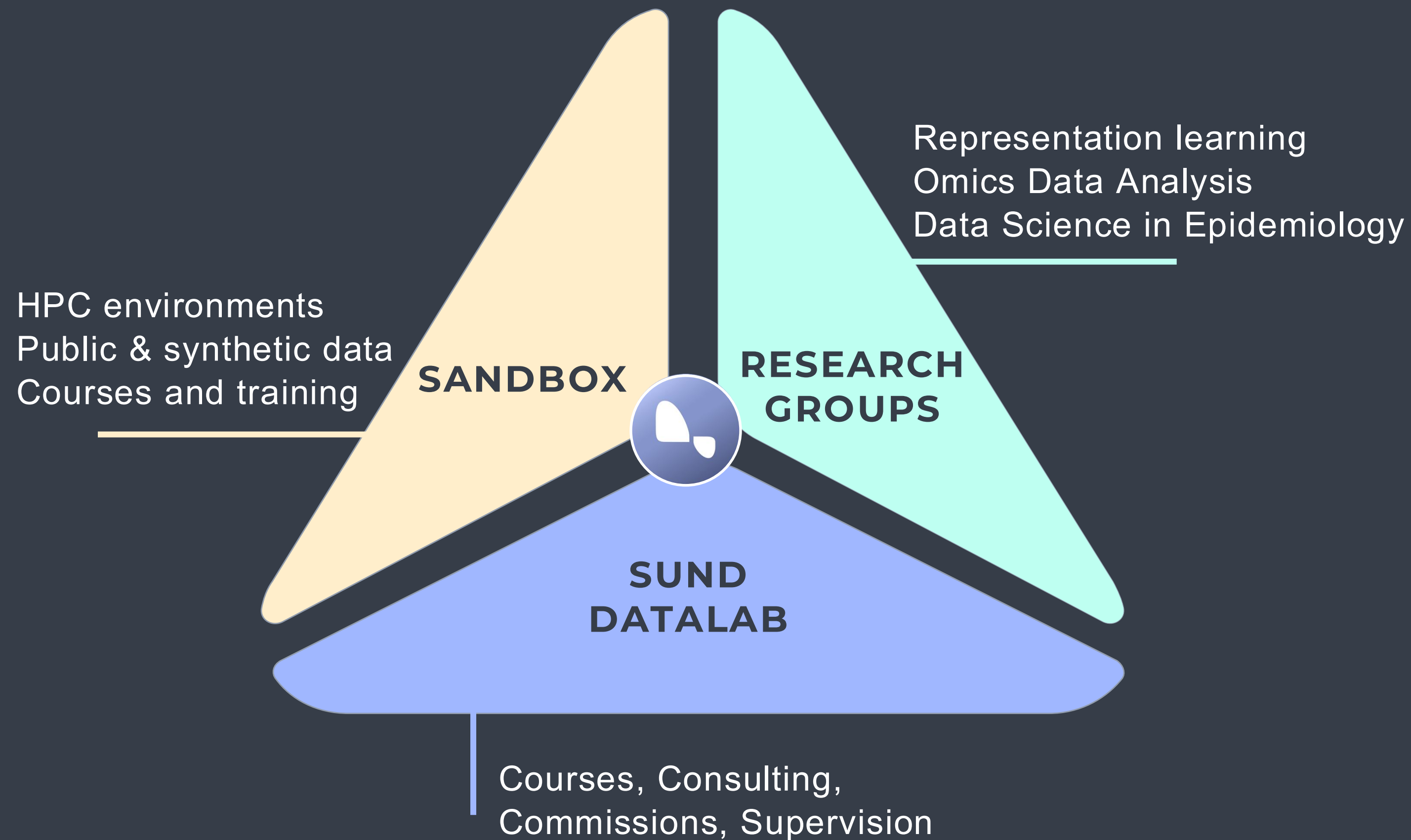


WHO ARE WE?

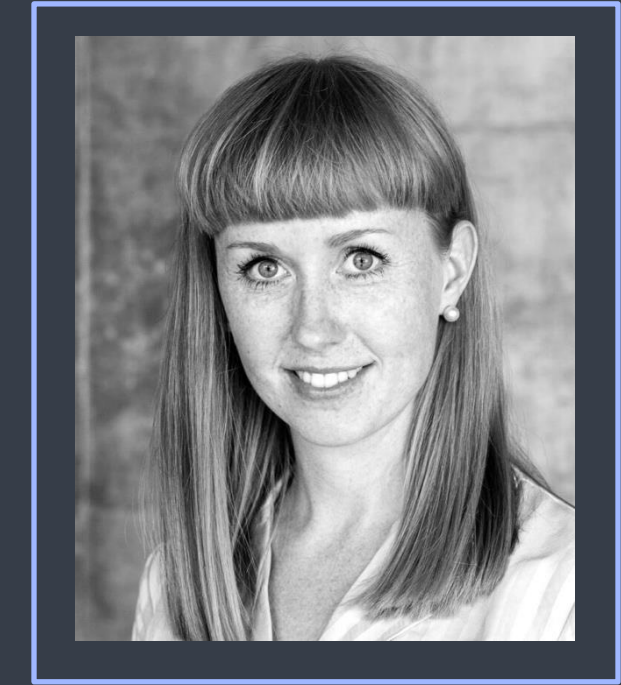


HEADS (SUND) DATA LAB

WEBSITE: <https://heads.ku.dk/>



Henrike Zschach



Thilde Terkelsen



Helene Wegener

PROGRAM

Day 1

08:45	MORNING COFFEE
09:00	INTRODUCTION TO COMMANDLINE
09:15	PART 1: NAVIGATING FILES & DIRECTORIES
09:35	EXERCISE 1
09:55	PART 2: FILE OPERATIONS
10:05	EXERCISE 2
10:20	COFFEE BREAK
10:30	PART 3: PROJECT ORGANIZATION & BACKUP
10:50	EXERCISE 3
11:10	PART 4: VIEWING AND EDITING FILES
11:35	EXERCISE 4
12:00	LUNCH
13:00	PART 5: DATA WRANGLING
13:35	EXERCISE 5
13:55	PART 6: DATA WRANGLING 2
14:24	COFFEE BREAK
14:35	EXERCISE 6

Day 2

08:30	MORNING COFFEE + Q&A
09:00	PART 7: REDIRECTION & PIPES
09:20	EXERCISE 7
10:20	COFFEE BREAK
10:30	PART 8: SHELL SCRIPTS & LOOPS
11:00	EXERCISE 8
12:00	LUNCH
13:00	PART 9: INSTALLATIONS & CONFIG FILES
13:20	EXERCISE 9
13:50	PART 10: COMPUTING ON HPCs
14:15	COFFEE BREAK
14:30	EXERCISE 10
15:30	BYE BYE

COURSE MATERIALS:

<https://github.com/Center-for-Health-Data-Science/Just-Bash-It>





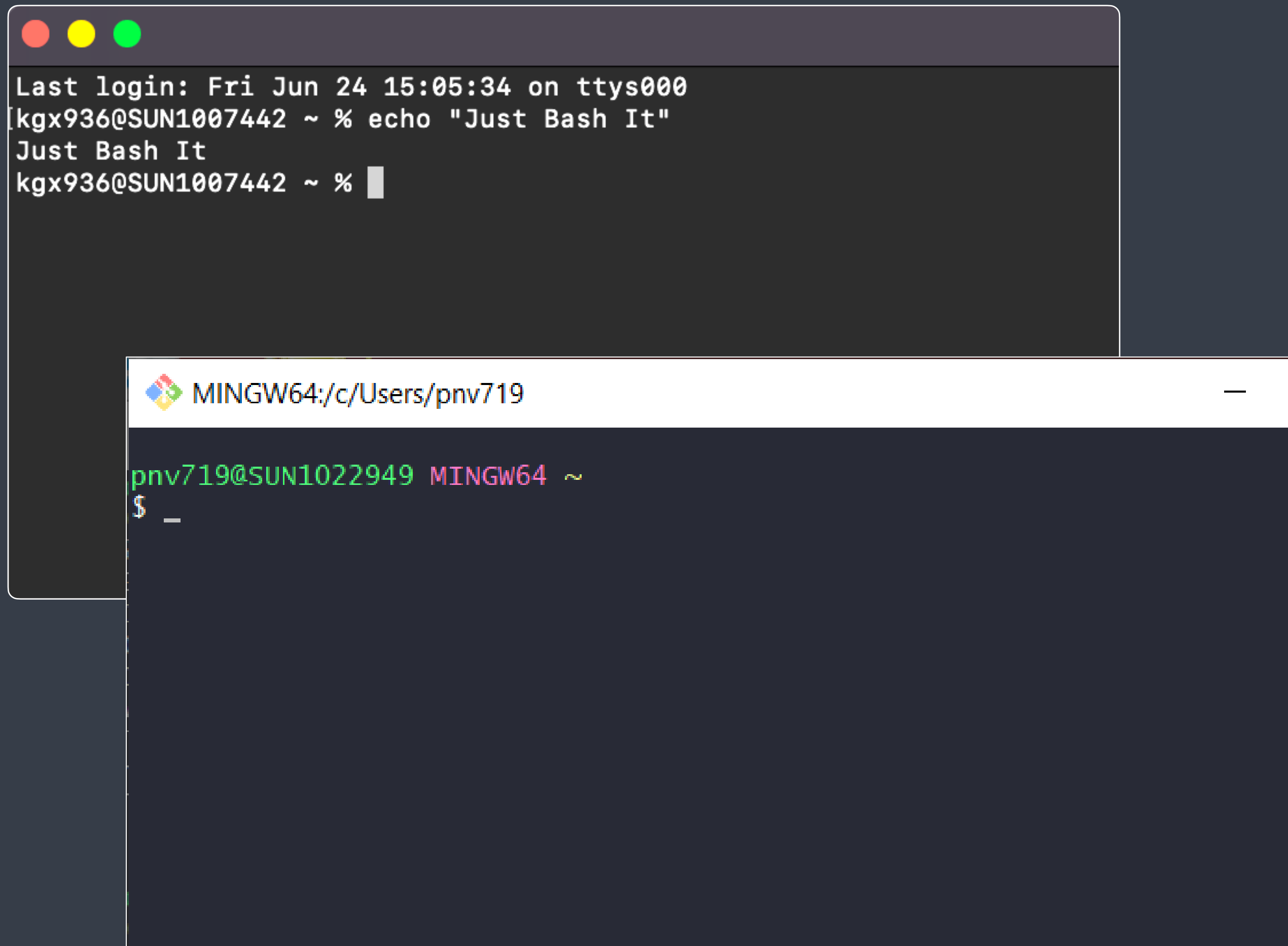
```
Last login: Fri Jun 24 15:05:34 on ttys000
[kgx936@SUN1007442 ~ % echo "Just Bash It"
Just Bash It
kgx936@SUN1007442 ~ % █
```



“What is a shell, a terminal, a command-line, and what is bash?”

“How do these concepts connect to my computer?”

SET UP CHECK



The image shows two terminal windows. The top window is a standard Linux terminal with a dark background and white text. It shows a login session for user 'kgx936' on 'SUN1007442'. The user has run the command 'echo "Just Bash It"' and the output is 'Just Bash It'. The bottom window is a Windows terminal (MINGW64) with a dark background and white text. It shows a session for user 'pnv719' on 'SUN1022949'. The prompt is '\$ _'.

```
Last login: Fri Jun 24 15:05:34 on ttys000
kgx936@SUN1007442 ~ % echo "Just Bash It"
Just Bash It
kgx936@SUN1007442 ~ %
```

```
MINGW64:/c/Users/pnv719
pnv719@SUN1022949 MINGW64 ~
$ _
```

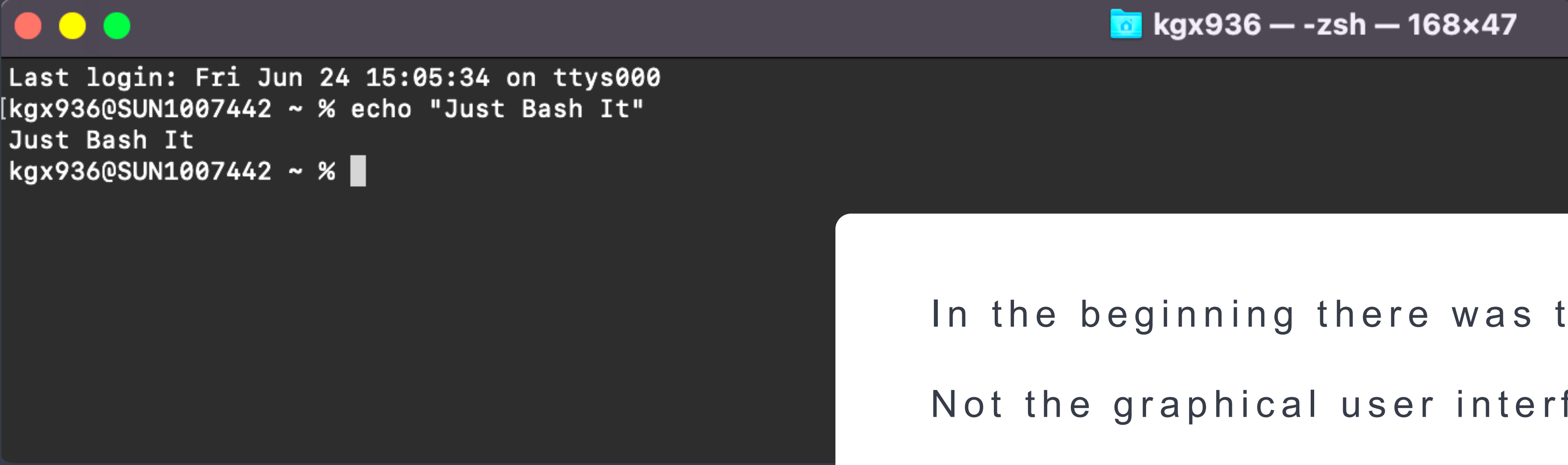
You should all have a terminal.
It looks like this...

During this course we will do
exercises inside the terminal.

We will also show commands
during the lecture.

We encourage you to open your
terminal and try them out as we
go along.

WHAT IS BASH?

A screenshot of a terminal window with a dark background. The title bar at the top shows three colored window control buttons (red, yellow, green) on the left and a folder icon followed by the text 'kgx936 — -zsh — 168x47' on the right. The terminal content shows a login message 'Last login: Fri Jun 24 15:05:34 on ttys000', a prompt '[kgx936@SUN1007442 ~ %' followed by the command 'echo "Just Bash It"', the output 'Just Bash It', and another prompt 'kgx936@SUN1007442 ~ %' with a cursor.

```
Last login: Fri Jun 24 15:05:34 on ttys000
[kgx936@SUN1007442 ~ % echo "Just Bash It"
Just Bash It
kgx936@SUN1007442 ~ % █
```

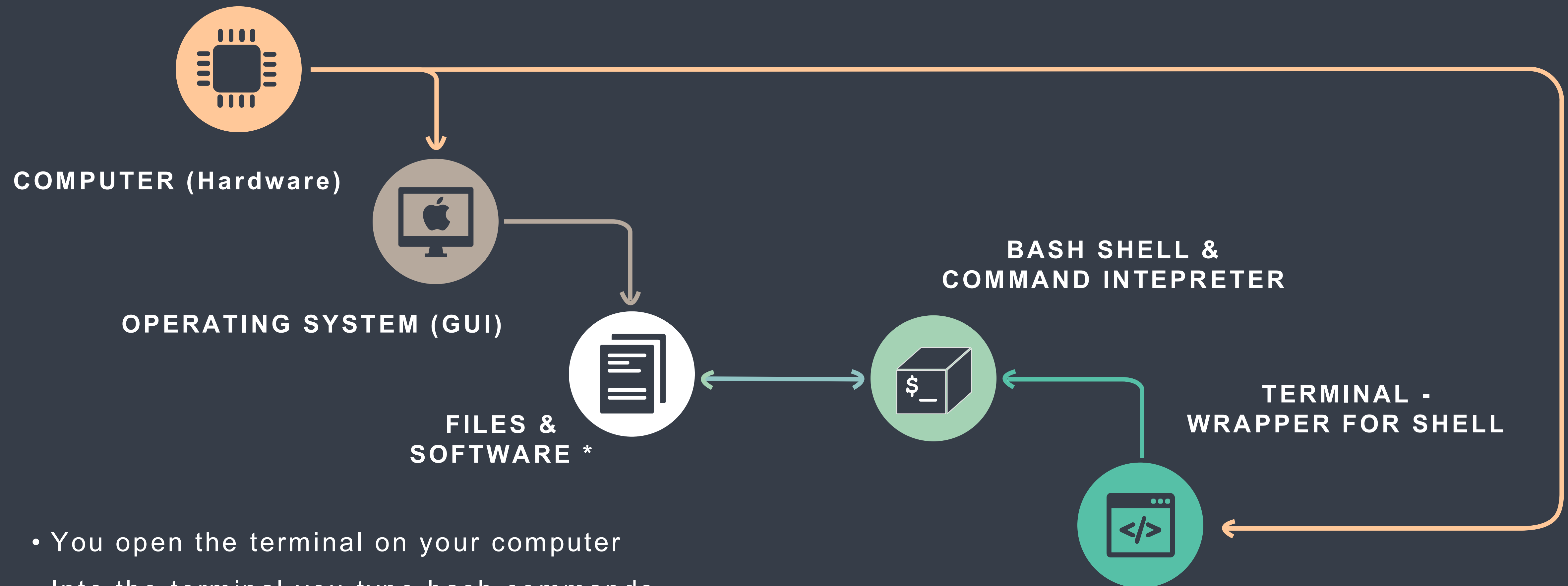
In the beginning there was the word –
Not the graphical user interface!

The first computers had no graphical user interfaces (GUI) where you could click. Instead, you interacted with the computer via the terminal.

There are still computing environments that do not have a GUI.

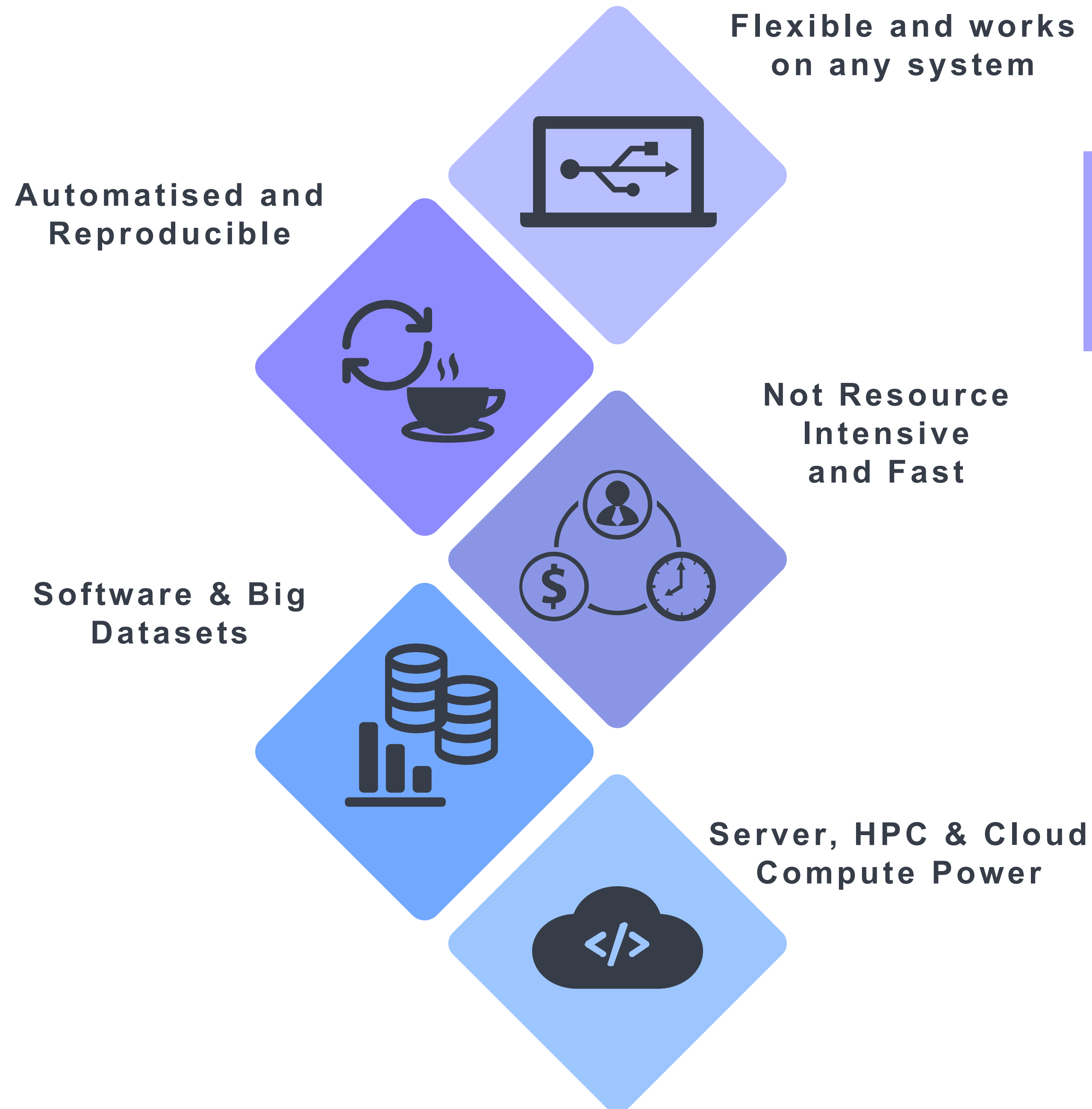
Some scientific software do not have GUIs either. You need to execute them via the terminal.

TERMINOLOGY



- You open the terminal on your computer
- Into the terminal you type bash commands
- Commands are interpreted by the shell

* software == executable file



SELL IT TO ME!

Benefits of bash & command line:

- Your analysis will be reproducible, automated and parallelised.
- Fast processes, less computationally intensive.
- Command line software.
- Power to handle big data and heavy computations.

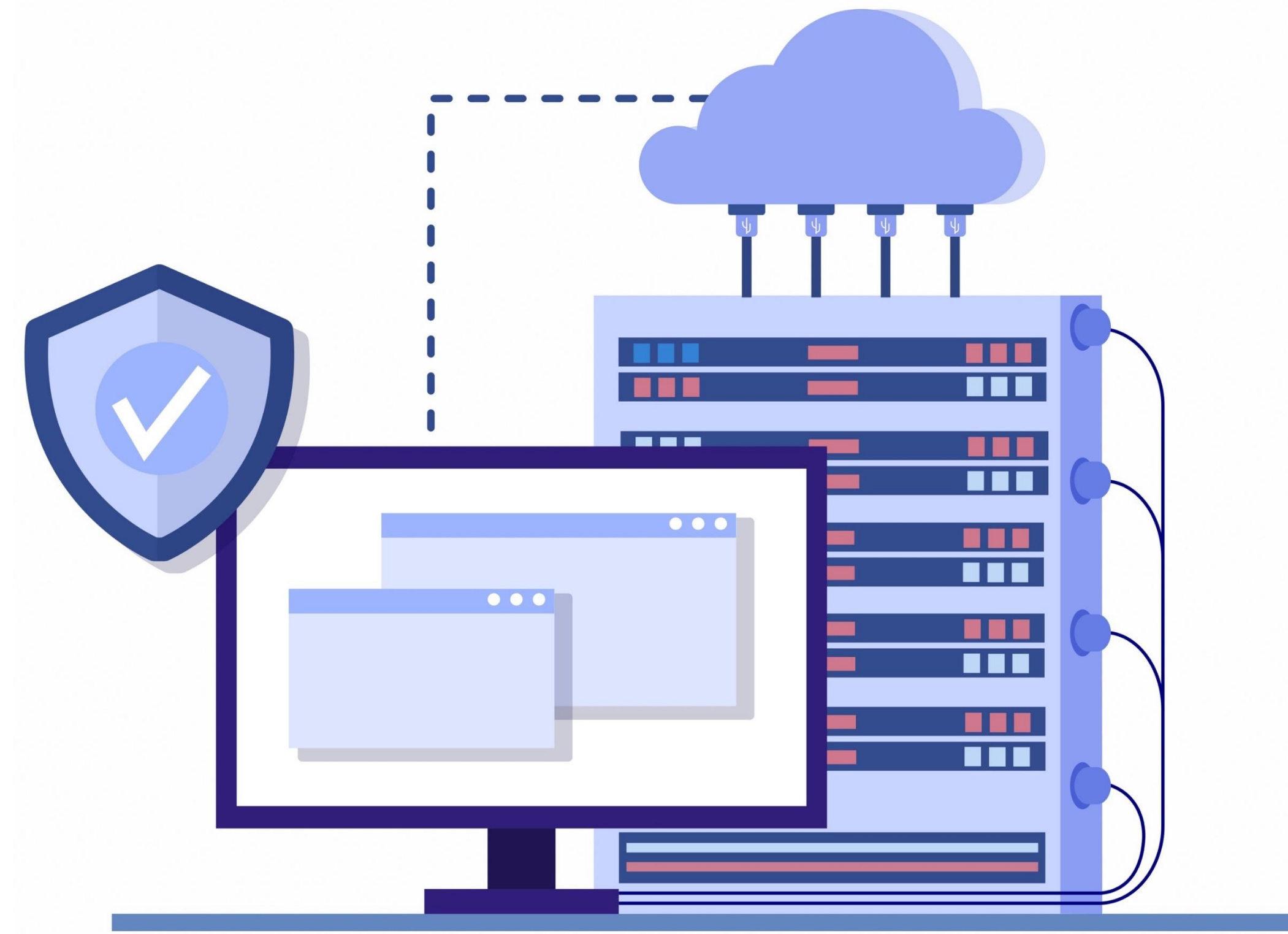
WORKING ON A REMOTE SERVER

Server or High Performance Computing (*HPC*) System.

When:

- 01 Not enough computational power (CPU, GPU)
- 02 Not enough memory (RAM) or storage (HDD)
- 03 Not secure, i.e. person sensitive data
- 04 Software is complicated to install and set up

Most servers and many HPCs do not have GUIs, you interact with them through the **command line**



A COUPLE OF EXAMPLES



REGISTRY DATA

Drug adverse effects and mortality:

Person sensitive
Huge files, many variables
Different formats



SEQUENCING DATA

Single Cell RNA from Cystic Fibrosis:

Paired-end sequencing
300 patients
Two fastq files per patient



DATABASES

Drug Databases - molecular structure & interaction:

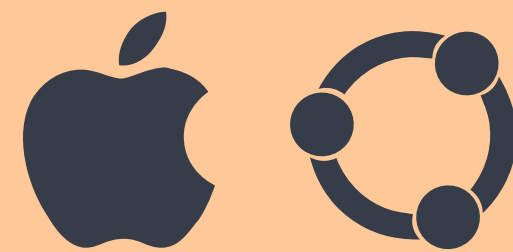
Multiple large databases
Different formats
Maybe not downloadable

HOW DO I GET A BASH SHELL?

YEAH!

YOU HAVE A BASH SHELL &
TERMINAL ALREADY.

Search for terminal on your
laptop and open it.



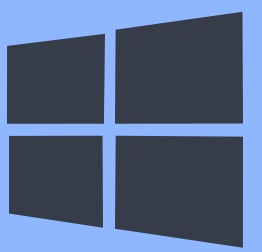
OS X or UBUNTU

>= WINDOWS 10 - Windows Subsystem for Linux:

<https://www.laptopmag.com/articles/use-bash-shell-windows-10>

OR INSTALL A BASH SHELL/TERMINAL:

- **Gitbash:** <https://gitforwindows.org>
- **MobaXterm:** <https://mobaxterm.mobatek.net/download.html>
- **Cygwin:** <https://www.cygwin.com/index.html>
- **Cmder:** <https://cmder.net/>
- **PuTTY:** <https://www.putty.org/>

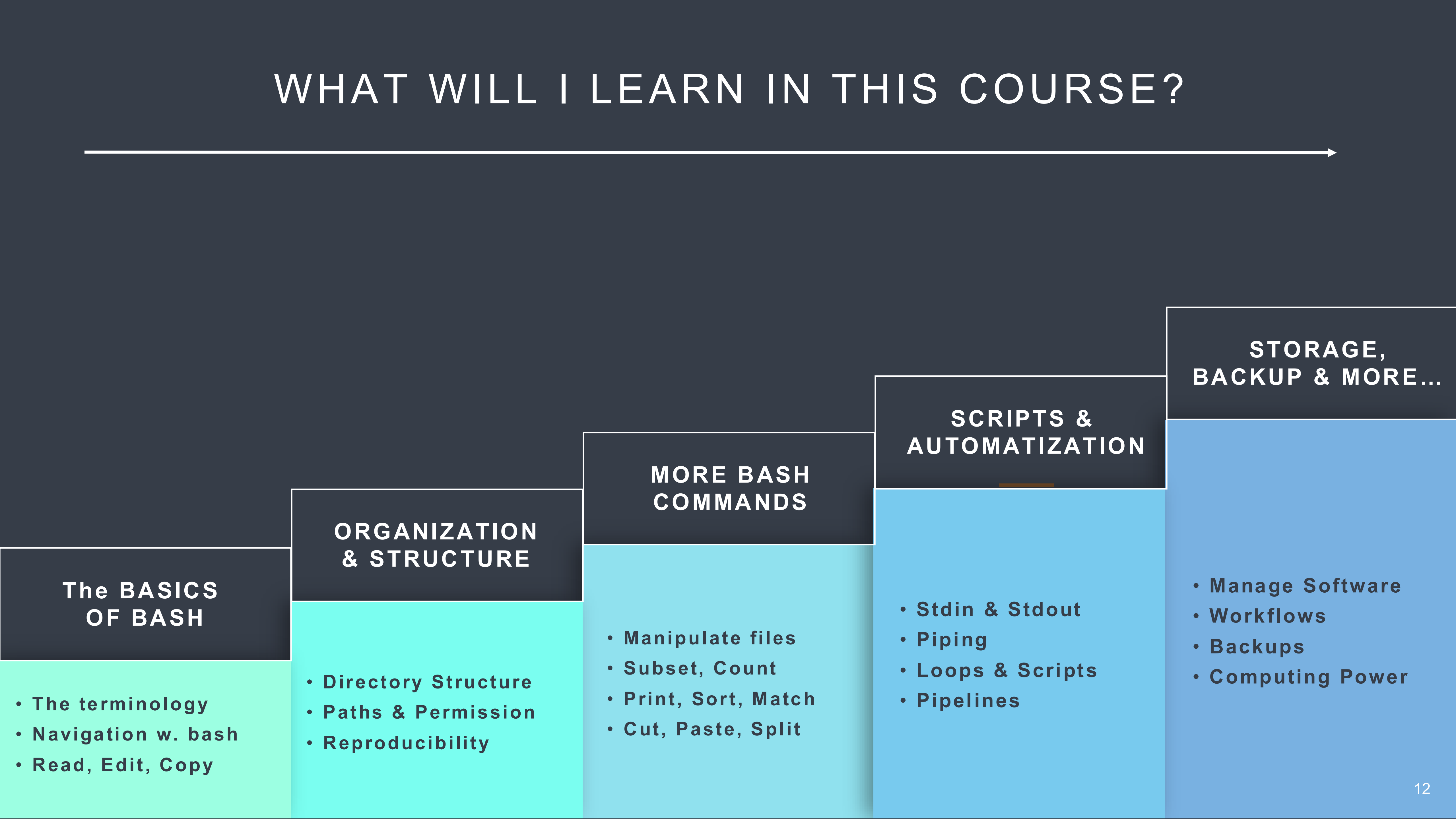


WINDOWS

In this course Windows users will be working on **gitbash**



WHAT WILL I LEARN IN THIS COURSE?



The BASICS OF BASH

- The terminology
- Navigation w. bash
- Read, Edit, Copy

ORGANIZATION & STRUCTURE

- Directory Structure
- Paths & Permission
- Reproducibility

MORE BASH COMMANDS

- Manipulate files
- Subset, Count
- Print, Sort, Match
- Cut, Paste, Split

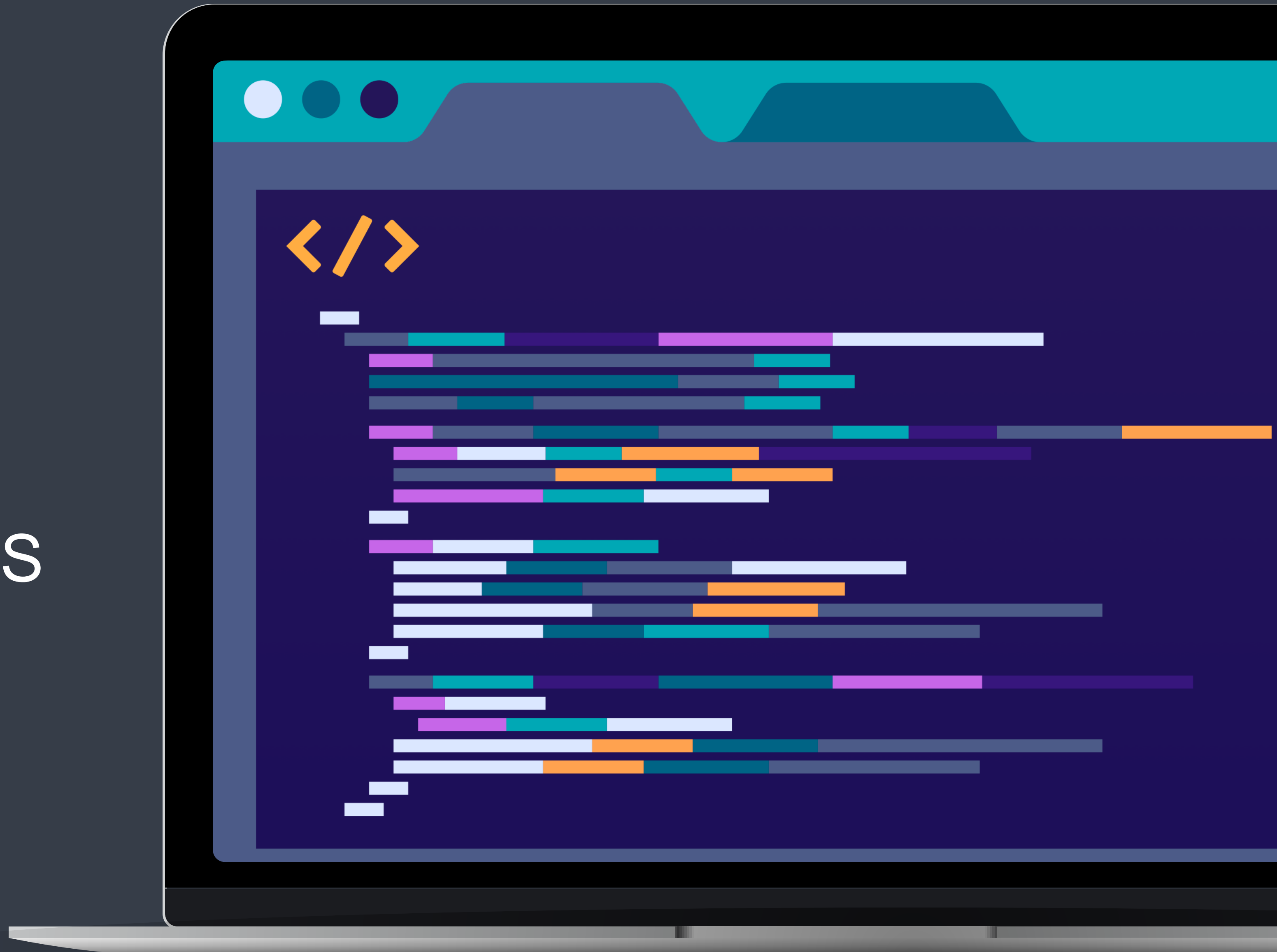
SCRIPTS & AUTOMATIZATION

- Stdin & Stdout
- Piping
- Loops & Scripts
- Pipelines

STORAGE, BACKUP & MORE...

- Manage Software
- Workflows
- Backups
- Computing Power

1. NAVIGATING FILES & DIRECTORIES

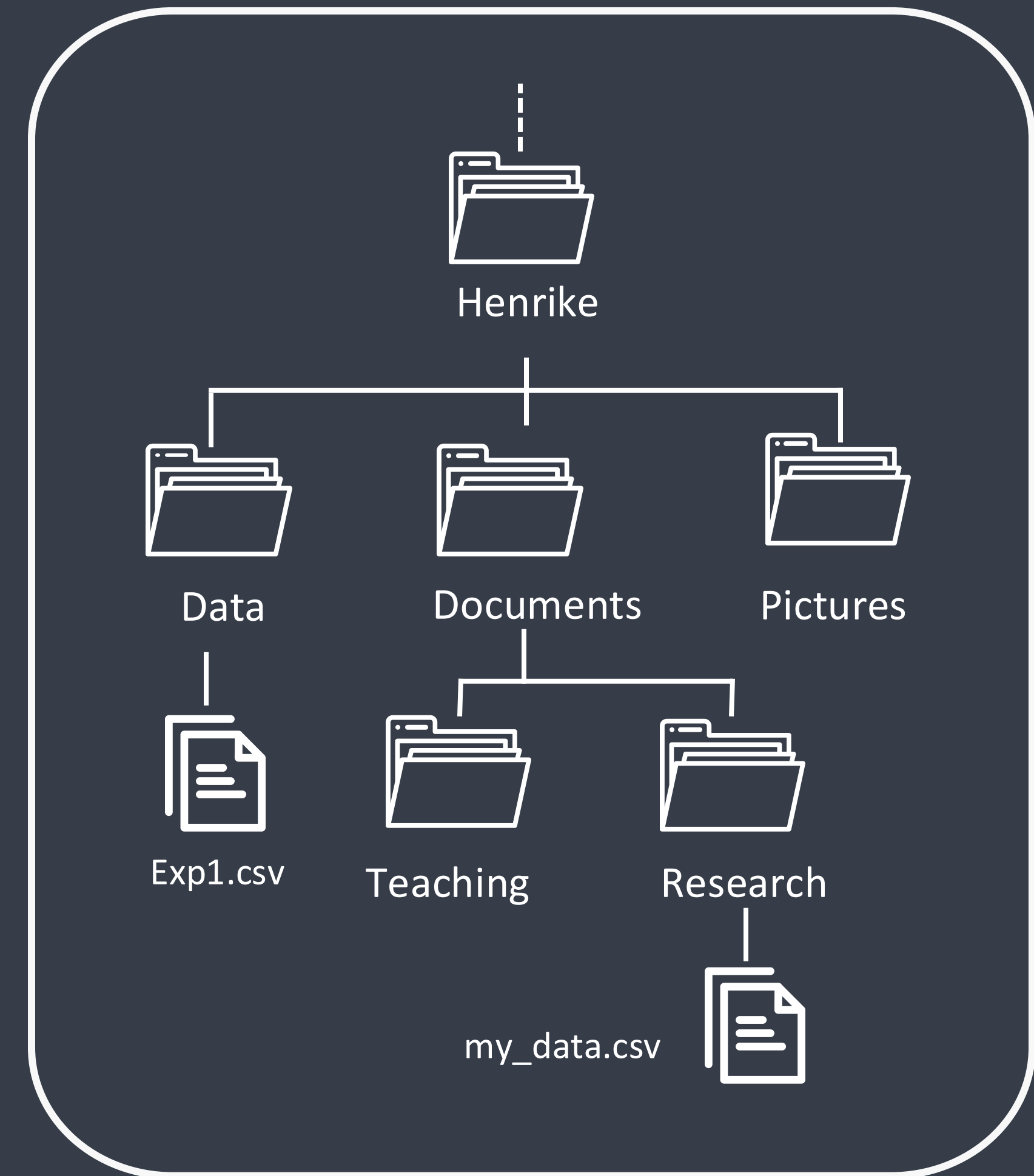
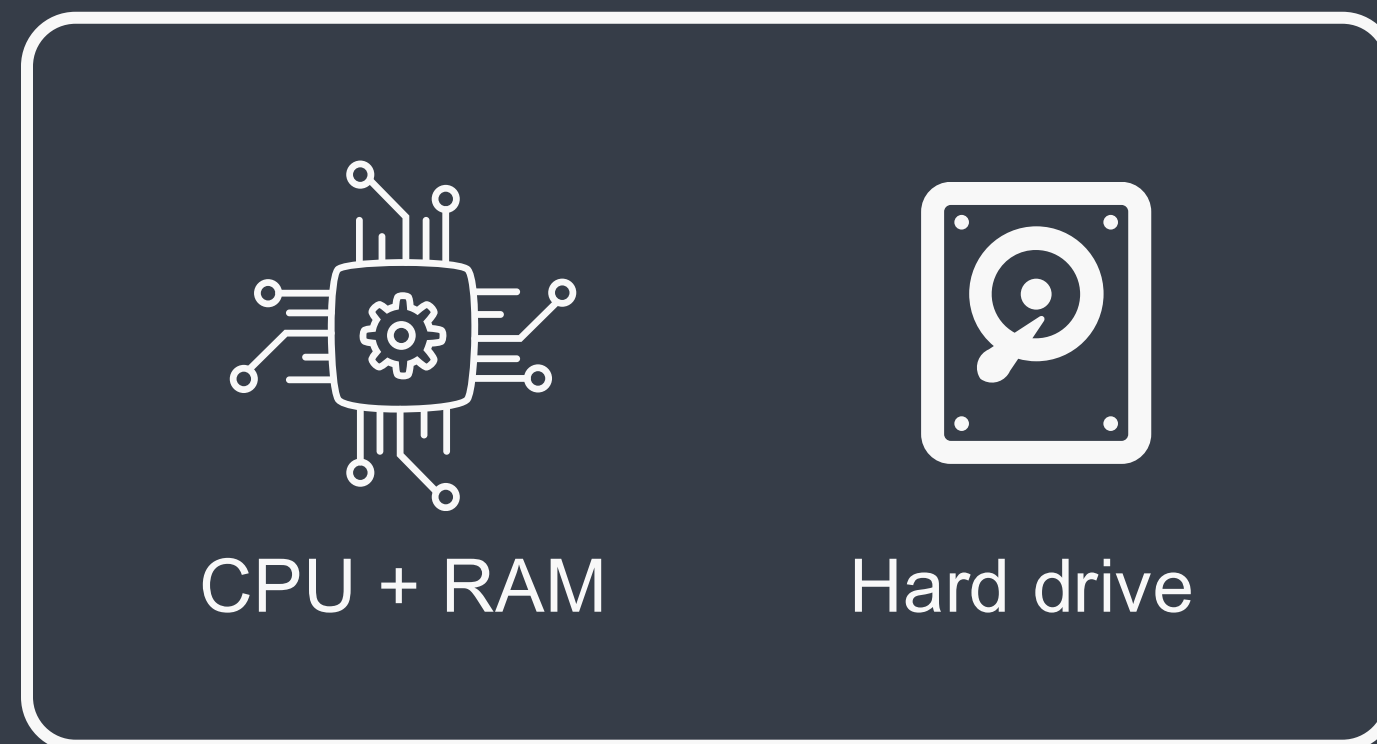


THE ANATOMY OF A COMPUTER

Software



Hardware



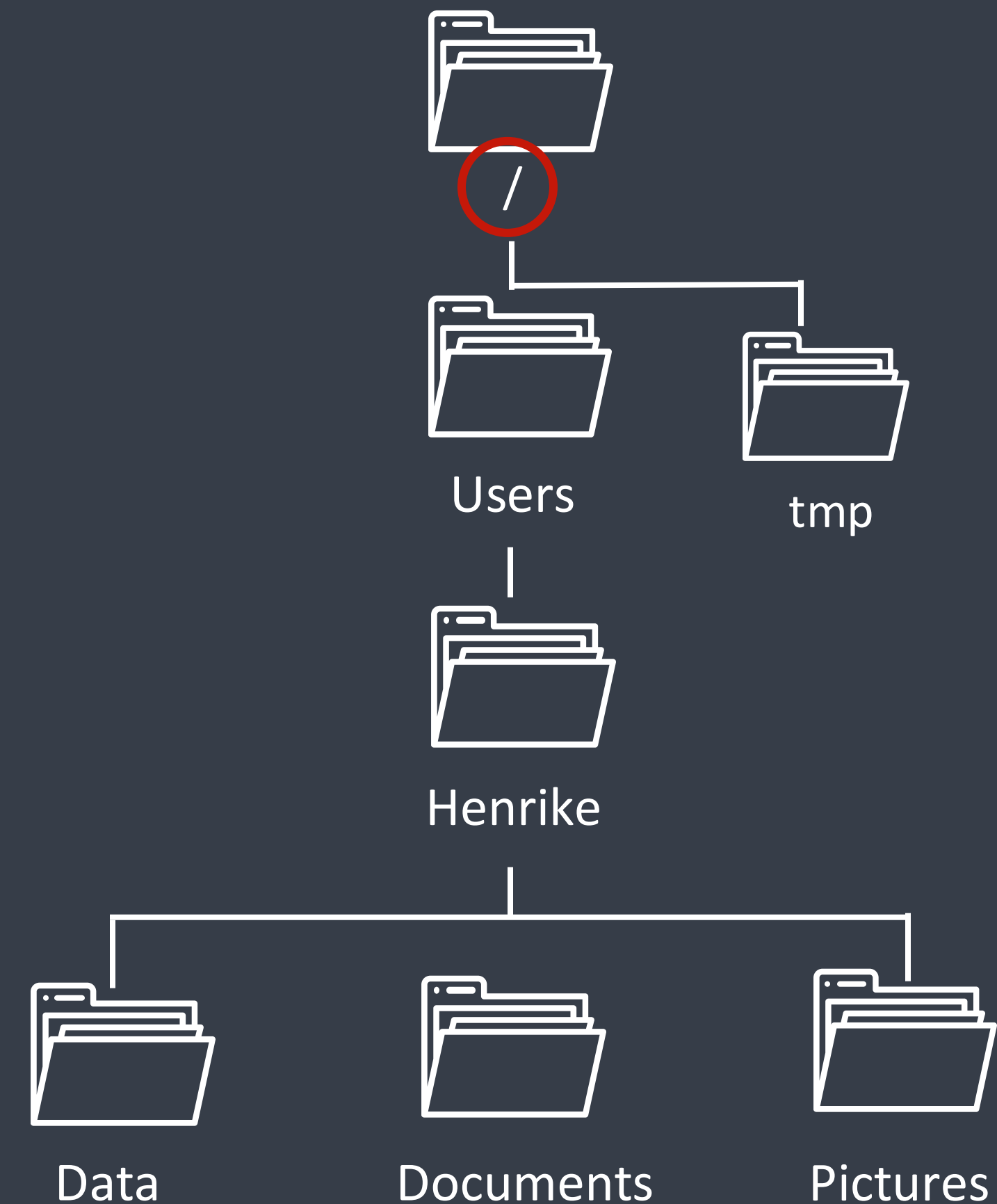
File tree

THE FILE TREE

- All data is stored inside the file tree.
- Branching and hierarchical structure
- The topmost place is called the '**root**' - written as '/'.

The scheme shown here and on the slides corresponds to what you have in 'Files' of the git hub repo.

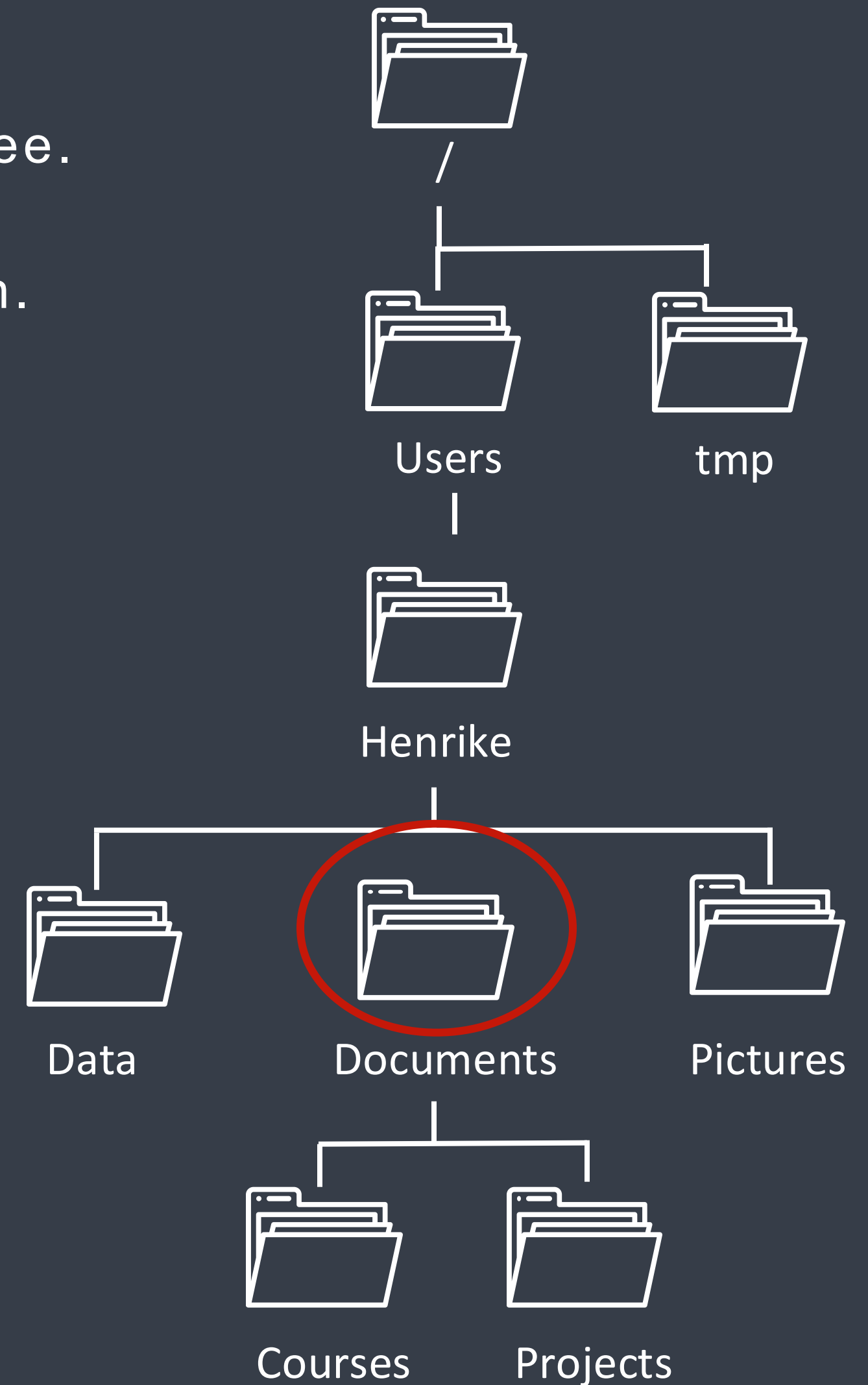
Sometimes not all folders are shown on the scheme.



PATHS

- Each file and directory exist in one specific place on the file tree.
- One could say they have an 'address' which is called their path.
- Paths can be full/absolute or relative.

`/Users/Henrike/Documents`

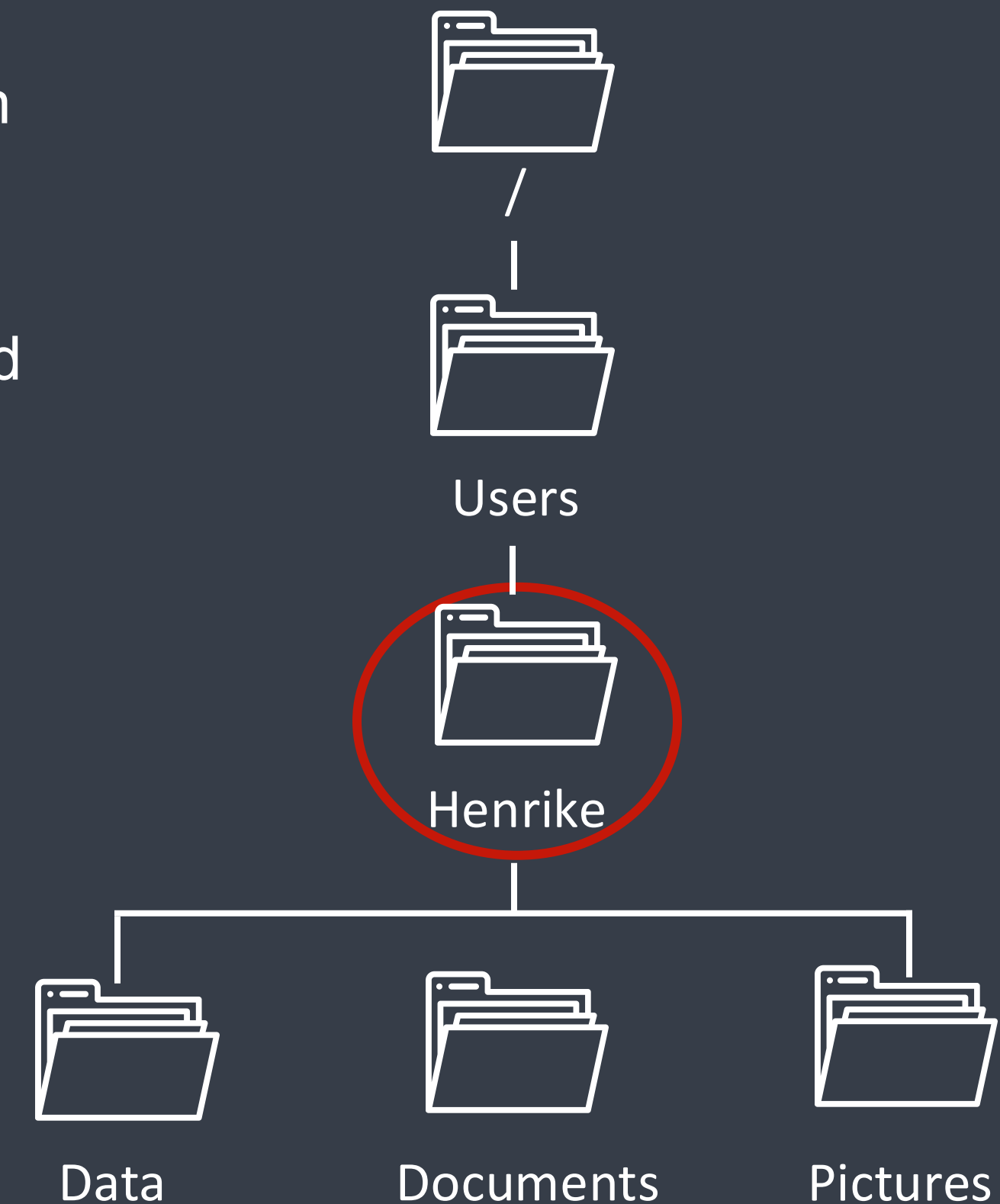


CURRENT DIRECTORY

- When you operate on the terminal you are always in a certain directory.
- The directory you are currently in is the working directory and you can print it with:

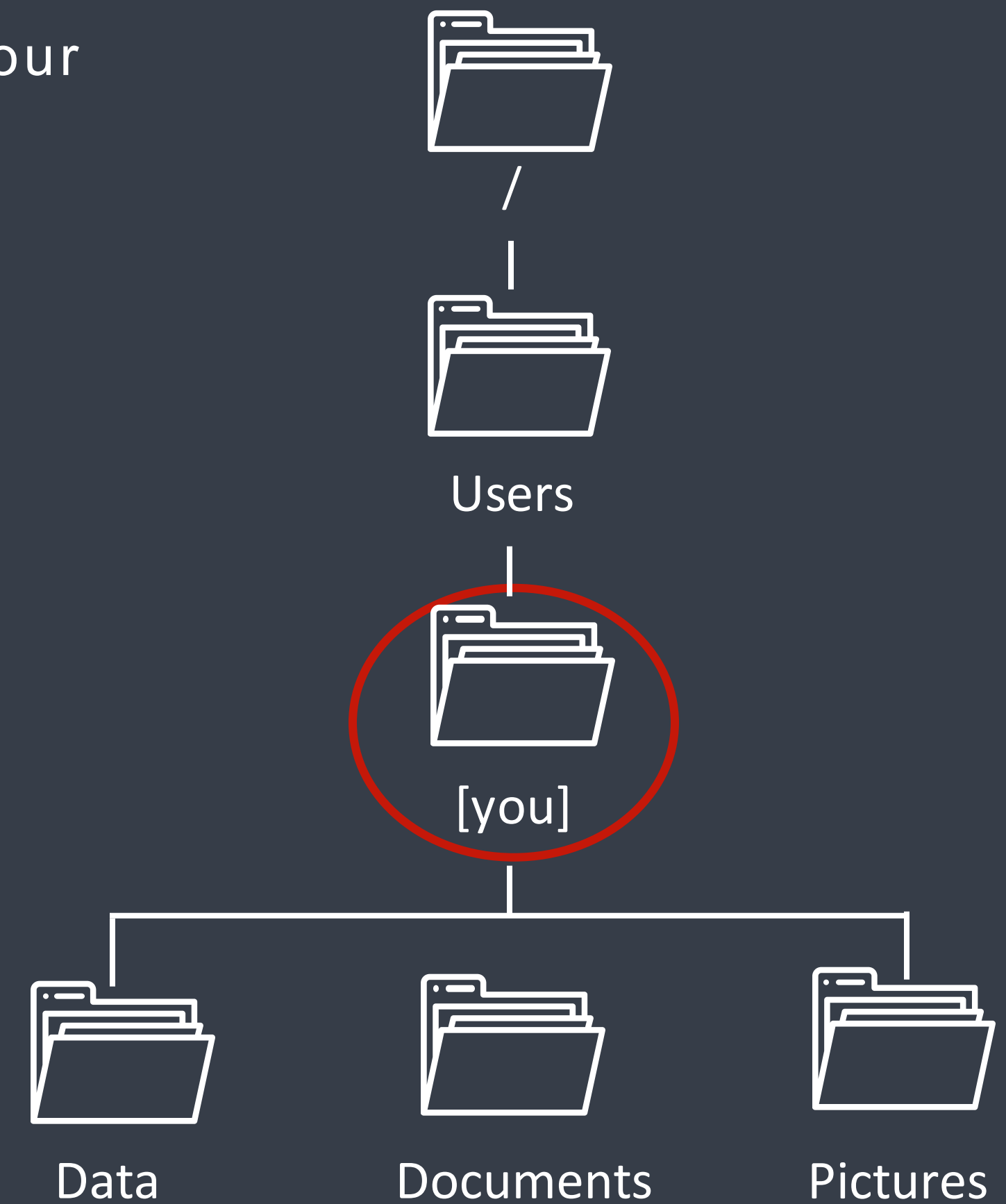
```
$ pwd
```

```
/Users/Henrike
```



THE HOME DIRECTORY

- The home directory is a special place in the file tree. It is your **user name** and is typically a subfolder of the 'users' folder.
- This is where you will be when you open your terminal
- The path to the home directory can be abbreviated with the symbol: ~



THE HOME DIRECTORY

What the home directory is called can differ between operating systems and computers.
All home directories are beautiful <3!

```
kgx936 — -zsh — 95x19
Last login: Tue Oct 31 13:56:40 on ttys000
kgx936@SUN1007442 ~ % pwd
/Users/kgx936
kgx936@SUN1007442 ~ %
```

```
11/11/2022 14:38.08 /home/mobaxterm pwd
/home/mobaxterm
```

```
henrike@henrike-ThinkPad-X390: ~
henrike@henrike-ThinkPad-X390: ~ 80x24
(base) henrike@henrike-ThinkPad-X390:~$ pwd
/home/henrike
(base) henrike@henrike-ThinkPad-X390:~$
```

```
MINGW64:/c/Users/pnv719
pnv719@SUN1022949 MINGW64 ~
$ pwd
/c/Users/pnv719
pnv719@SUN1022949 MINGW64 ~
$ _
```

As a rule of thumb, the home directory is generally two steps from the root '/' and it is personal.

When there are several users on the same machine each has their own home directory.

FULL PATHS

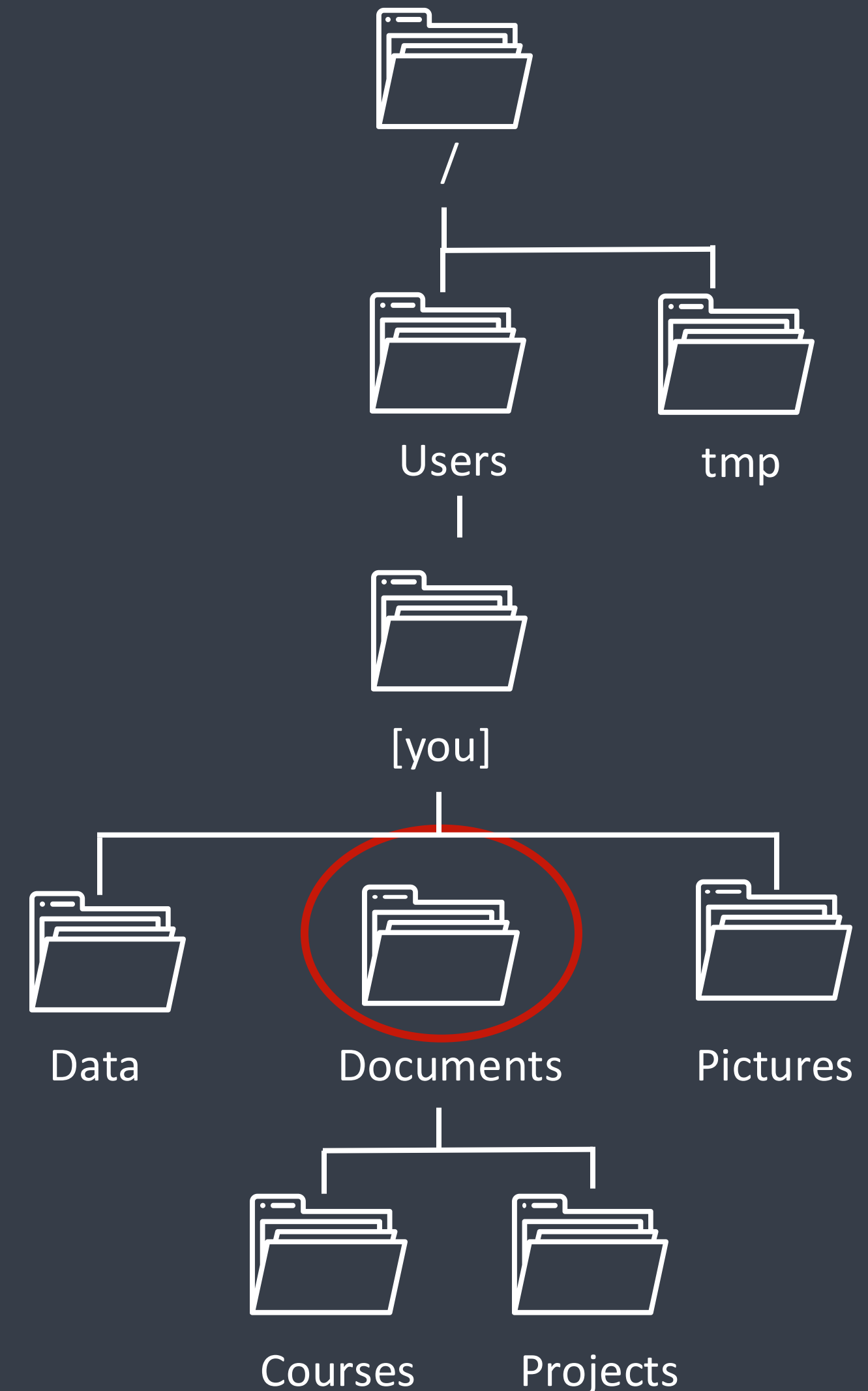
- Full paths start at the **root**. They are unambiguous because they describe the entire 'journey' starting from the root.
- Remember, ~ represents the path to your home directory.

`/Users/[you]/Documents`

and

`~/Documents`

- Are the same place and the same path and are **both** full paths.
- Any path starting with '/' or '~' is a full path.



RELATIVE PATHS

- A relative path is an address *relative* to the current wd.
- If in 'Documents', I address the file 'Commandline.pptx' as:

```
Courses/Just-Bash-It/Commandline.ppt
```

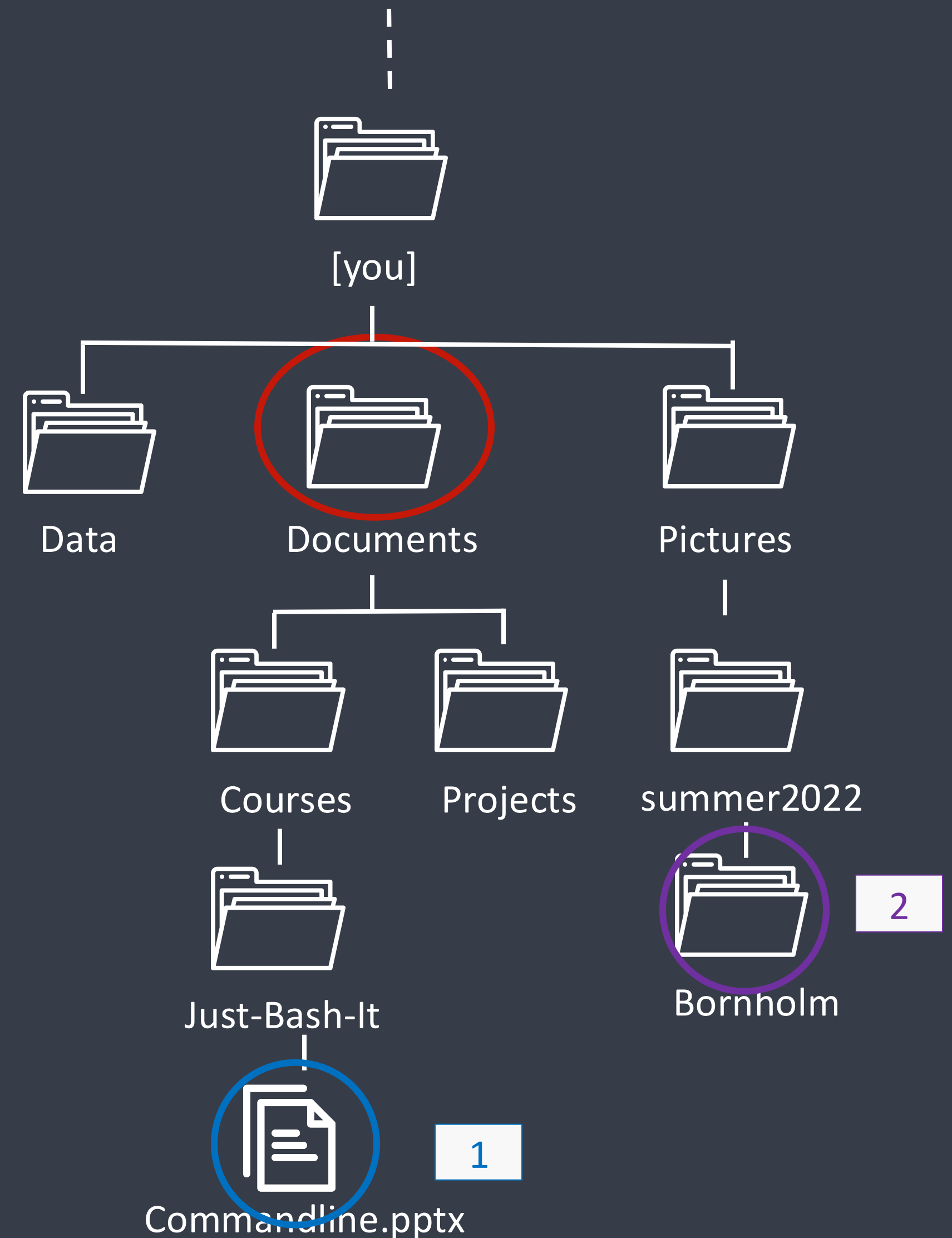
1

- Relative paths can include going up the directory tree:

```
../Pictures/summer2022/Bornholm
```

2

- This notation does not make sense for full paths as they start from the root and go downwards.



LISTING CONTENT

- To list the contents of a directory:

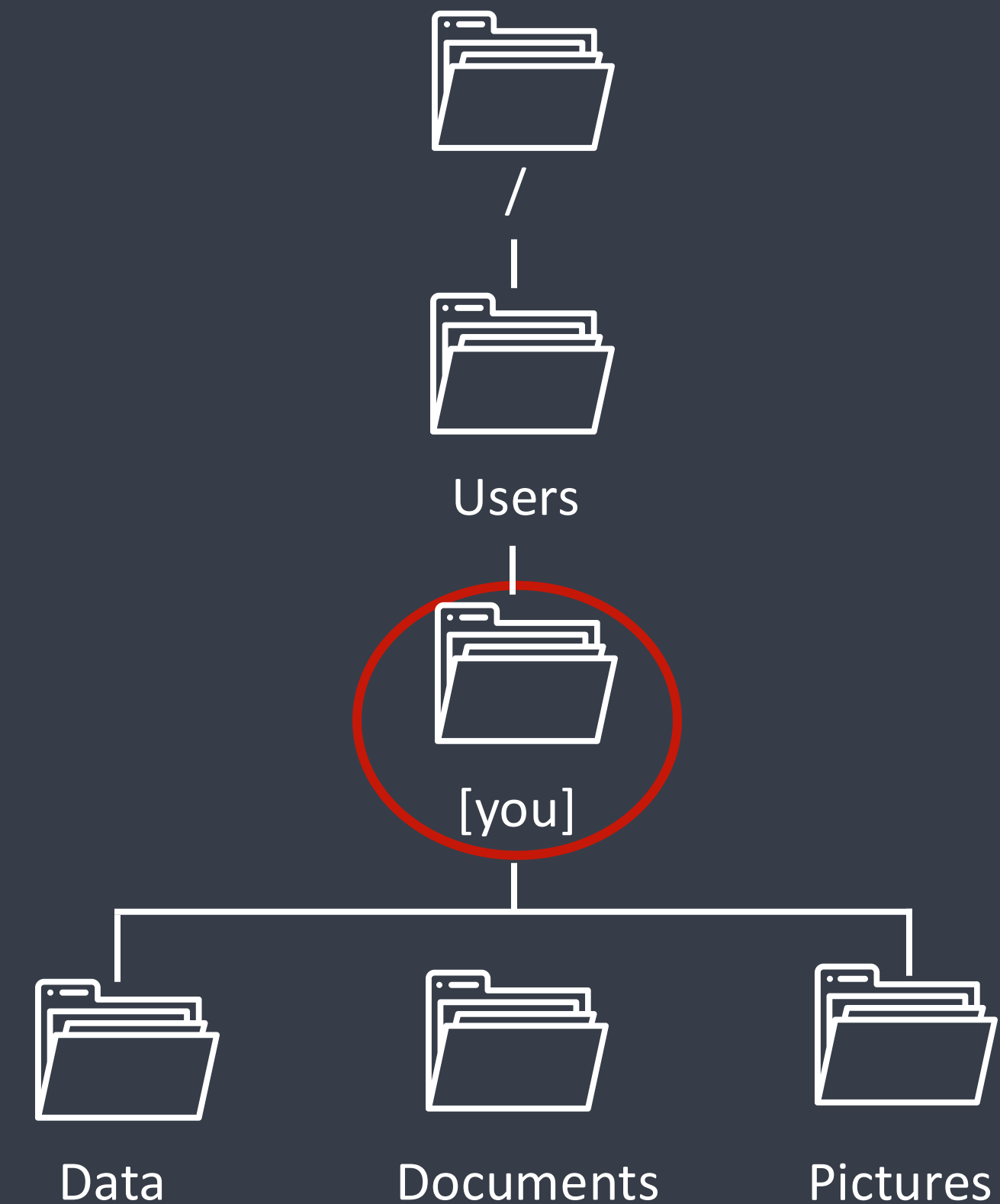
```
$ ls
```

- Example on the right will give you the Data, Documents and Pictures folders.
- List directories you are NOT currently inside of by specifying the **path** to them:

```
$ ls /Users
```

```
$ ls ~/Data
```

- ls has many useful options (**-l**, **-t** and **-h**, **-F**).



CHANGING DIRECTORIES

- From your **wd**, move into any subdirectory:

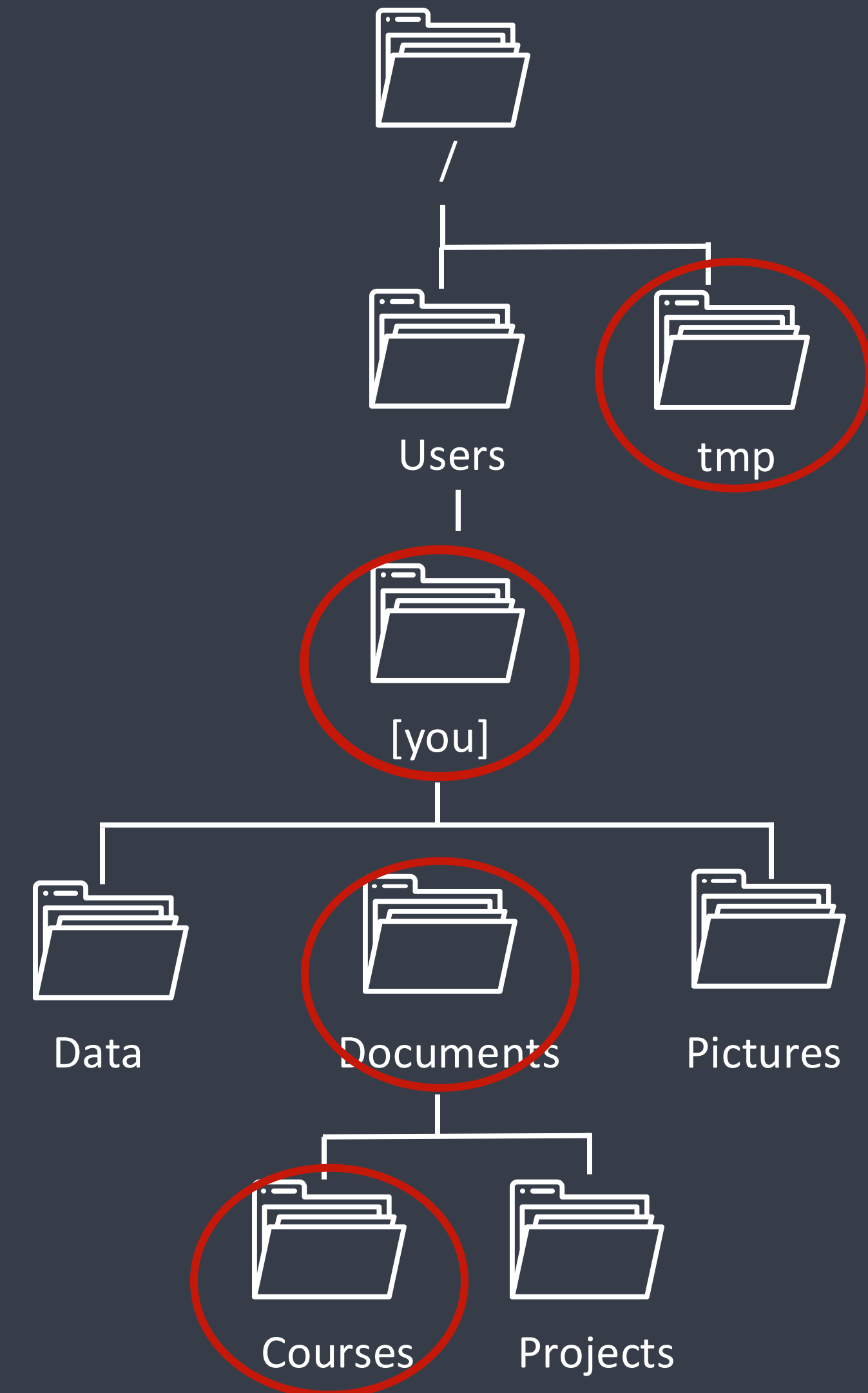
```
$ cd Documents
```

- You can also move several directories down:

```
$ cd Documents/Courses
```

- To move 'up' in the directory tree you use '../' .
Each time you write this, you go one level up.

```
$ cd ../ ../tmp
```



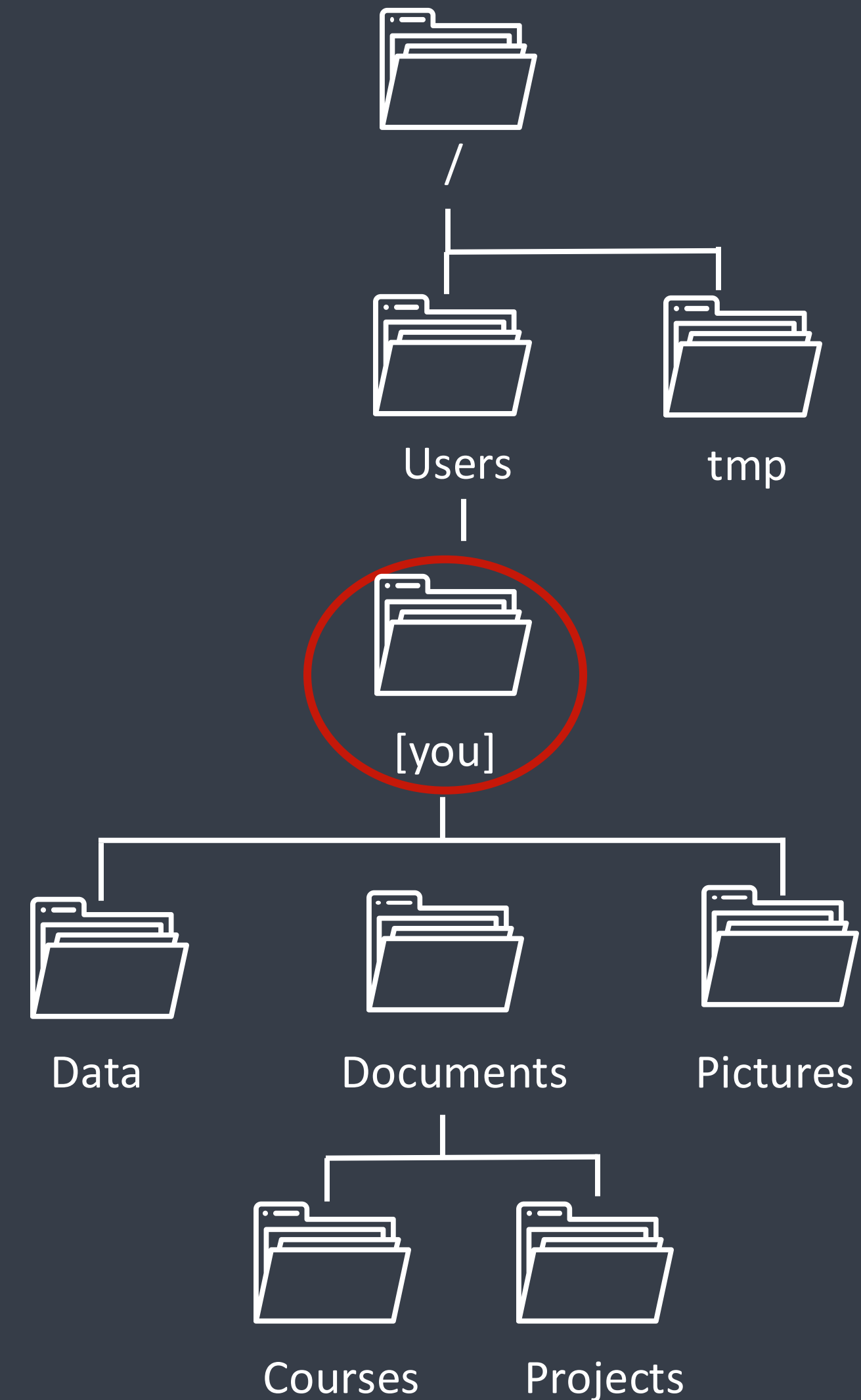
CHANGING DIRECTORIES

- `cd` without any arguments will always bring you to your home directory

```
$ cd
```

- `cd` with a dash will bring you to the last directory you were in:

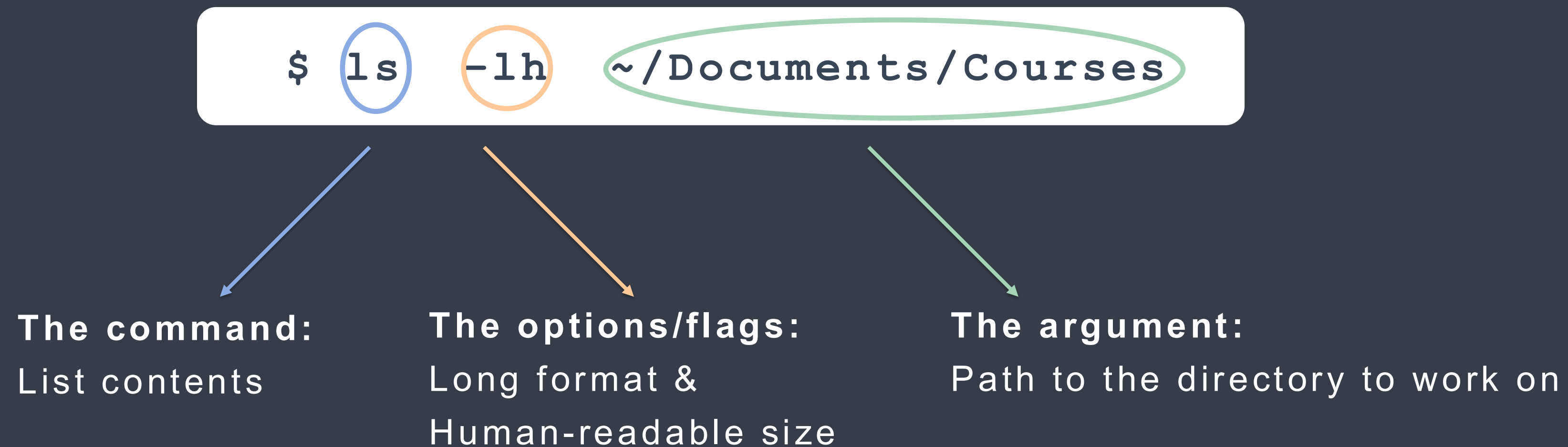
```
$ cd -
```



COMMANDS

We interact with the computer by issuing **commands** to it. You have already tried some like **ls**, **pwd**.

Some commands stand by themselves like **pwd**, but often they have arguments and options/flags:



We put a space it means we are proceeding to new part of the command.
File and directory names cannot contain spaces, they will be misinterpreted!

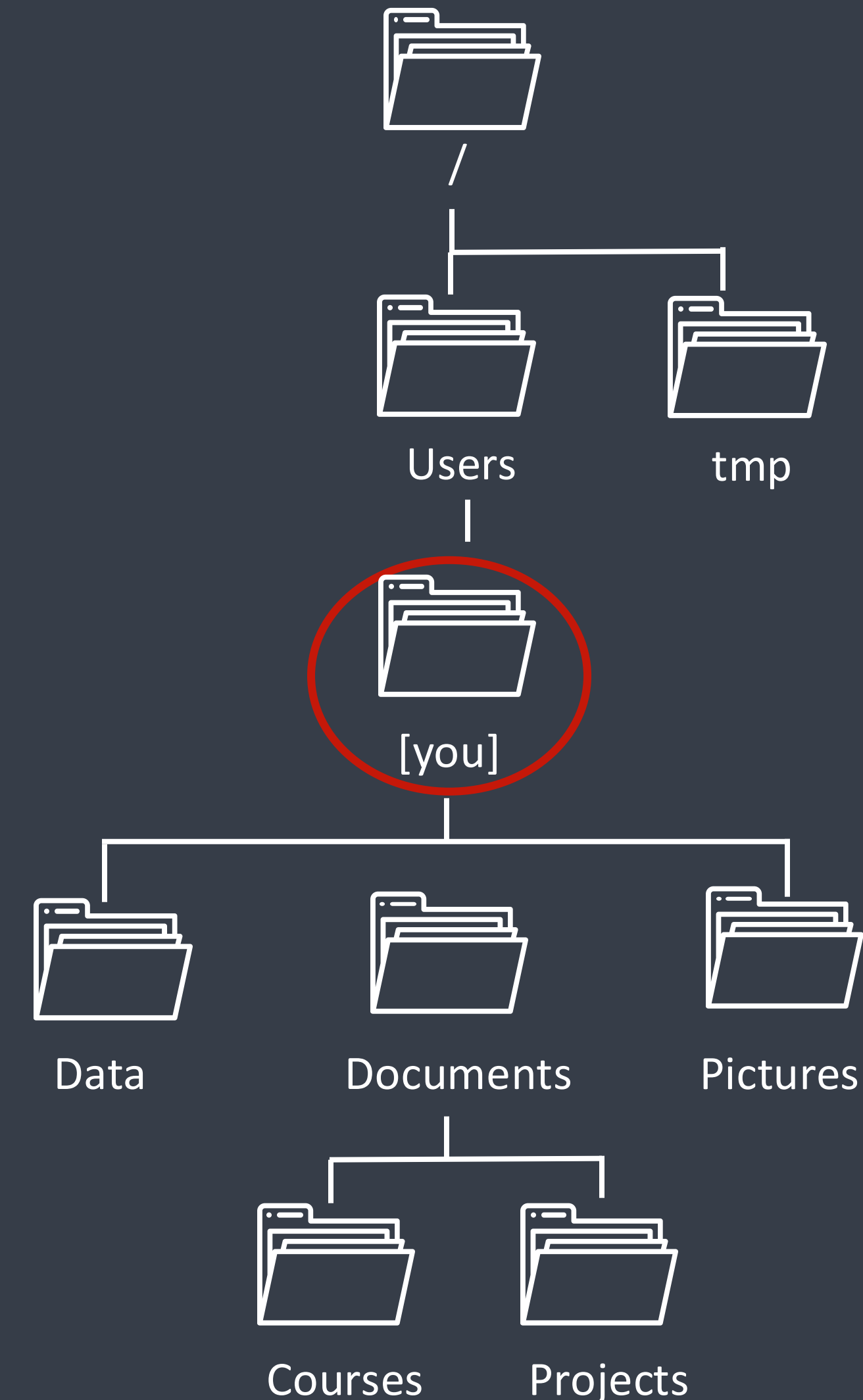
WHEN THINGS GO WRONG

Assuming I am in my home directory and I write:

```
$ cd Courses
```

What will happen?

```
MINGW64:/c/Users/pnv719/Documents/Just-Bash-It/Files
pnv719@SUN1022949 MINGW64 ~
$ cd Documents/Just-Bash-It/Files/
pnv719@SUN1022949 MINGW64 ~/Documents/Just-Bash-It/Files (june_2023)
$ ls
Data/  Documents/  Pictures/  Scripts/  readme.txt
pnv719@SUN1022949 MINGW64 ~/Documents/Just-Bash-It/Files (june_2023)
$ cd Courses
bash: cd: Courses: No such file or directory
pnv719@SUN1022949 MINGW64 ~/Documents/Just-Bash-It/Files (june_2023)
$ _
```



WHEN THINGS GO WRONG

1

Don't panic!

2

Read the error message

3

Try to correct your error

4

If lost, google or read the help page

```
MINGW64:/c/Users/pnv719/Documents/Just-Bash-It/Files

pnv719@SUN1022949 MINGW64 ~
$ cd Documents/Just-Bash-It/Files/

pnv719@SUN1022949 MINGW64 ~/Documents/Just-Bash-It/Files (june_2023)
$ ls
Data/  Documents/  Pictures/  Scripts/  readme.txt

pnv719@SUN1022949 MINGW64 ~/Documents/Just-Bash-It/Files (june_2023)
$ cd Courses
bash: cd: Courses: No such file or directory

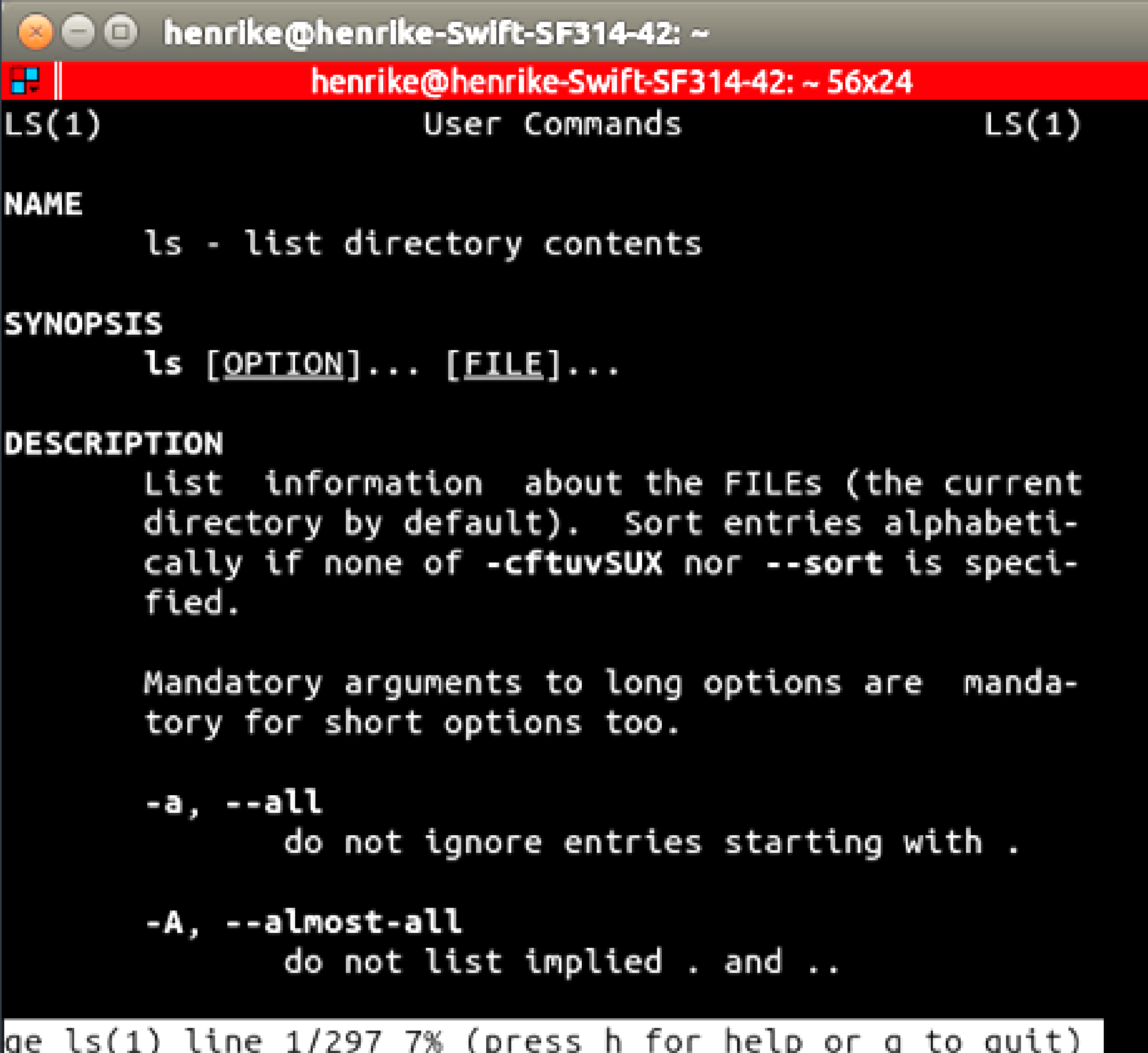
pnv719@SUN1022949 MINGW64 ~/Documents/Just-Bash-It/Files (june_2023)
$ _
```

GETTING HELP

- Many commands have a help page, i.e. a **manual**:

```
$ man ls
```

- The **man** page includes:
 - A description of what the command does
 - Explanation for arguments
 - Usage examples (the syntax of the command, what goes where)
- **man** is **not included in gitbash** but you can try **--help** or **-h** instead.
- You can always search the web if all else fails!



```
henrike@henrike-Swift-SF314-42: ~
henrike@henrike-Swift-SF314-42: ~ 56x24
LS(1)                                User Commands                                LS(1)

NAME
    ls - list directory contents

SYNOPSIS
    ls [OPTION]... [FILE]...

DESCRIPTION
    List information about the FILES (the current
    directory by default).  Sort entries alphabeti-
    cally if none of -cftuvSUX nor --sort is speci-
    fied.

    Mandatory arguments to long options are manda-
    tory for short options too.

    -a, --all
        do not ignore entries starting with .

    -A, --almost-all
        do not list implied . and ..

ge ls(1) line 1/297 7% (press h for help or q to quit)
```

CHEAT SHEET 1

WHERE & WHAT

pwd # print working dir

cd # go to home dir

cd [path] # change dir (remember path)

ls # list dir content

man [cmd] # get info about command

[cmd] --help # view the help for command



```
Last login: Fri Jun 24 15:05:34 on ttys000
[kgx936@SUN1007442 ~ % echo "Just Bash It"
Just Bash It
[kgx936@SUN1007442 ~ % █
```

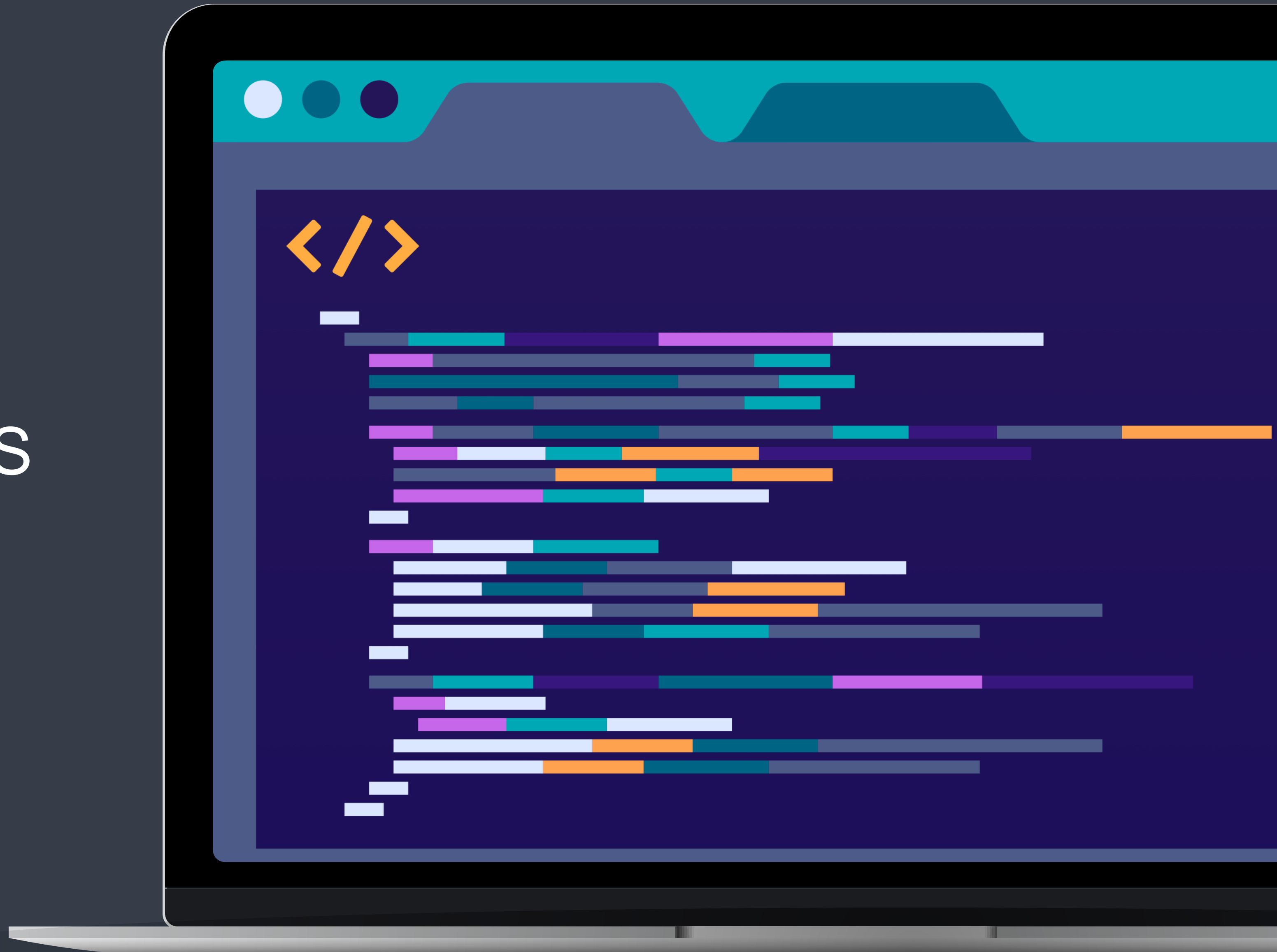


Let's Bash It!

Open your terminal on your computer and do **Exercise 1**.

We're here to help if you get stuck!

2. FILE OPERATIONS



MOVING FILES AND DIRS

Go into the '*Documents*' folder under the 'Files' of the github repo.

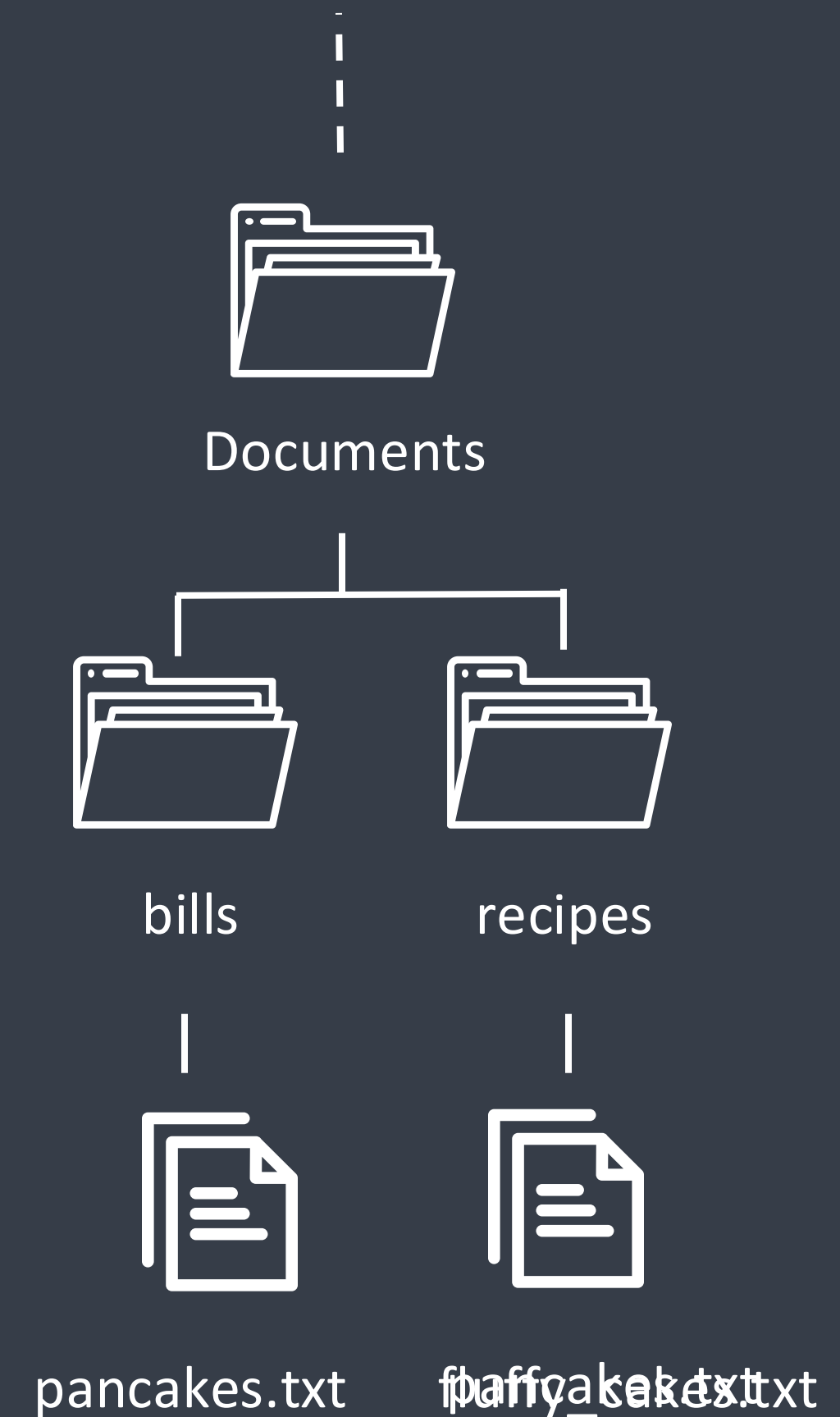
- You can move files (directories) with **mv**:

```
$ mv pancakes.txt ../recipes
```

- **mv** can also be used for renaming:

```
$ mv pancakes.txt fluffy_cakes.txt
```

- Do NOT move or remove your home directory or other system critical dirs. This may have unintended consequences.



COPY

- Files can be copied with **cp**:
- First argument is the file and the second argument is the destination

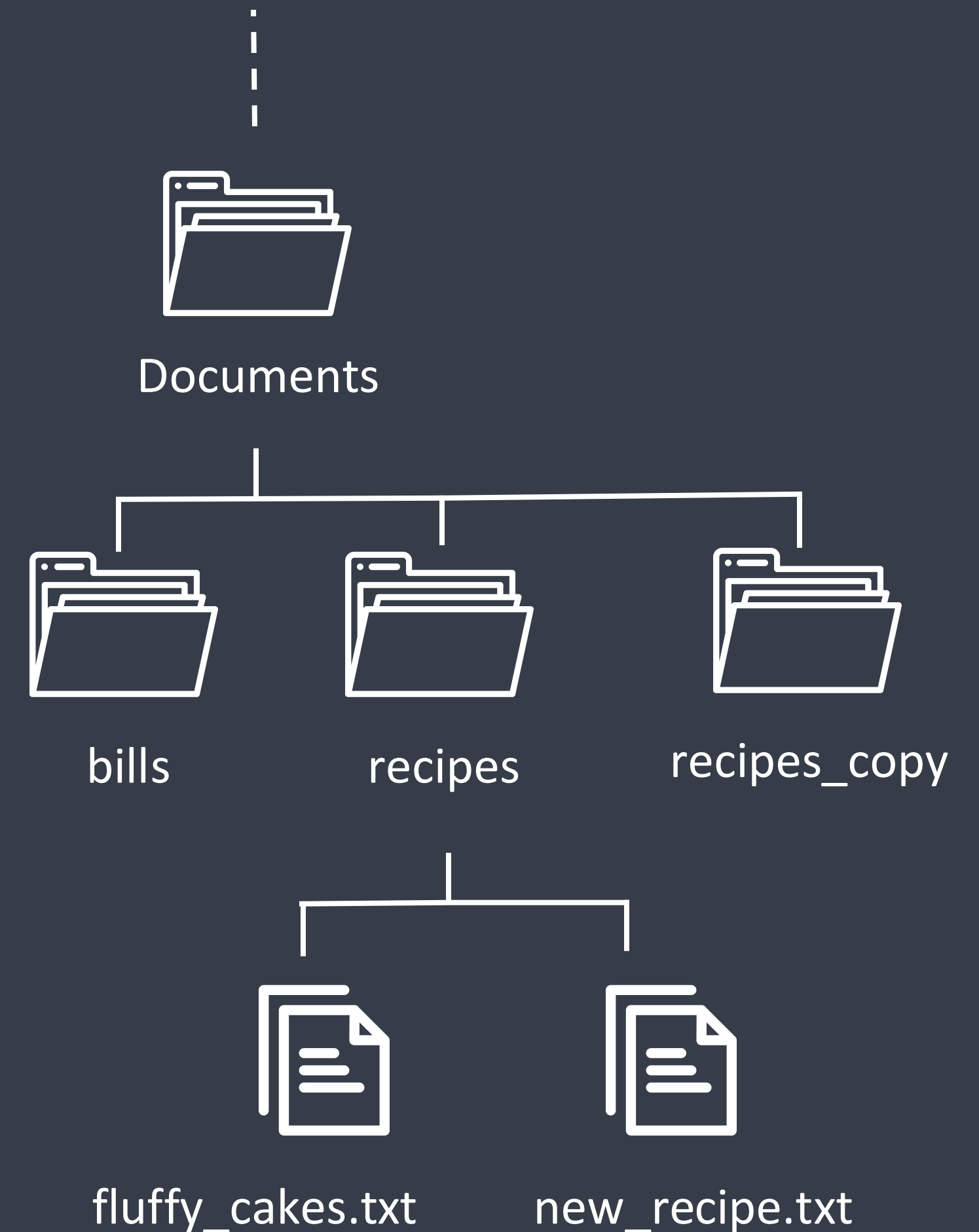
```
$ cp fluffy_cakes.txt new_recipe.txt
```

- To copy a directory (and its contents!) use **-r**:

```
$ cp -r recipes recipes_copy
```

- You must specify a destination when you copy.
- The destination can be the name of the new file, or another directory:

```
$ cp fluffy_cakes.txt ../recipes_copy/
```



REMOVE

- To remove files and dirs, use `rm`

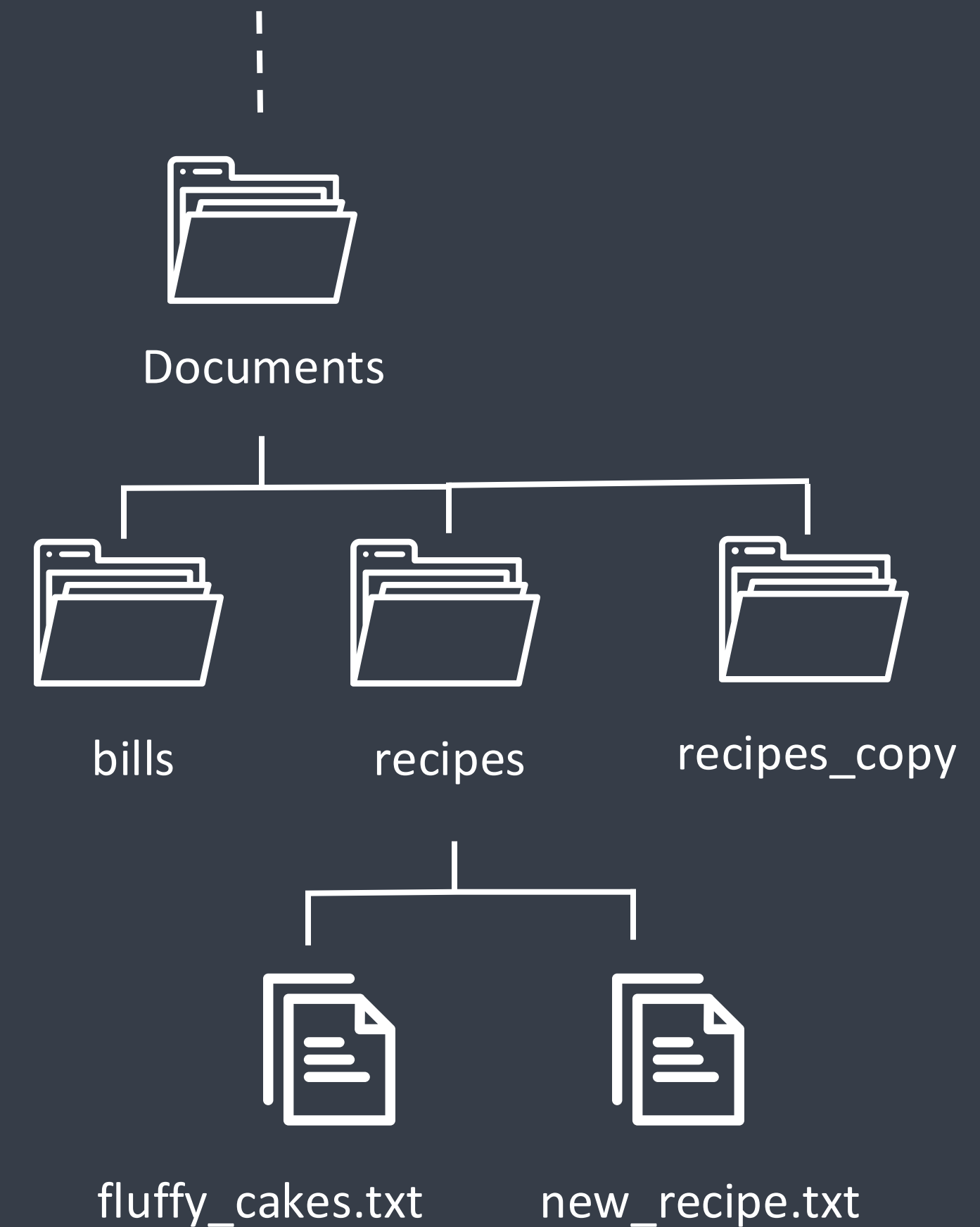
```
$ rm new_recipe.txt
```

- Just like when copying, you need the `-r` flag when removing a directory:

```
$ rm -r recipes_copy
```



- **rm is permanent!** You cannot recover a removed file and you will not be asked whether you are sure you want it gone.



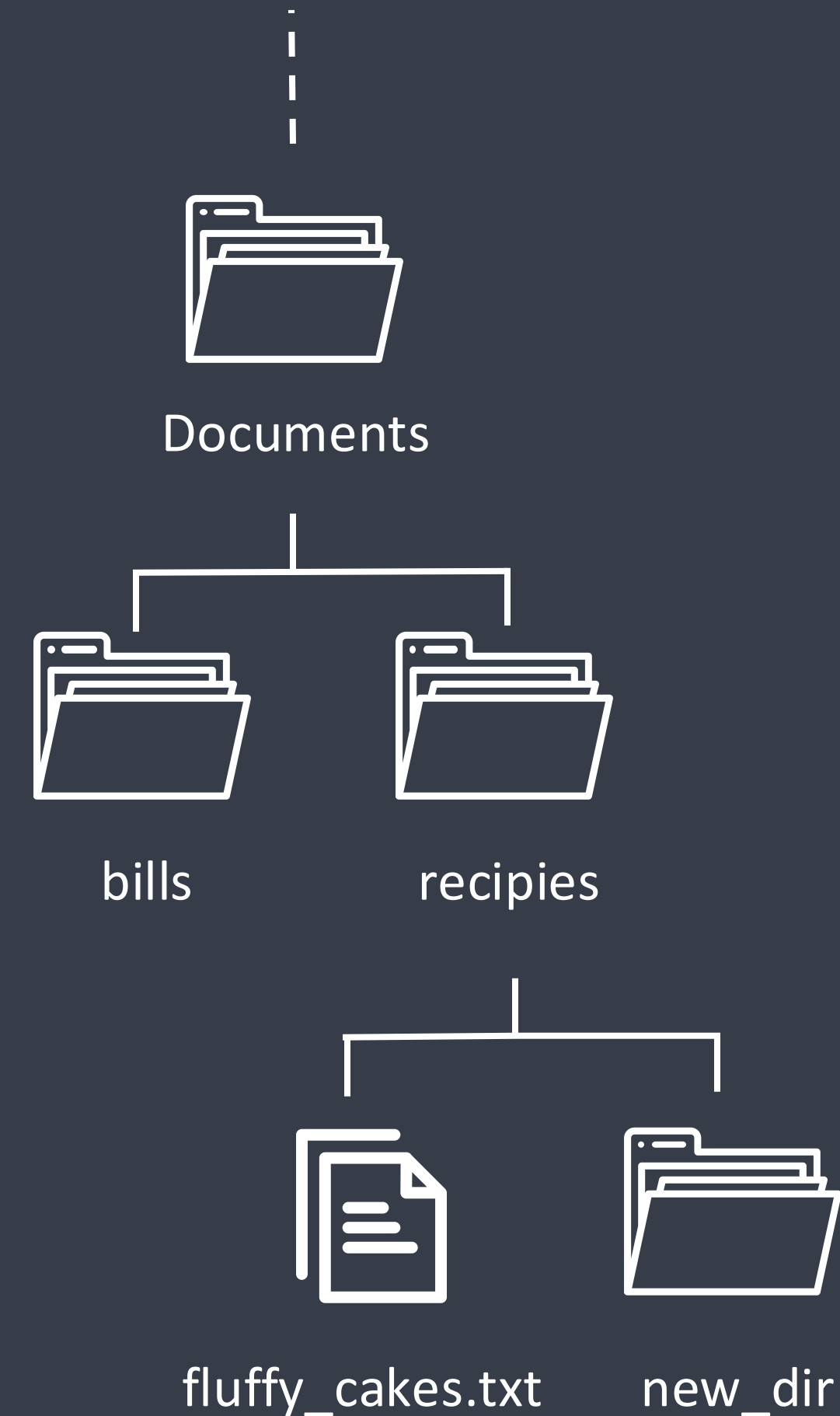
CREATING FILES AND DIRECTORIES

- To create a new directory we use **mkdir**:

```
$ mkdir new_dir
```

- The new directory will not have any content.
- Many ways to create files, mostly as output from programs, but you can always create an empty with **touch**:

```
$ touch a_new_file.txt
```

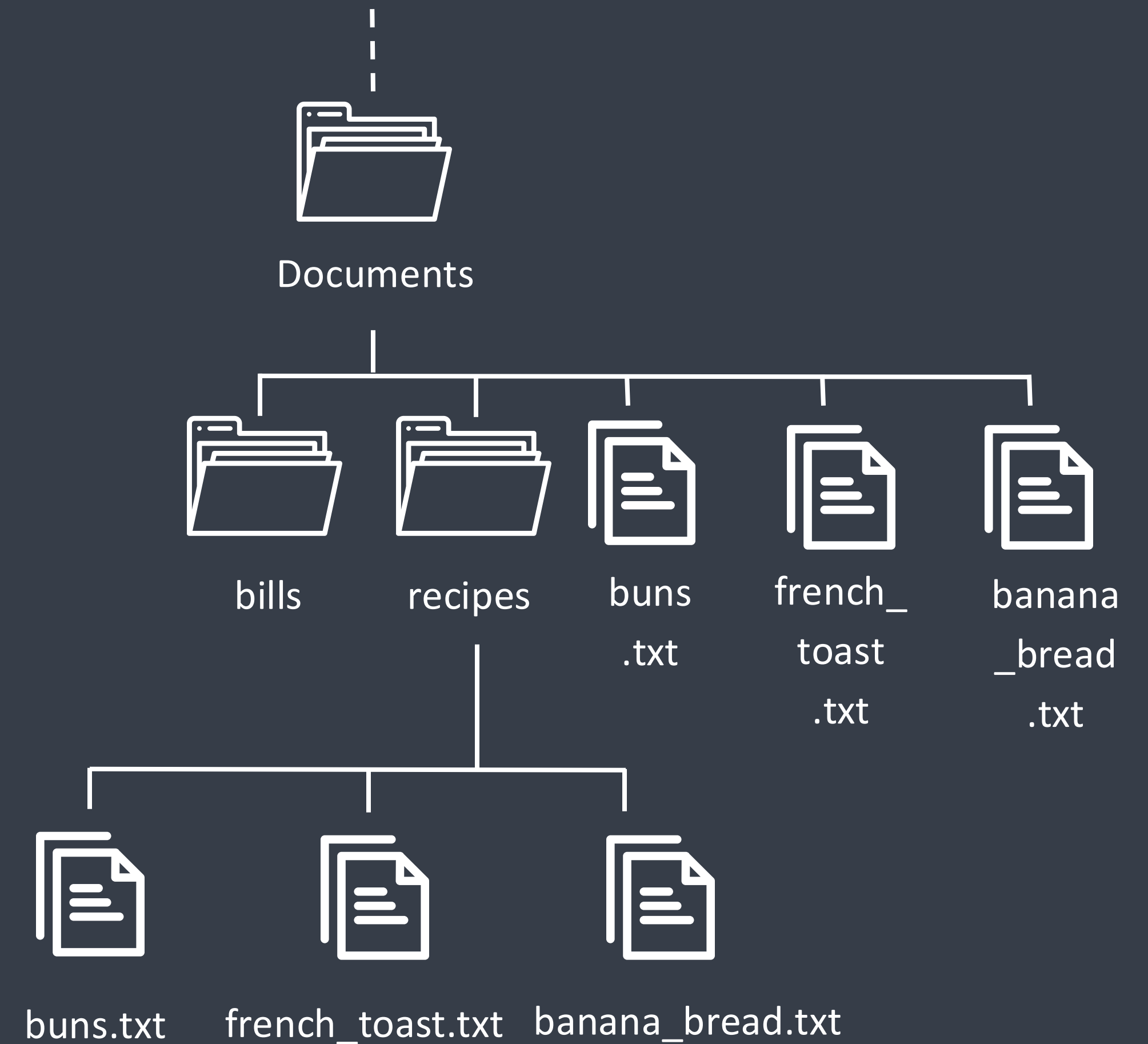


OPERATING ON SEVERAL FILES

- We can also interact with several at once!
- This would take all files named something .txt and move them one directory up:

```
$ mv *.txt ../
```

- The star '*' is called a wildcard. It matches every character (letter, number, ect).



OPERATING ON SEVERAL FILES

- Use the wildcard '*' to restrict actions to certain files and directories.
- By default 'ls' lists the entire directory.
- If we only want to see files with names ending in *.md* we can specify that:

```
$ ls *.md
```



CHEAT SHEET 2

WHERE & WHAT

pwd # print working dir

cd # go to home dir

cd [path] # change dir (remember path)

ls # list dir content

man [cmd] # get info about command

[cmd] --help # view the help for command

FILE/DIR BASICS

rm [name] # remove file

rm -r [name] # remove dir

cp [name] # copy a file/dir

mv [name] [path] # move file/dir

mkdir [name] # make dir

touch [name] # create file



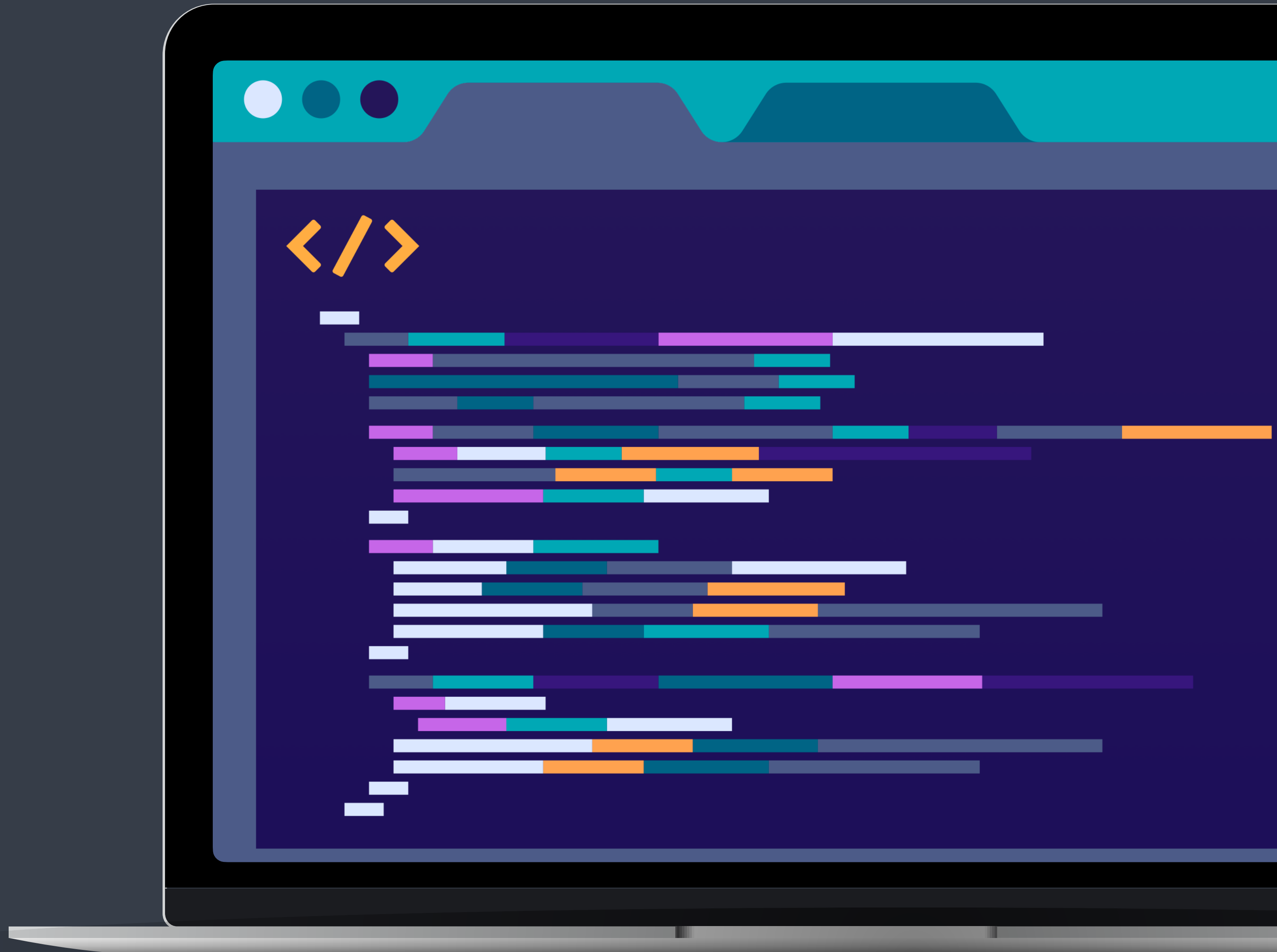
```
Last login: Fri Jun 24 15:05:34 on ttys000
[kgx936@SUN1007442 ~ % echo "Just Bash It"
Just Bash It
[kgx936@SUN1007442 ~ % █
```



Let's Bash It!

Time for **Exercise 2!**

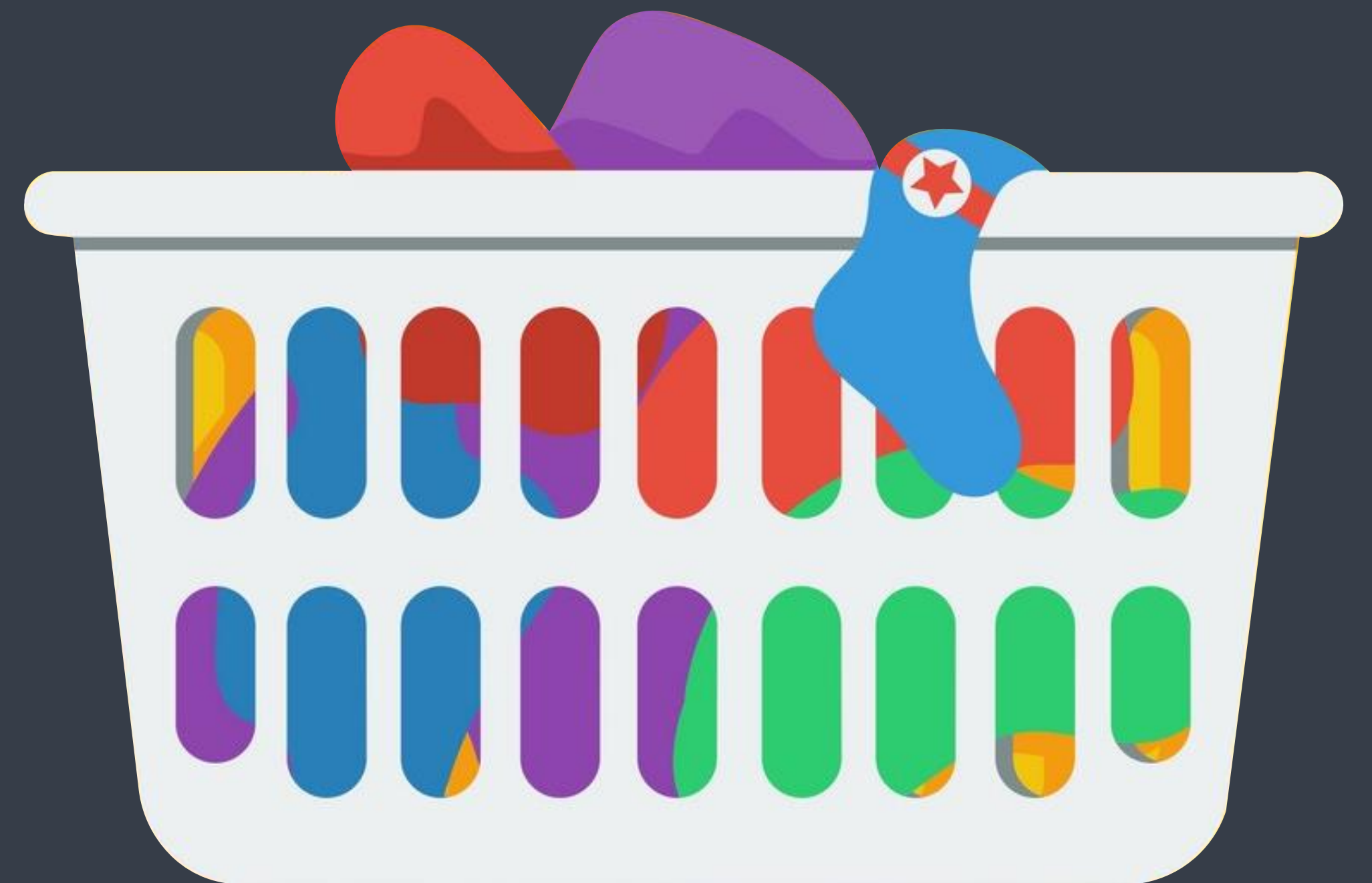
3. PROJECT ORGANIZATION & BACKUP



IS YOUR COMPUTER A LAUNDRY BASKET?

Why should I care about directory structure and file naming?

- Not nice to not “work” in a mess.
- If your computer crashes it is MUCH easier to recover work if files/directories are structured.
- If you apply a command (bash, R, Python) to a group of files/directories, naming is important!
- Large files should be stored smartly to save space (storage & memory).



DO'S and DON'TS



Consistent and Logical Directory Structure



Consistant & Non-redundant File Naming



**Version Control System
git/GitHub & Back-up**



Clean and Update Your Computer Regularly



Don't Use Symbols or Spaces in File/Directory Names



Don't store multiple copies of file, instead point to it (path).



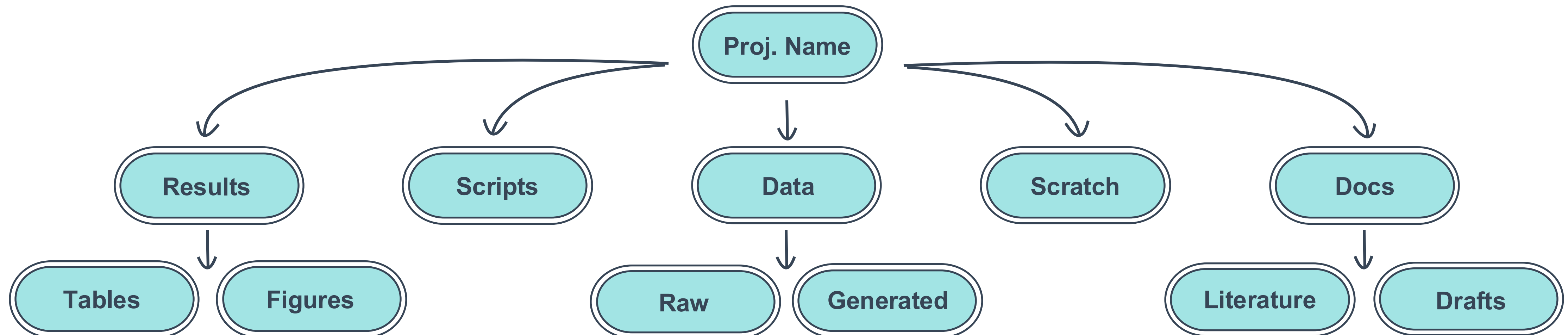
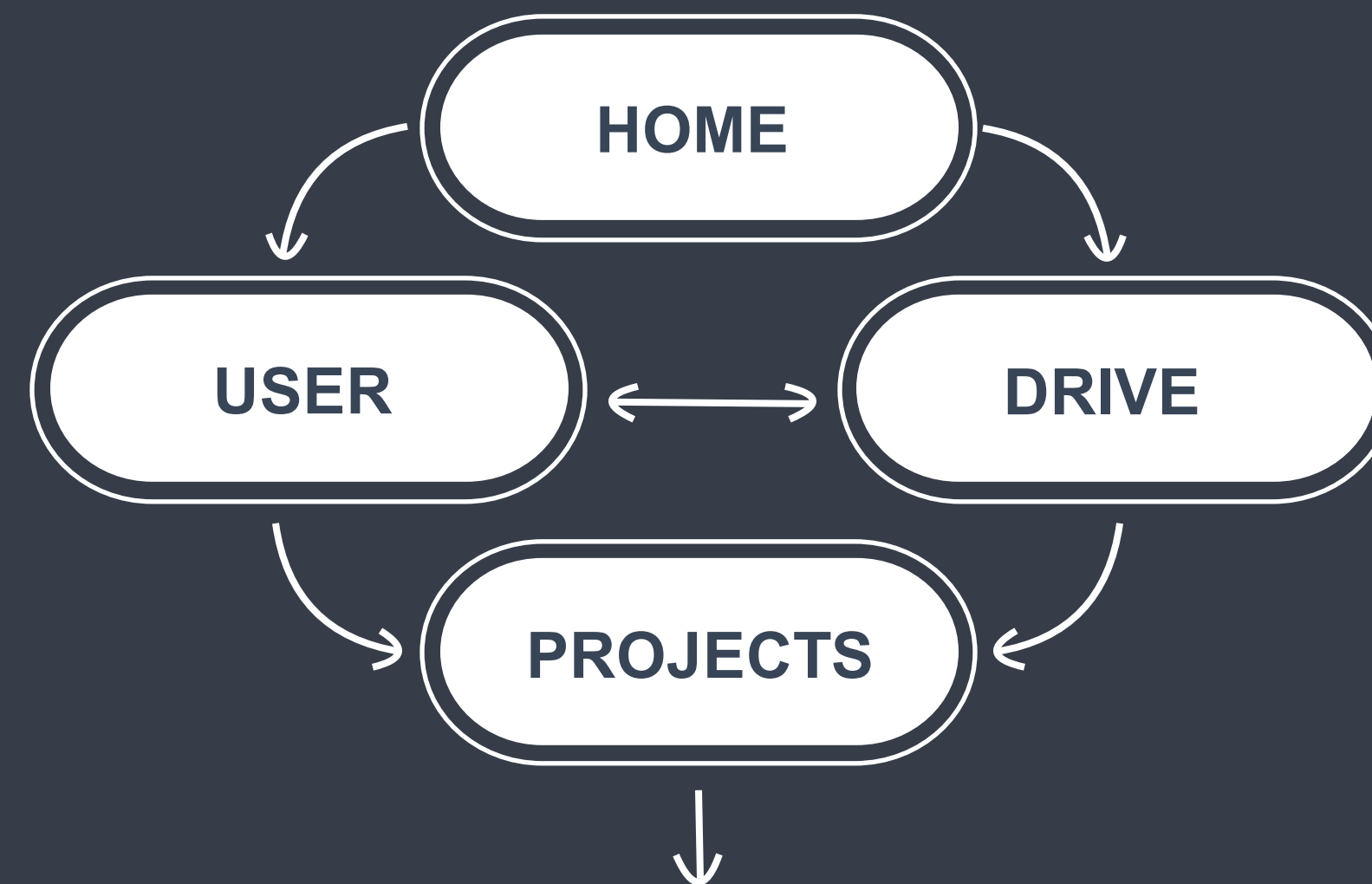
Don't Store Large Data/Files on Your Local Computer



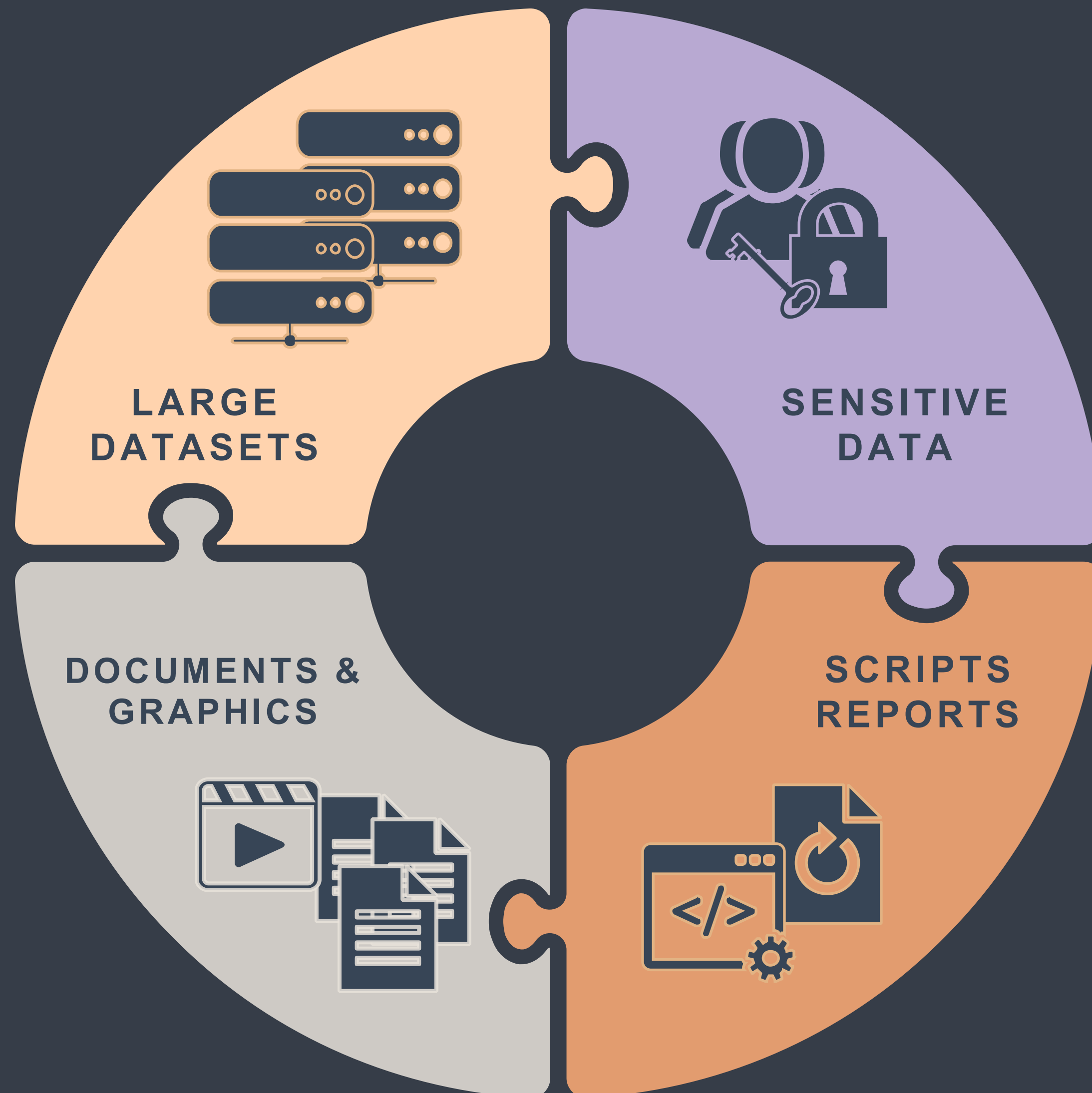
Don't Store Sensitive Datasets on Your Computer

A SUGGESTION FOR STRUCTURE

LOCAL COMPUTER



STORAGE & BACKUP



- **BIG DATA & SENSITIVE DATA:**
 - Your own KU drives (S-drive)
 - ERDA & SIF (KU data storage)
 - Lab or department server
 - Lab or department cloud solution
- **DOCX, EXCEL, POWERPOINT:**
 - Google Drive
 - Dropbox
 - KU drives
- **SCRIPTS:**
 - git/GitHub**
 - (KU drives)

GIT & GITHUB

Git is a version control software, created by Linus Torvalds for the management of the Linux kernel.

Other clients exist.

<https://git-scm.com>

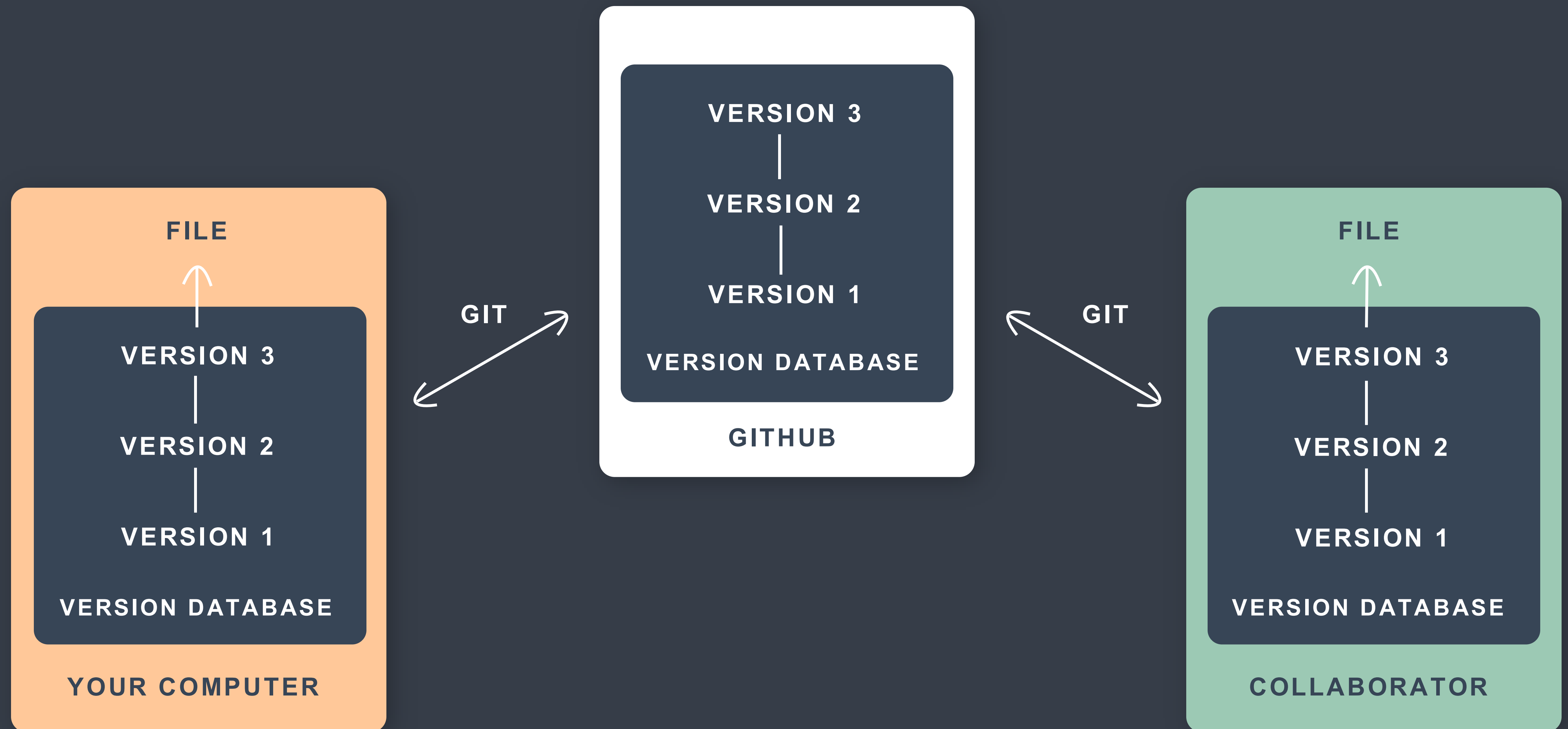


GitHub is a web-based hosting service for version control that uses git

Other services exist.

<https://github.com>

WHAT IS A VERSION CONTROL SYSTEM?



GIT WORKS FROM THE COMMAND LINE

Git is used from the command line to edit directories & files in a version controlled way

A git command always begins with `git`

Git enables a local computer to interact with a cloud service (GitHub) to back up your work



Sounds cool, yes!
Take our 'Introduction to git & GitHub'
Workshop and learn how to git

```
git clone [your repository]
```

```
Username: [your user]
```

```
Password: [your password]
```

```
git status (status of the repo)
```

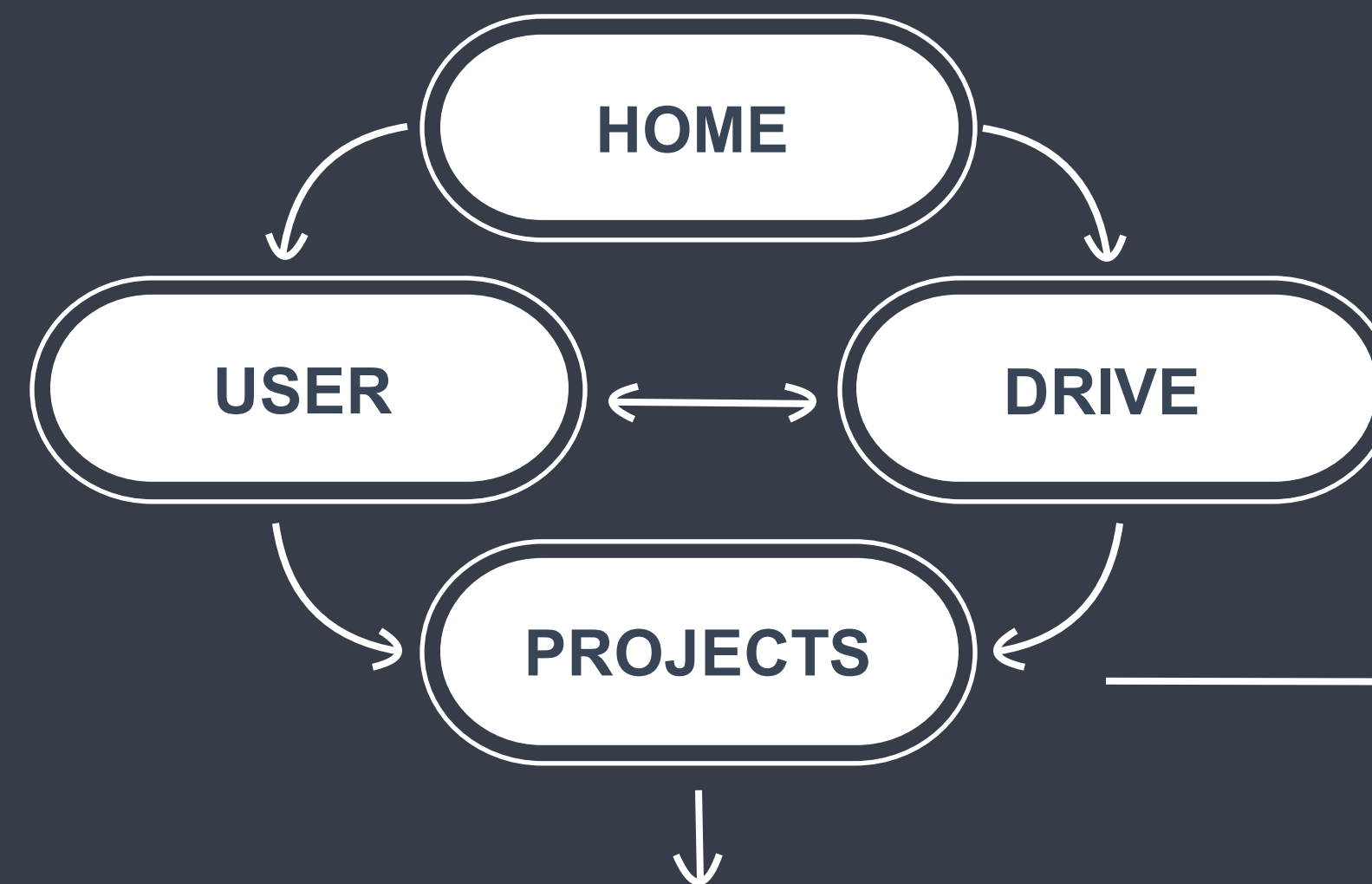
```
git add (add new files or changes)
```

```
git commit -m (commit the changes)
```

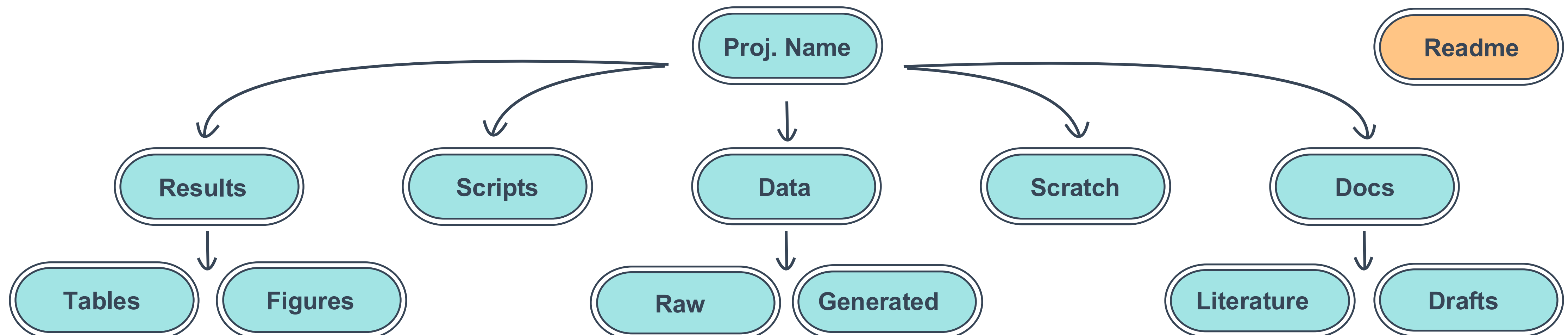
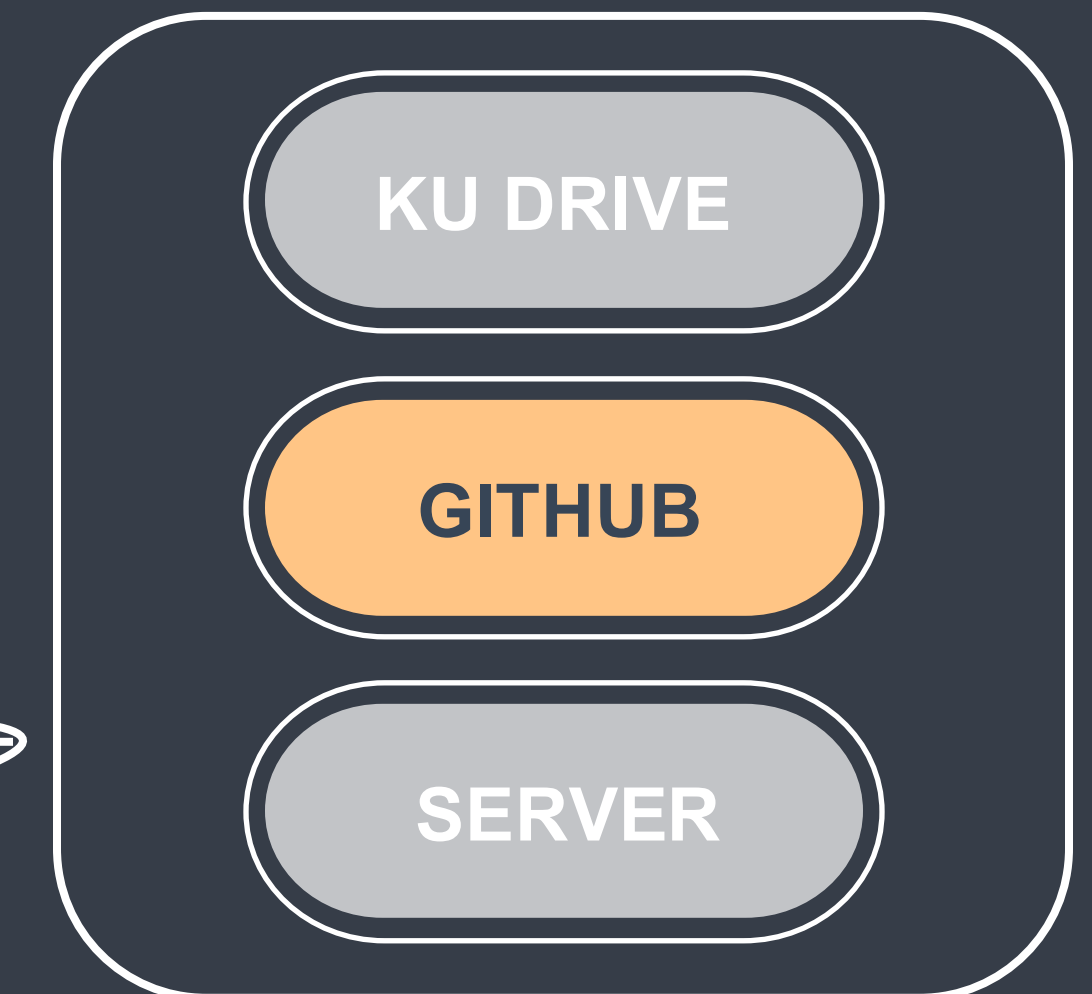
```
git push (push the changes to GitHub)
```

A SUGGESTION FOR STRUCTURE

LOCAL COMPUTER

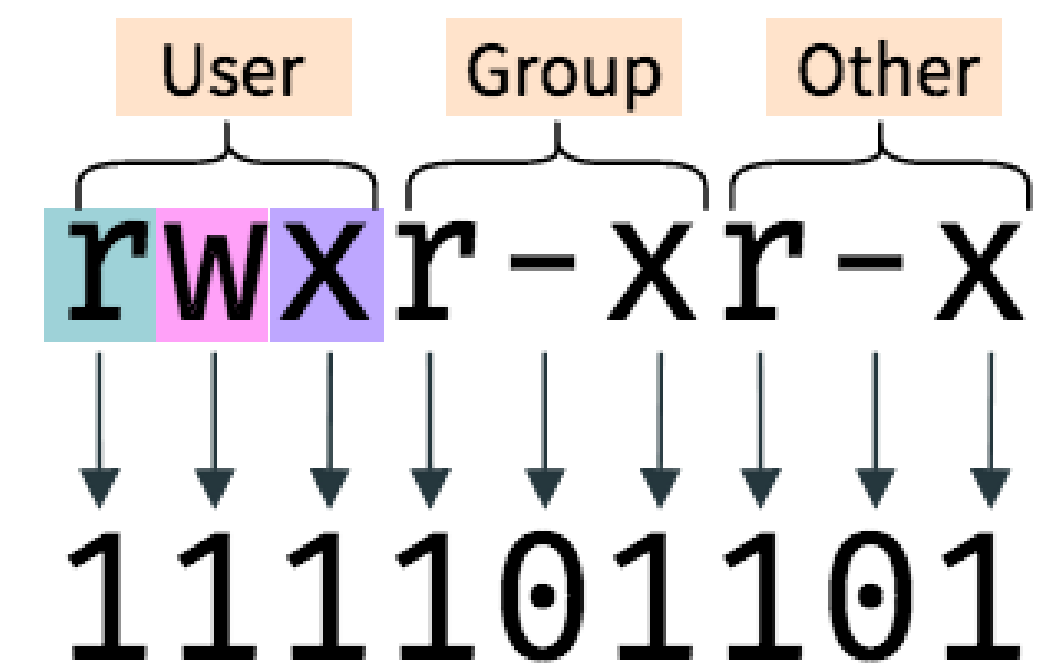


EXTERNAL BACKUP



USERS, GROUPS & PERMISSIONS

- Permission settings denote who can **read**, **write** (*edit*) and **execute** a file/dir
- On linux-based systems (Mac, Ubuntu) you can use the `chmod` command to set permissions **IF** you are a system administrator (sys admin).
 - **Private computer** : You are sys admin
 - **KU computer** : You may be the system admin
 - **Shared KU drives** : You are not sys admin
 - **HPCs, Servers & Clouds** : You are not sys admin
- On Windows there are alternatives to set permissions with powershell (you can google this if needed).



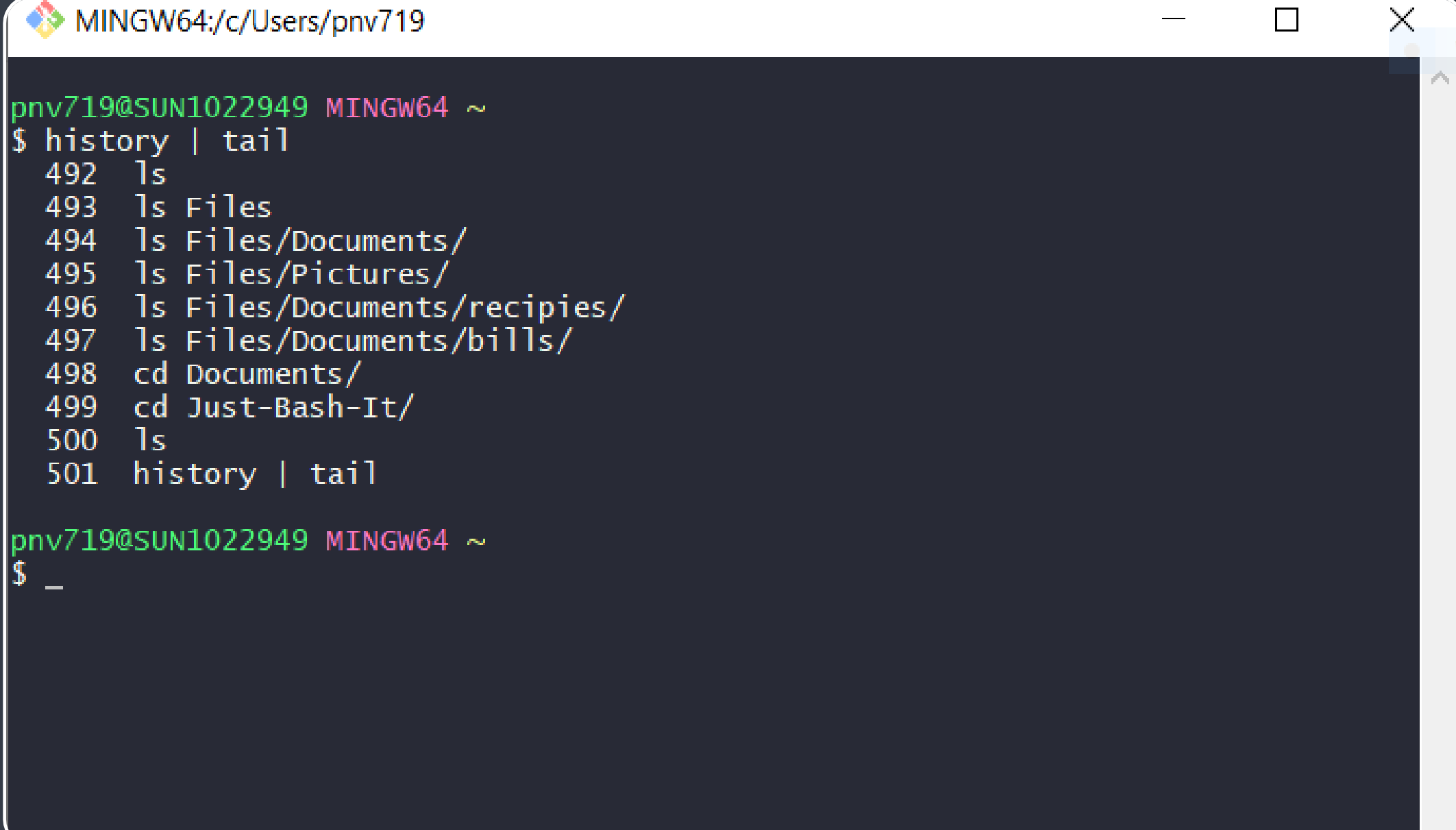
```
chmod +rwx [filename]
chmod -wx [directoryname]

chmod u+rw [filename]
chmod g-w [filename]
chmod o-rw [filename]
```

BACK TO THE PAST

“Those who don’t know history are doomed to retype their commands.”

- **history** command brings up a list of previously used commands
- The most recent command is last, the oldest command is first
- Can use head/tail to only see the beginning/end



```
MINGW64:/c/Users/pnv719
pnv719@SUN1022949 MINGW64 ~
$ history | tail
492  ls
493  ls Files
494  ls Files/Documents/
495  ls Files/Pictures/
496  ls Files/Documents/recipes/
497  ls Files/Documents/bills/
498  cd Documents/
499  cd Just-Bash-It/
500  ls
501  history | tail

pnv719@SUN1022949 MINGW64 ~
$ _
```

Pro tip:
Use the **'grep'** command to search your history or bring up the interactive **'reverse-i-search'** with **ctrl+r**.

CHEAT SHEET 3

WHERE & WHAT

```
pwd # print working dir
cd # go to home dir
cd [path] # change dir (remember path)
ls # list dir content
man [cmd] # get info about command
[cmd] --help # view the help for command
```

FILE/DIR BASICS

```
rm [name] # remove file or dir
cp [name] # copy a file/dir
mv [name] [path] # move file/dir

touch [name] # make a file
mkdir [name] # make a dir
```

SPECIAL CHARACTERS

```
* # select everything
/ # forward slash paths
\ # escape character (don't use for now)
.. # one dir up/back
. # current dir
- # denotes a flag/argument
```

SIZE & PERMISSION

```
ls -lh * # all sizes
ls -lh [name] # file/dir size
du -hA # disk space
chmod +rwx [name] # add permission
chmod -rwx [name] # remove permission
chmod ugo+rwx [name] # specify who, add permission
```

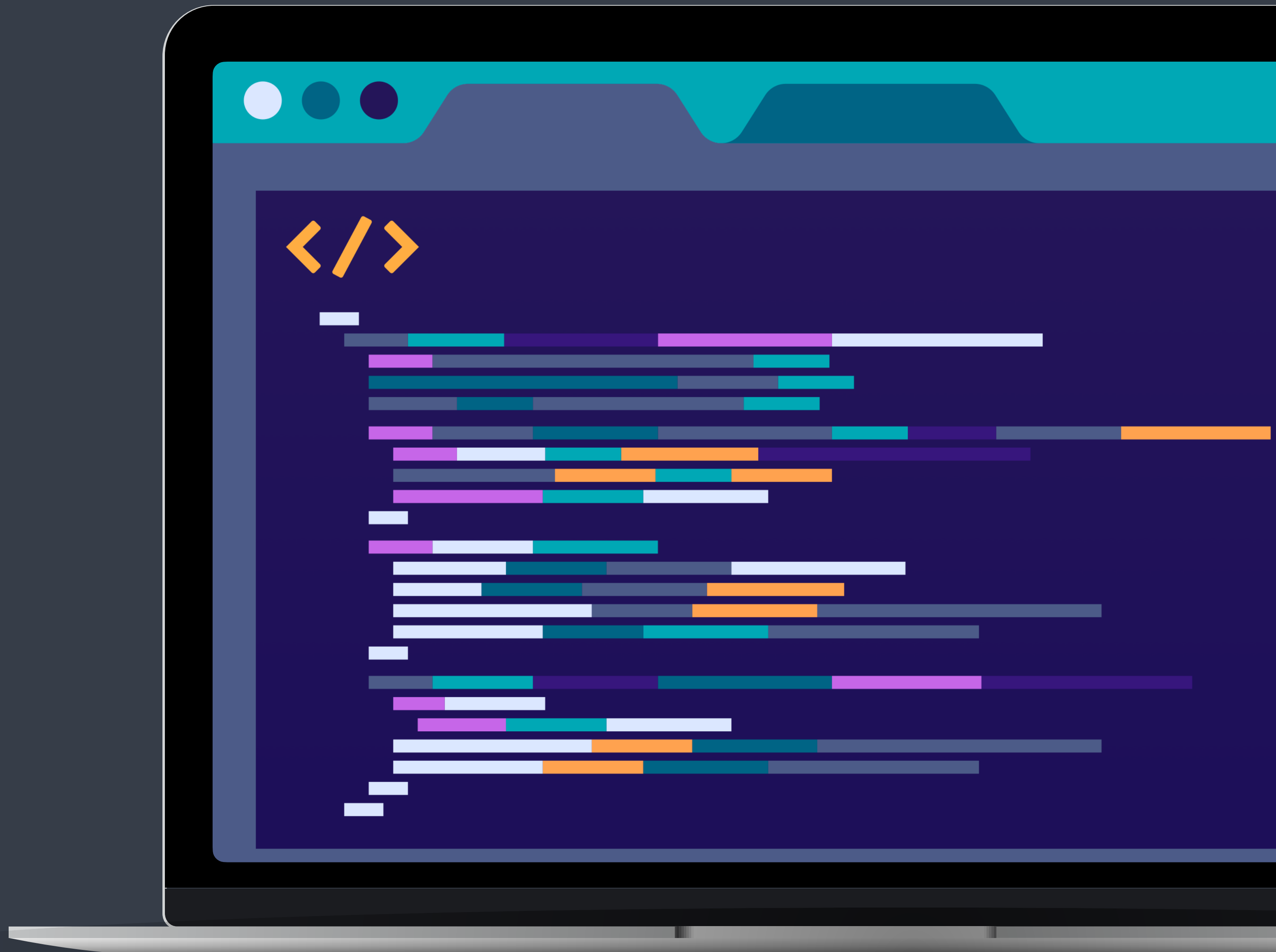


```
Last login: Fri Jun 24 15:05:34 on ttys000
[kgx936@SUN1007442 ~ % echo "Just Bash It"
Just Bash It
[kgx936@SUN1007442 ~ % █
```



Exercise 3, you know the drill...

4. VIEWING AND EDITING FILES



VIEWING FILES



FILE CONTENT

- Many roads lead to Rome... For viewing files we can use:
- **less [file]:**
 - Prints N first lines of file. You can change the default N. Interactive viewing.
 - Exit with **q**
- **cat [file]:**
 - (concatenation), prints all lines of file.
 - Interrupt with **ctrl + c**
- **head/tail [file]:**
 - Prints **n first / last** lines of file. You can change the default **n**. Static viewing.

VIEW - SUBSET & RENAME

head
&
tail

```
head myfile.txt
```

```
tail -n 20 myfile.txt
```

less
&
cat

```
less myfile.txt
```

<i>Patient</i>	<i>Age</i>	<i>Sex</i>	<i>Smoker</i>	<i>Grade</i>
<i>ID1</i>	<i>61</i>	<i>Female</i>	<i>No</i>	<i>G2</i>
<i>ID2</i>	<i>58</i>	<i>Female</i>	<i>Yes</i>	<i>G3</i>
<i>.</i>	<i>.</i>	<i>.</i>	<i>.</i>	<i>.</i>

```
cat myfile.txt
```

echo

```
echo 'Hello, World!' > hello.txt
```

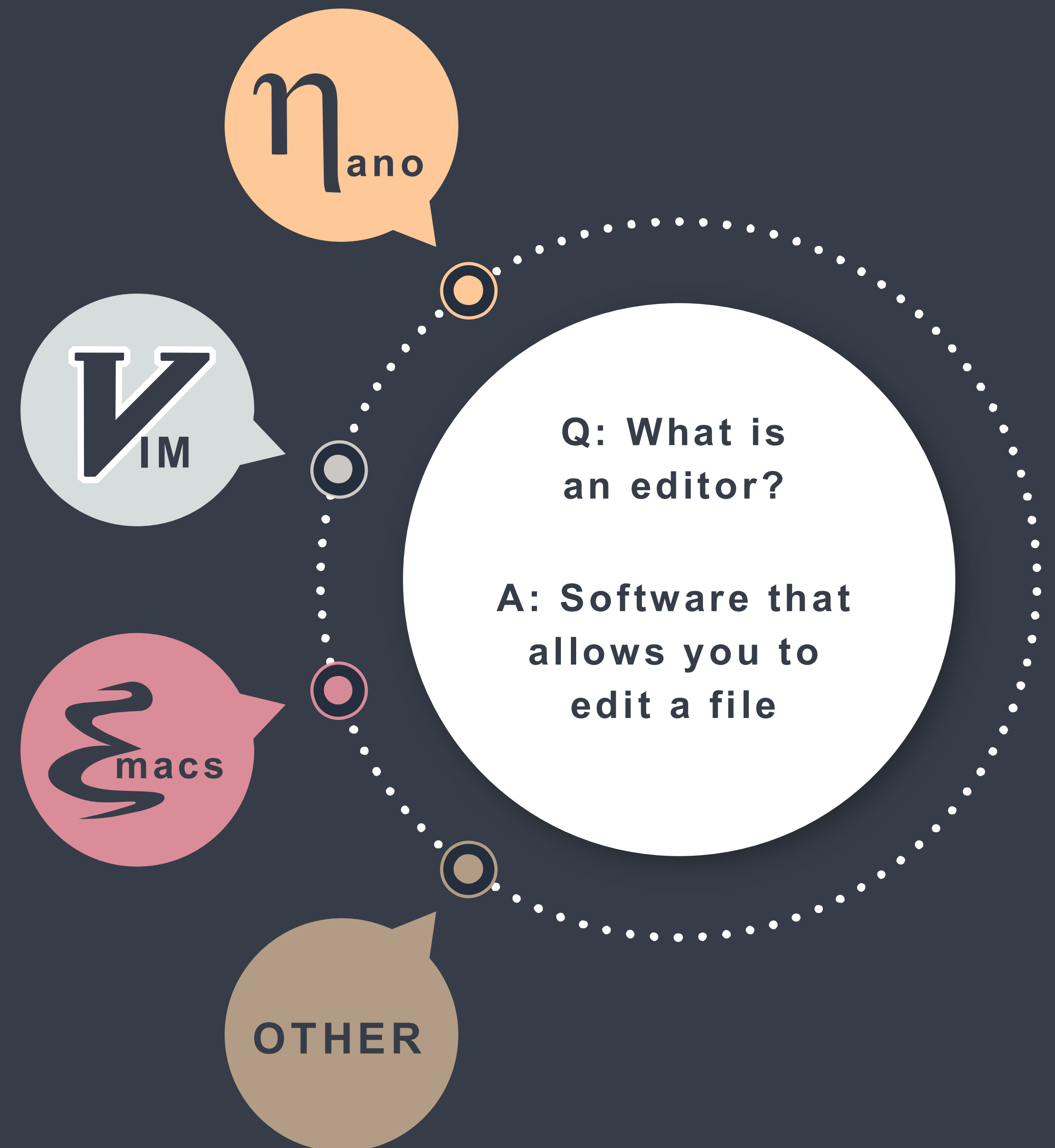
```
echo 'More text' >> hello.txt
```

> = Redirect - Save
output to new file.

More on this later.

EDITING FILES

- **Nano** [file]:
 - Simple and easy to use
 - Limited functionality
 - Exit with **Ctrl + x** (**Y** + enter)
- **VIM** or **VI** [file]:
 - Many functionalities, i.e. complex
 - Keyboard shortcuts
 - Exit with **:q** (**:wq** or **:q!**)
- **EMACS** [file]:
 - Oldest editor
 - Keyboard shortcuts, **Ctrl**, **Alt/Esc** + **[L]**
 - Exit with **Ctrl + x** + **Ctrl + c** (**Ctrl + s**)
- **OTHER** [file]:
 - There are many other editors ...



COMPRESSED FILES

.gz

Compresses a file with standard gzip (GNU zip) compression.

.tar

Combines all files within a directory in one, single archive file.

.tar.gz (.tgz)

gzip and tar together. Compresses multiple files and puts them into one, single file archive.



Fun fact: tar stands for tape archive



CHEAT SHEET 4

View Files

```
less [file] # view file content
cat [file] # view file content (full)
head / tail -n 10 [file] # view n first/last lines
nano [file] # https://www.nano-editor.org/dist/latest/cheatsheet.html
vim | vi [file] # https://vim.rtorr.com/
```

Quit & Save

```
Ctrl + c # quit process
q # quit less command

:q (:w) # quit (and save) vim editor
Ctrl + x (Y + enter) # quit (and save) nano editor
```

Compressed Files

```
gzip [file] # Create compressed gzip file
gunzip [file] # Unpack compressed gzip file
gunzip -k [file] # Unpack but also keep compressed file
zip/unzip [file] # create/decompress .zip files
zless [file] # view compressed file w/o decompression
tar -czvf [file.tar.gz] [dir] # Create a compressed tar archive of [dir] at [file.tar.gz]
tar -xvf [file.tar.gz] # Unpack a compressed tar archive
tar -tf [file.tar.gz] # List files inside a compressed tar archive
```

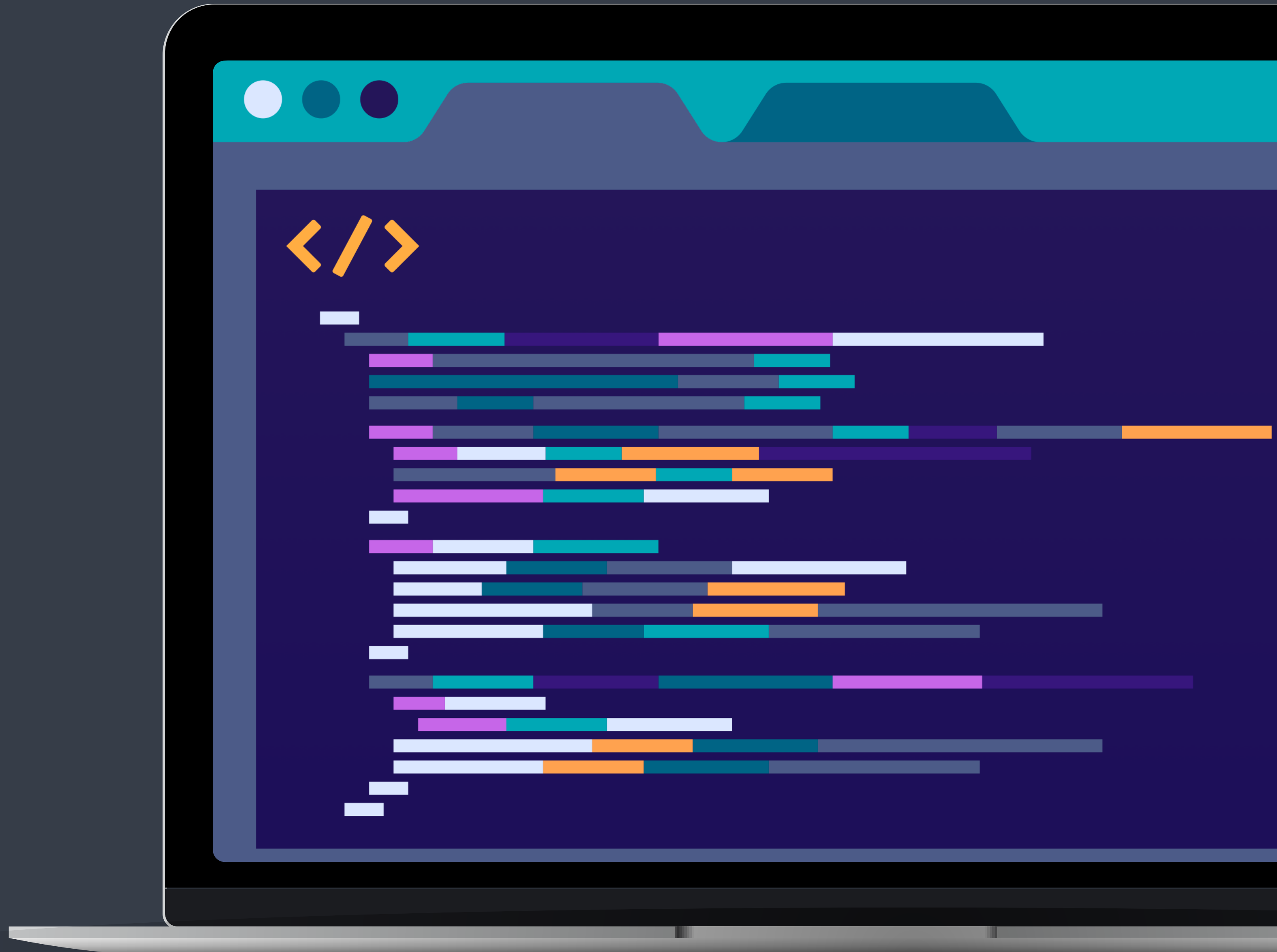


```
Last login: Fri Jun 24 15:05:34 on ttys000
[kgx936@SUN1007442 ~ % echo "Just Bash It"
Just Bash It
[kgx936@SUN1007442 ~ % █
```



The only thing between you
and lunch is **Exercise 4!**

5. DATA WRANGLING 1



FROM FILE NAVIGATION TO MANIPULATION

- **FILES & DIRECTORIES:**

- Move & Copy
- Make & Remove
- Open & Read
- Edit & Save
- Subset & Rename
- View & Change Permissions



NAVIGATE THE TERMINAL
&
FILE MANAGEMENT



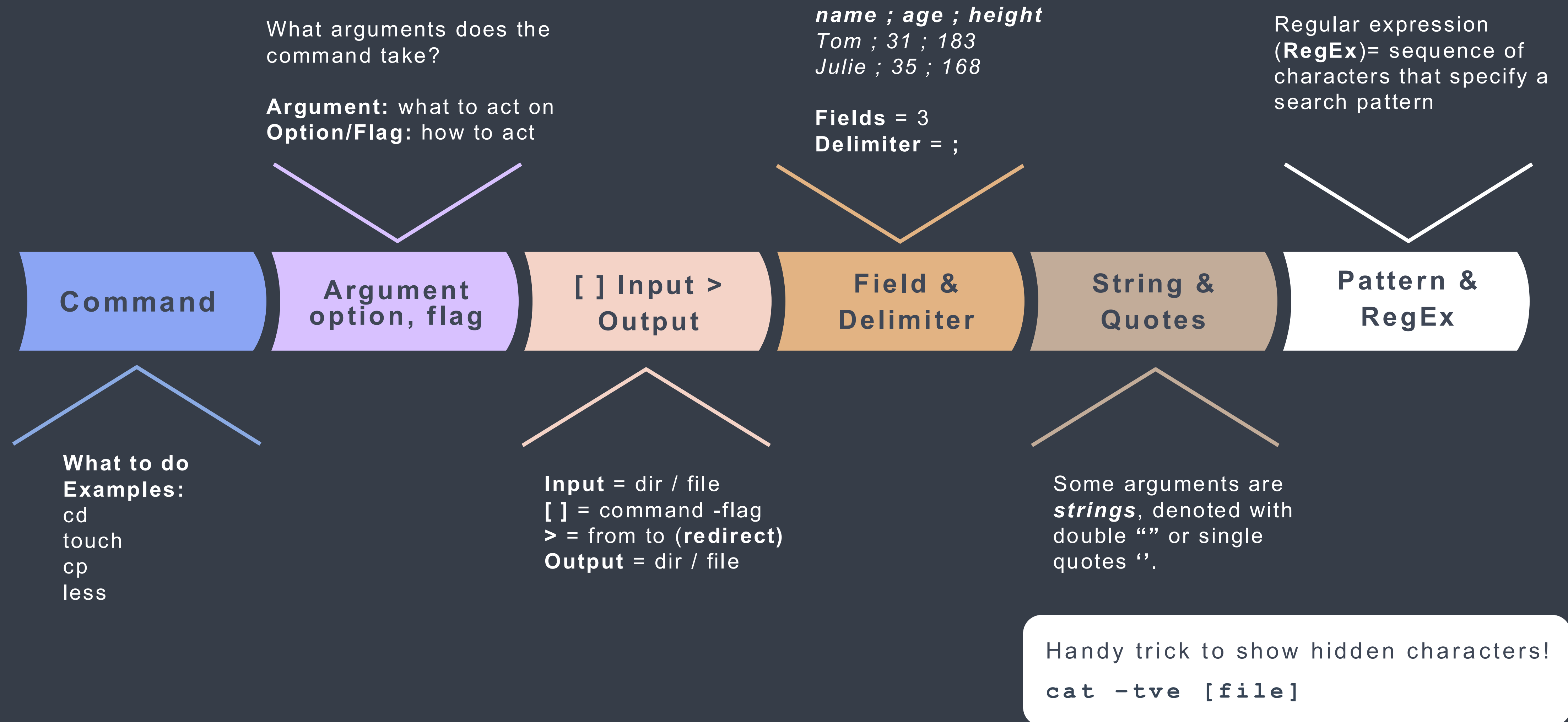
- **FILES:**

- Sort
- Count Lines & Entries
- Merge & Concatenate
- Find & Replace Patterns
- Cut & Paste
- Insert & Delete

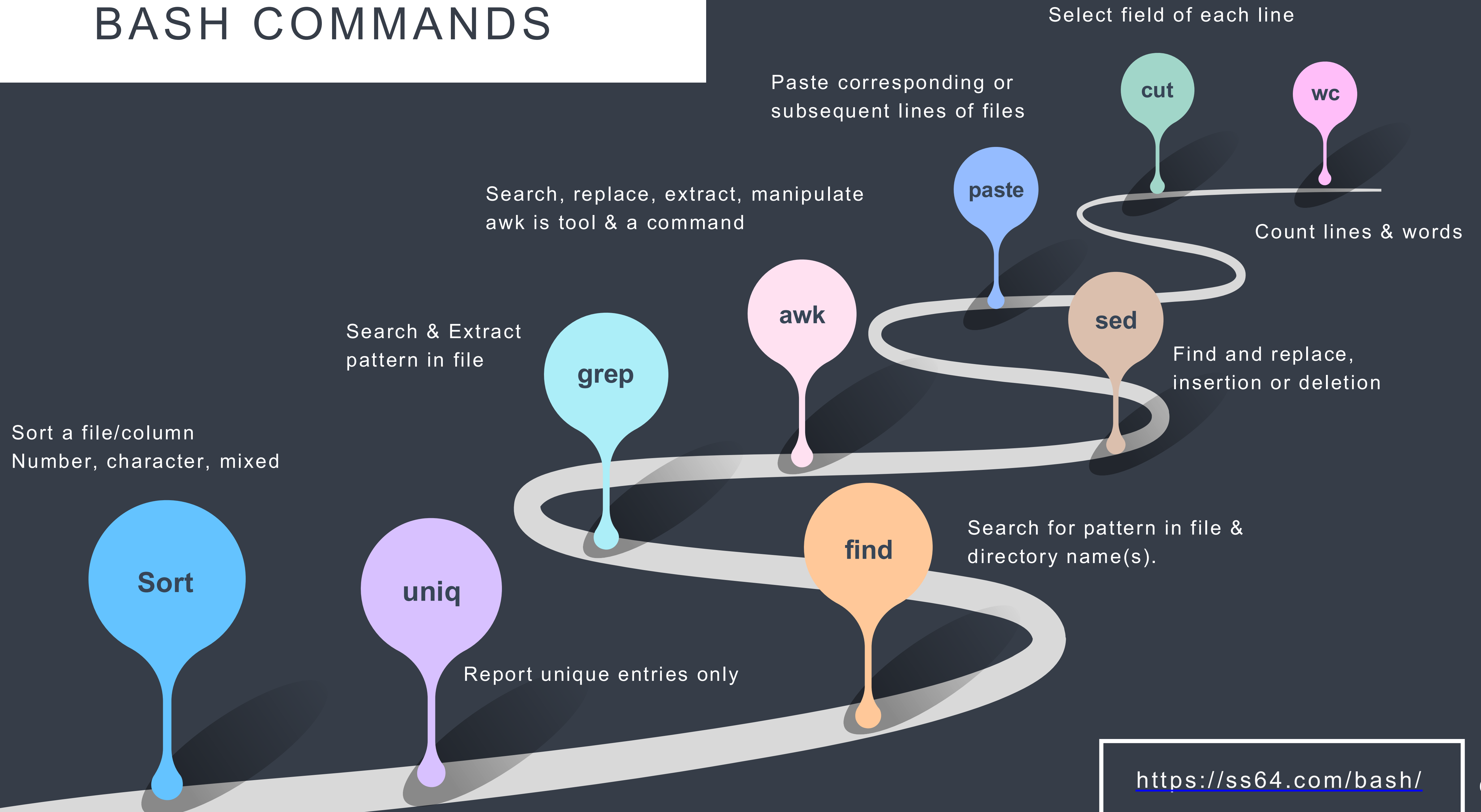


FILE MANIPULATION,
WRANGLING, SUMMATION

A LITTLE TERMINOLOGY



BASH COMMANDS



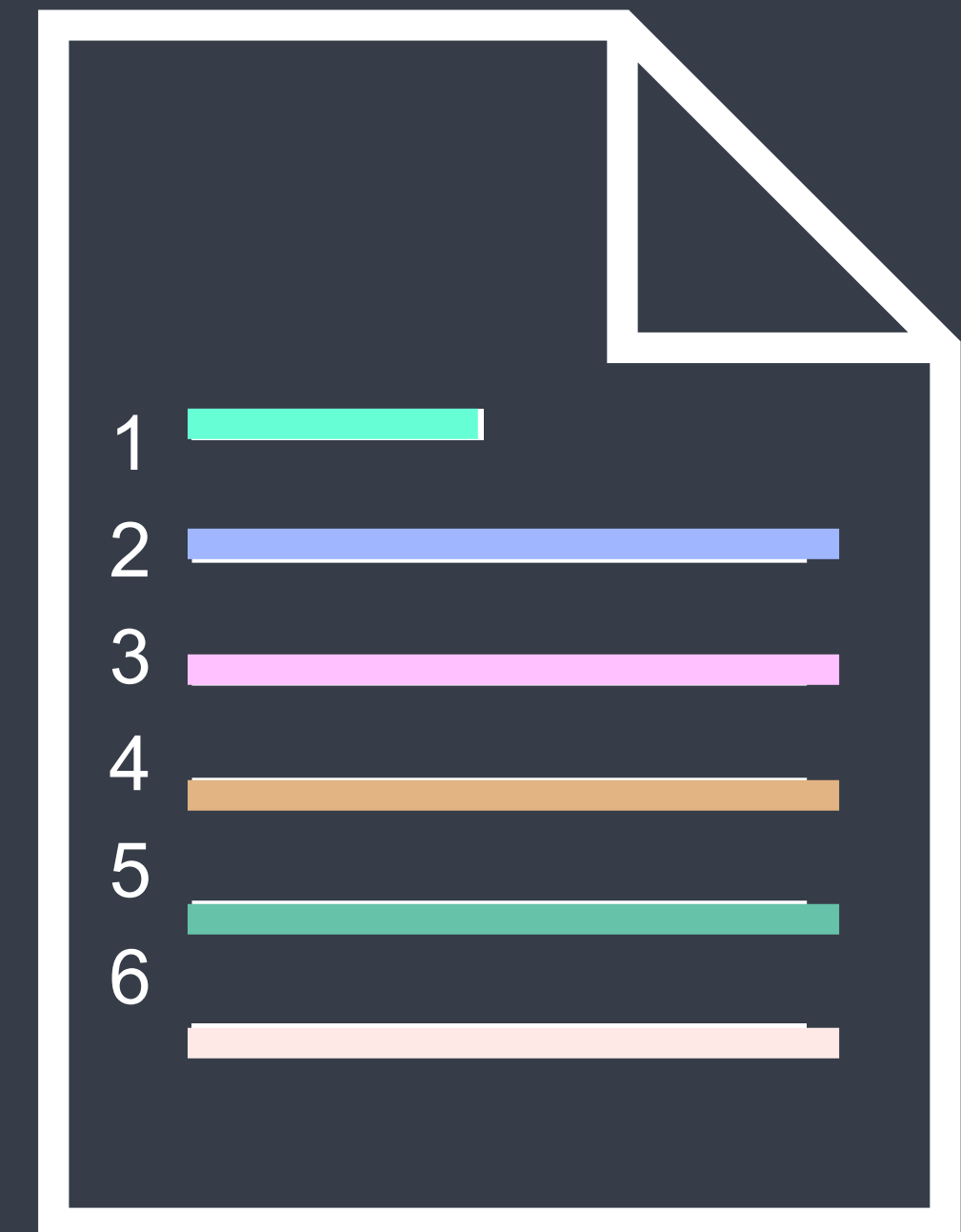
WC – NO..., NOT THAT ONE

- Word count (wc) lets us know how many words, lines and bytes there are in a file.

```
$ wc patients.txt
```

- We can also specify which count we want. For example only the lines:

```
$ wc -l patients.txt
```



CUT – GETTING COLUMNS OUT

- Cut slices a column out of a file. What a 'column' is, depends on the field separator:

```
$ cut -f 1 -d ',' patients.txt
```

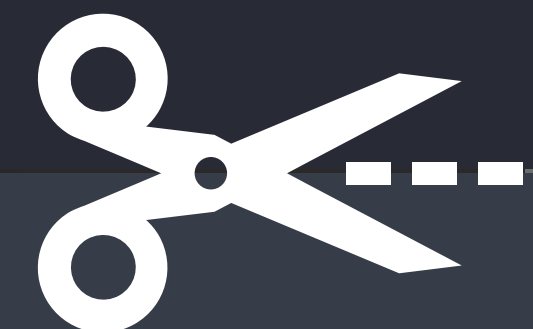
- We can also specify a range of columns:

```
$ cut -f 1-3 -d ',' patients.txt
```

- The opposite of **cut** is **paste**, putting columns together.

```
MINGW64:/c/Users/pnv719/Documents/Just-Bash-It/Fi...
pnv719@SUN1022949 MINGW64 ~/Documents/Just-Bash-It/Files/Data
(june_2023)
$ head patients.txt
patient_ID,technique,hospital,conv_days,age,satisfaction
402109,A,Rigshospitalet,15,68,3
092070,A,Rigshospitalet,13,74,5
994082,B,Herlev,27,76,2
843094,A,Herlev,30,65,5
369360,B,Rigshospitalet,21,68,5
688213,B,Rigshospitalet,29,77,3
197347,A,Herlev,25,65,5
759063,B,Rigshospitalet,16,75,4
121219,B,Herlev,27,68,4

pnv719@SUN1022949 MINGW64 ~/Documents/Just-Bash-It/Files/Data
(june_2023)
$ cut -f 1 -d ',' patients.txt
patient_ID
402109
092070
994082
843094
369360
688213
```



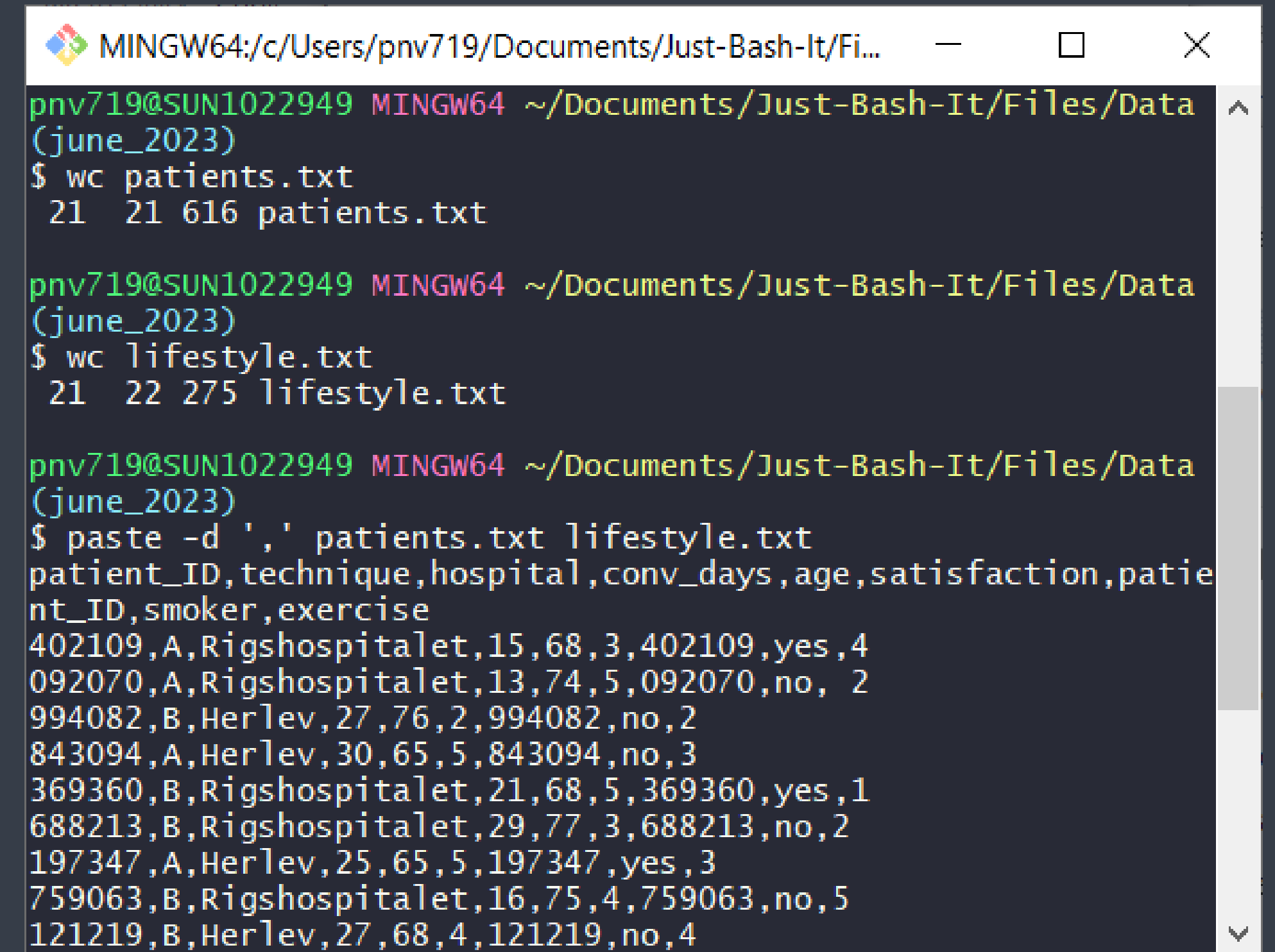
PASTE – GLUE THINGS TOGETHER

- Paste two or more files together line by line. Default delimiter is space:

```
$ paste patients.txt lifestyle.txt
```

- Specify another delimiter for pasting together files:

```
$ paste -d ',' patients.txt lifestyle.txt
```

A terminal window titled 'MINGW64:/c/Users/pnv719/Documents/Just-Bash-It/Fi...' showing the execution of the 'paste' command. The user first checks the line counts of 'patients.txt' (21 lines) and 'lifestyle.txt' (22 lines). Then, they run 'paste -d ',' patients.txt lifestyle.txt', which outputs the contents of both files side-by-side, separated by commas. The output shows patient data from 'patients.txt' and lifestyle data from 'lifestyle.txt' for each patient ID.

```
MINGW64:/c/Users/pnv719/Documents/Just-Bash-It/Fi...
pnv719@SUN1022949 MINGW64 ~/Documents/Just-Bash-It/Files/Data (june_2023)
$ wc patients.txt
21 21 616 patients.txt

pnv719@SUN1022949 MINGW64 ~/Documents/Just-Bash-It/Files/Data (june_2023)
$ wc lifestyle.txt
21 22 275 lifestyle.txt

pnv719@SUN1022949 MINGW64 ~/Documents/Just-Bash-It/Files/Data (june_2023)
$ paste -d ',' patients.txt lifestyle.txt
patient_ID,technique,hospital,conv_days,age,satisfaction,patient_ID,smoker,exercise
402109,A,Rigshospitalet,15,68,3,402109,yes,4
092070,A,Rigshospitalet,13,74,5,092070,no,2
994082,B,Herlev,27,76,2,994082,no,2
843094,A,Herlev,30,65,5,843094,no,3
369360,B,Rigshospitalet,21,68,5,369360,yes,1
688213,B,Rigshospitalet,29,77,3,688213,no,2
197347,A,Herlev,25,65,5,197347,yes,3
759063,B,Rigshospitalet,16,75,4,759063,no,5
121219,B,Herlev,27,68,4,121219,no,4
```

SED – THE STREAM EDITOR

`sed` is a powerful tool used to manipulate files without the manual labor of a text editor.

- A common use it to display only specific lines:

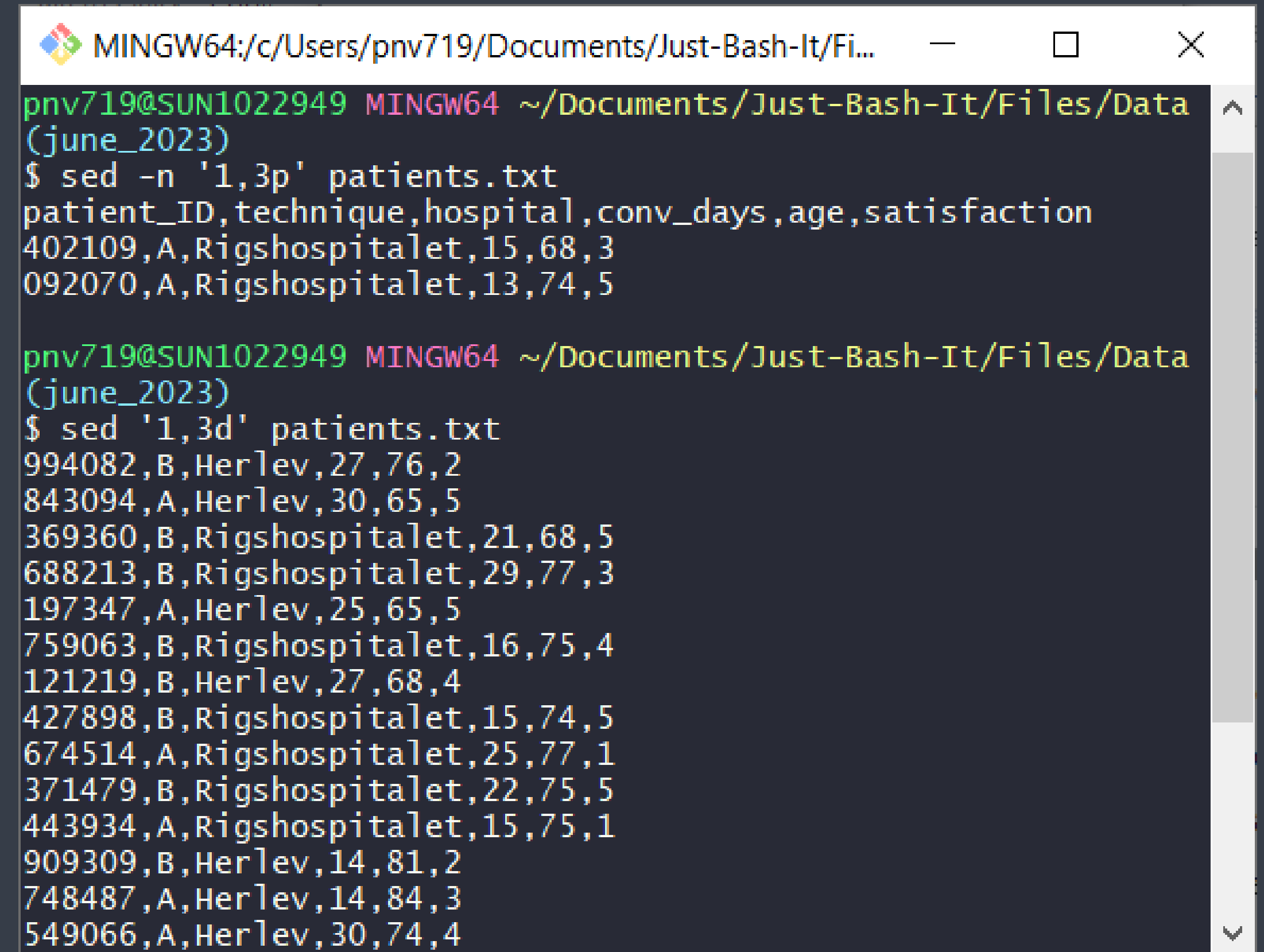
```
$ sed -n '1,3p' patients.txt
```

- Or to delete them!

```
$ sed '1,3d' patients.txt
```

NB! We haven't saved the output, it is just displayed.

One very powerful use of `sed` is to edit files with *regular expressions*



```
MINGW64:/c/Users/pnv719/Documents/Just-Bash-It/Fi...
pnv719@SUN1022949 MINGW64 ~/Documents/Just-Bash-It/Files/Data (june_2023)
$ sed -n '1,3p' patients.txt
patient_ID,technique,hospital,conv_days,age,satisfaction
402109,A,Rigshospitalet,15,68,3
092070,A,Rigshospitalet,13,74,5

pnv719@SUN1022949 MINGW64 ~/Documents/Just-Bash-It/Files/Data (june_2023)
$ sed '1,3d' patients.txt
994082,B,Herlev,27,76,2
843094,A,Herlev,30,65,5
369360,B,Rigshospitalet,21,68,5
688213,B,Rigshospitalet,29,77,3
197347,A,Herlev,25,65,5
759063,B,Rigshospitalet,16,75,4
121219,B,Herlev,27,68,4
427898,B,Rigshospitalet,15,74,5
674514,A,Rigshospitalet,25,77,1
371479,B,Rigshospitalet,22,75,5
443934,A,Rigshospitalet,15,75,1
909309,B,Herlev,14,81,2
748487,A,Herlev,14,84,3
549066,A,Herlev,30,74,4
```

REGULAR EXPRESSIONS

Regular expression (RegEx) = sequence of characters that specify a search pattern

Anchors

<code>^</code>	Start of string, or start of line in multi-line pattern
<code>\A</code>	Start of string
<code>\$</code>	End of string, or end of line in multi-line pattern
<code>\Z</code>	End of string
<code>\b</code>	Word boundary
<code>\B</code>	Not word boundary
<code>\<</code>	Start of word
<code>\></code>	End of word

Character Classes

<code>\c</code>	Control character
-----------------	-------------------

Quantifiers

<code>*</code>	0 or more	<code>{3}</code>	Exactly 3
<code>+</code>	1 or more	<code>{3,}</code>	3 or more
<code>?</code>	0 or 1	<code>{3,5}</code>	3, 4 or 5

Add a `?` to a quantifier to make it ungreedy.

Escape Sequences

<code>\</code>	Escape following character
<code>\Q</code>	Begin literal sequence
<code>\E</code>	End literal sequence

"Escaping" is a way of treating characters which have a special meaning in regular expressions literally, rather than as special characters.

Groups and Ranges

<code>.</code>	Any character except new line (<code>\n</code>)
<code>(a b)</code>	a or b
<code>(...)</code>	Group
<code>(?:...)</code>	Passive (non-capturing) group
<code>[abc]</code>	Range (a or b or c)
<code>[^abc]</code>	Not (a or b or c)
<code>[a-q]</code>	Lower case letter from a to q
<code>[A-Q]</code>	Upper case letter from A to Q
<code>[0-7]</code>	Digit from 0 to 7
<code>\x</code>	Group/subpattern number "x"

Ranges are inclusive.

<https://cheatography.com/davechild/cheat-sheets/regular-expressions/>

Regex Testers : <https://regexr.com/>
<https://regex101.com/>

SED – THE STEAM EDITOR

Search and replace patterns with `sed`.

- Changing 'pat' to 'bat':

```
$ sed 's/pat/bat/g' patients.txt
```

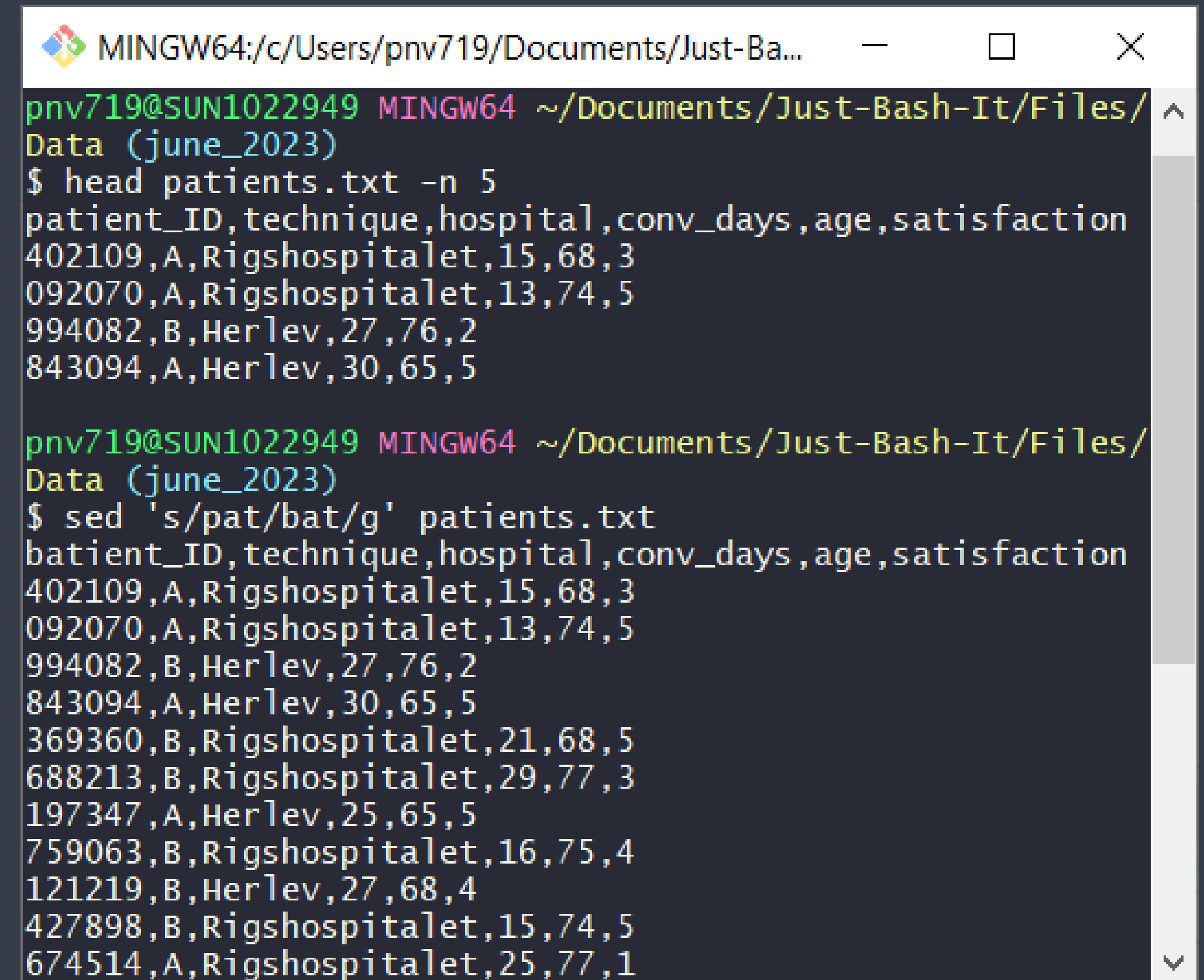
- Using *anchors* we can be specific about the exact pattern we want to hit:

```
$ sed 's/^40/XX/g' patients.txt
```

You should be as specific as possible to avoid changing the wrong part.

You can find many examples here:

<https://www.howtogeek.com/666395/how-to-use-the-sed-command-on-linux/>



```
MINGW64:/c/Users/pnv719/Documents/Just-Ba...  
pnv719@SUN1022949 MINGW64 ~/Documents/Just-Bash-It/Files/  
Data (june_2023)  
$ head patients.txt -n 5  
patient_ID,technique,hospital,conv_days,age,satisfaction  
402109,A,Rigshospitalet,15,68,3  
092070,A,Rigshospitalet,13,74,5  
994082,B,Herlev,27,76,2  
843094,A,Herlev,30,65,5  
  
pnv719@SUN1022949 MINGW64 ~/Documents/Just-Bash-It/Files/  
Data (june_2023)  
$ sed 's/pat/bat/g' patients.txt  
batient_ID,technique,hospital,conv_days,age,satisfaction  
402109,A,Rigshospitalet,15,68,3  
092070,A,Rigshospitalet,13,74,5  
994082,B,Herlev,27,76,2  
843094,A,Herlev,30,65,5  
369360,B,Rigshospitalet,21,68,5  
688213,B,Rigshospitalet,29,77,3  
197347,A,Herlev,25,65,5  
759063,B,Rigshospitalet,16,75,4  
121219,B,Herlev,27,68,4  
427898,B,Rigshospitalet,15,74,5  
674514,A,Rigshospitalet,25,77,1
```

CHEAT SHEET 5

View Files

```
less [file] # view file content
cat [file] # view file content (full)
head / tail -n 10 [file] # view n first/last lines
nano [file] # https://www.nano-editor.org/dist/latest/cheatsheet.html
vim [file] # https://vim.rtorr.com/
```

Compressed Files

```
tar -[f] [file] # .tar files
gzip -[f] [file] # (de)compress (.tar).gz files
unzip -[f] [file] # decompress .zip files
zless -[f][file] # view .gz file w/o decompression
gunzip -[f] [file] # uncompress and keep original

others: zcat, zmore, gzcat
```

Manipulating Files

```
wc -[f][file] # Count lines, characters, bits
sort -[f][file] # Sort file (by field/column)
uniq -[f][file] # Return unique values
cut -[f][file]: # Extract field/column
paste -[f][files]: # Merge file lines
```

```
sed -[f]'command'[file] # Insertion, deletion, ...
grep -[f]['pattern'][file] # Search for pattern
awk '{pattern}'[file] # Search, replace, extract, ...
find -[f][path]['pattern'] # Search pattern in file name
```

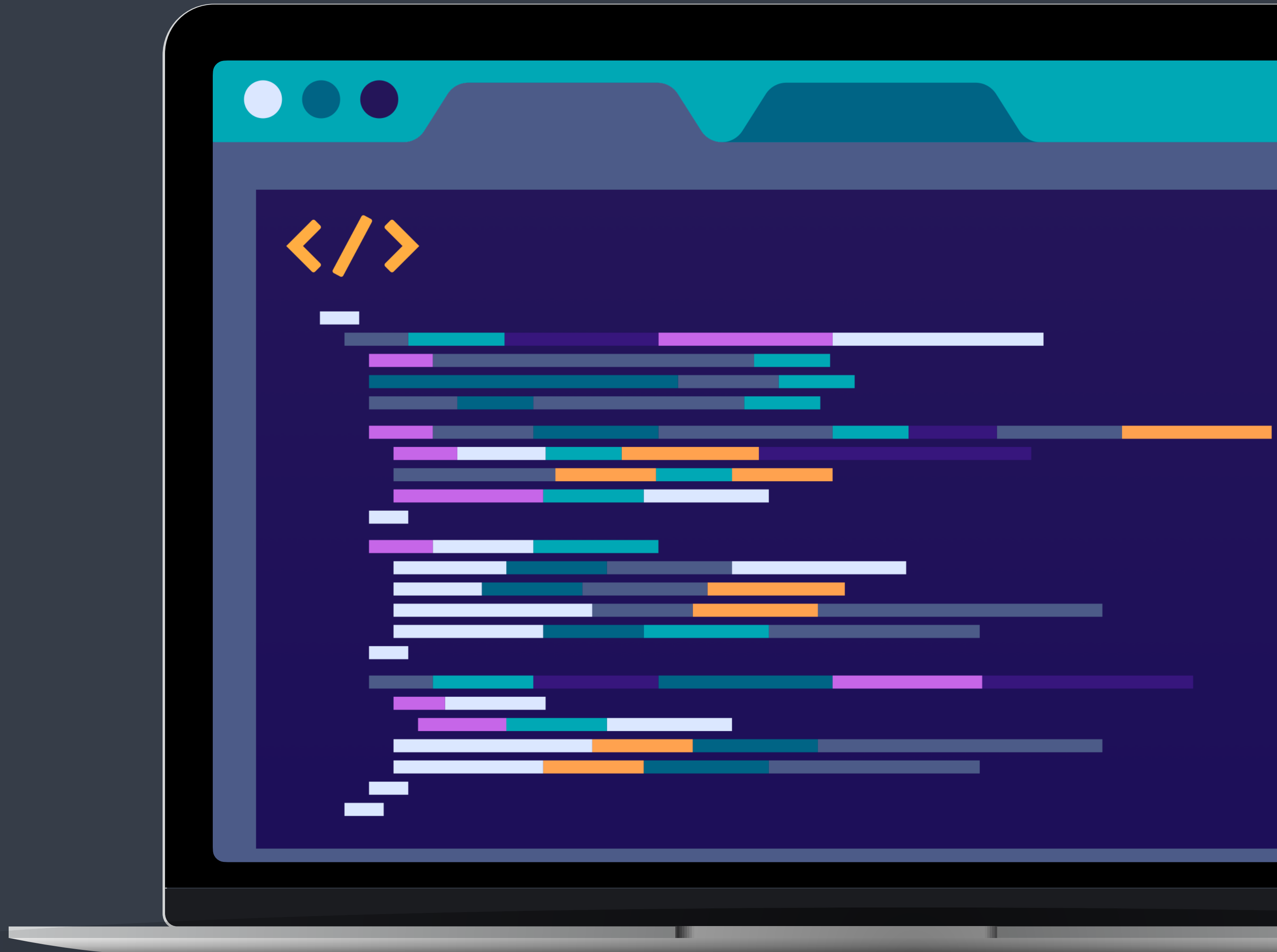


```
Last login: Fri Jun 24 15:05:34 on ttys000
[kgx936@SUN1007442 ~ % echo "Just Bash It"
Just Bash It
[kgx936@SUN1007442 ~ % █
```



Exercise 5, ready to
wrangle!

6. DATA WRANGLING 2



SORT - GUESS WHAT, IT SORTS!

Sort produces a sorted output.

We need to specify the field separator as the default separator in sort is a space.

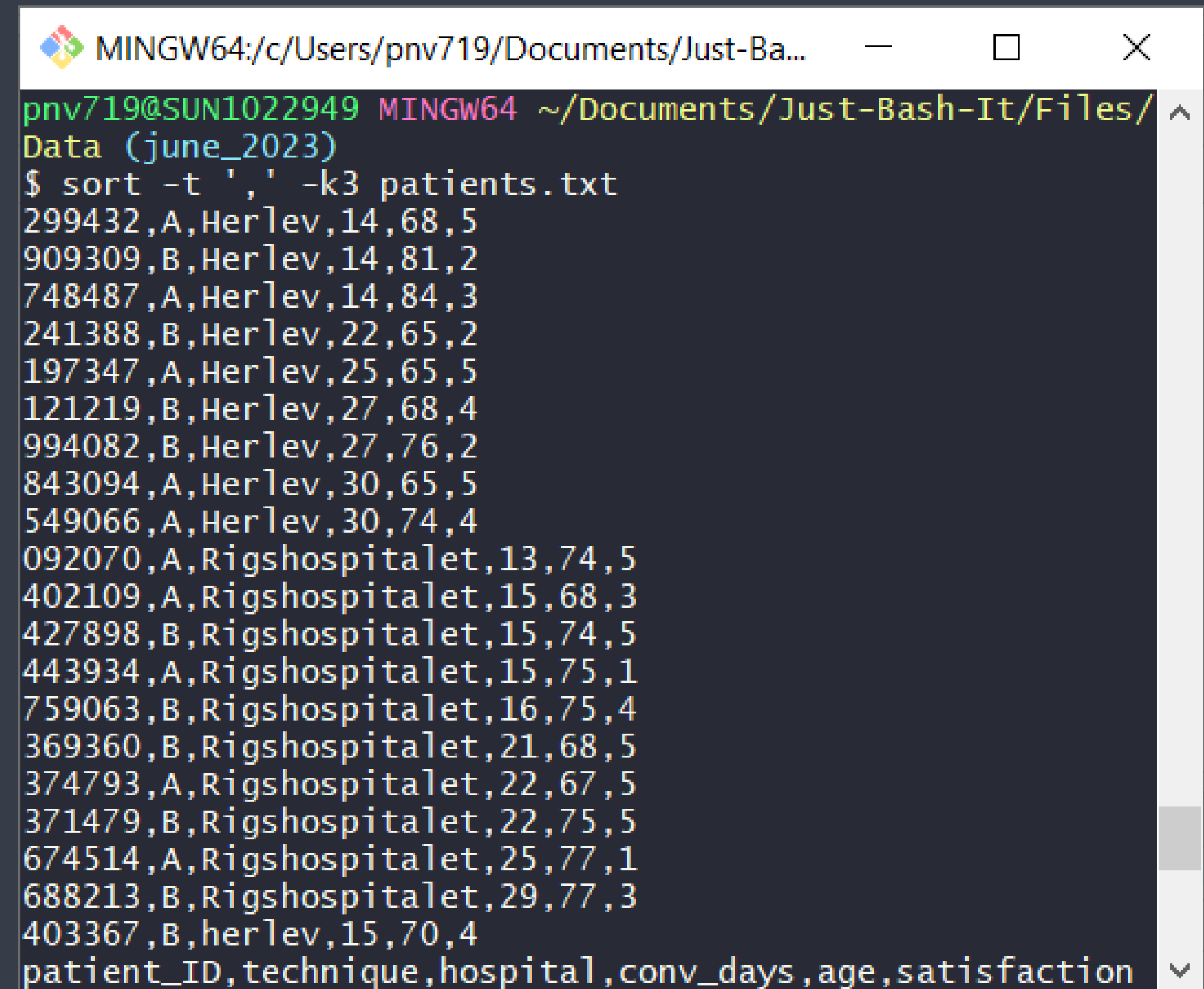
- The column to sort on is given by -k.

```
$ sort -t ',' -k3 patients.txt
```

Sort is alphanumeric, i.e. 10 is smaller than 9 since $1 < 9$.

- To switch to numeric sort, use the -n option:

```
$ sort -t ',' -n -k4 patients.txt
```



```
MINGW64:/c/Users/pnv719/Documents/Just-Ba...  
pnv719@SUN1022949 MINGW64 ~/Documents/Just-Bash-It/Files/  
Data (june_2023)  
$ sort -t ',' -k3 patients.txt  
299432,A,Herlev,14,68,5  
909309,B,Herlev,14,81,2  
748487,A,Herlev,14,84,3  
241388,B,Herlev,22,65,2  
197347,A,Herlev,25,65,5  
121219,B,Herlev,27,68,4  
994082,B,Herlev,27,76,2  
843094,A,Herlev,30,65,5  
549066,A,Herlev,30,74,4  
092070,A,Rigshospitalet,13,74,5  
402109,A,Rigshospitalet,15,68,3  
427898,B,Rigshospitalet,15,74,5  
443934,A,Rigshospitalet,15,75,1  
759063,B,Rigshospitalet,16,75,4  
369360,B,Rigshospitalet,21,68,5  
374793,A,Rigshospitalet,22,67,5  
371479,B,Rigshospitalet,22,75,5  
674514,A,Rigshospitalet,25,77,1  
688213,B,Rigshospitalet,29,77,3  
403367,B,herlev,15,70,4  
patient_ID,technique,hospital,conv_days,age,satisfaction
```

GREP – LOOKING FOR SOMETHING

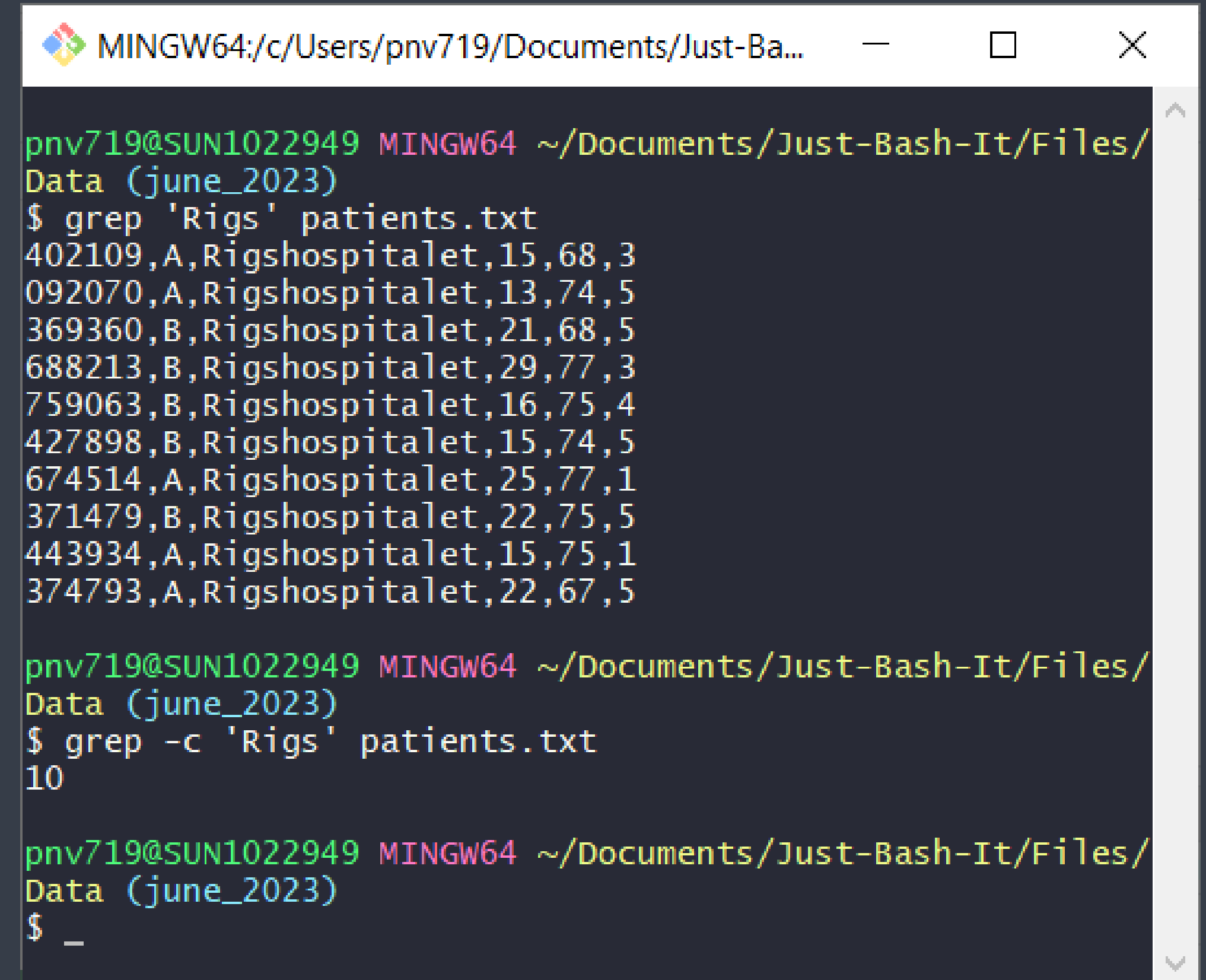
- The `grep` searches files for matches to a pattern and by default returns the entire line:

```
$ grep 'Rigs' patients.txt
```

- Has option for counting the number of matching lines:

```
$ grep -c 'Rigs' patients.txt
```

`grep` understands `regex` (like `sed`) so you can make exact patterns. You may need the `-P` option to get `grep` to compile the expression in Perl style



```
MINGW64:/c/Users/pnv719/Documents/Just-Ba...  
pnv719@SUN1022949 MINGW64 ~/Documents/Just-Bash-It/Files/  
Data (june_2023)  
$ grep 'Rigs' patients.txt  
402109,A,Rigshospitalet,15,68,3  
092070,A,Rigshospitalet,13,74,5  
369360,B,Rigshospitalet,21,68,5  
688213,B,Rigshospitalet,29,77,3  
759063,B,Rigshospitalet,16,75,4  
427898,B,Rigshospitalet,15,74,5  
674514,A,Rigshospitalet,25,77,1  
371479,B,Rigshospitalet,22,75,5  
443934,A,Rigshospitalet,15,75,1  
374793,A,Rigshospitalet,22,67,5  
  
pnv719@SUN1022949 MINGW64 ~/Documents/Just-Bash-It/Files/  
Data (june_2023)  
$ grep -c 'Rigs' patients.txt  
10  
  
pnv719@SUN1022949 MINGW64 ~/Documents/Just-Bash-It/Files/  
Data (june_2023)  
$ -
```

REGULAR EXPRESSIONS

Regular expression (Regex) = sequence of characters that specify a *flexible* search pattern

Anchors

<code>^</code>	Start of string, or start of line in multi-line pattern
<code>\A</code>	Start of string
<code>\$</code>	End of string, or end of line in multi-line pattern
<code>\Z</code>	End of string
<code>\b</code>	Word boundary
<code>\B</code>	Not word boundary
<code>\<</code>	Start of word
<code>\></code>	End of word

Character Classes

<code>\c</code>	Control character
-----------------	-------------------

Quantifiers

<code>*</code>	0 or more	<code>{3}</code>	Exactly 3
<code>+</code>	1 or more	<code>{3,}</code>	3 or more
<code>?</code>	0 or 1	<code>{3,5}</code>	3, 4 or 5

Add a `?` to a quantifier to make it ungreedy.

Escape Sequences

<code>\</code>	Escape following character
<code>\Q</code>	Begin literal sequence
<code>\E</code>	End literal sequence

"Escaping" is a way of treating characters which have a special meaning in regular expressions literally, rather than as special characters.

Groups and Ranges

<code>.</code>	Any character except new line (<code>\n</code>)
<code>(a b)</code>	a or b
<code>(...)</code>	Group
<code>(?:...)</code>	Passive (non-capturing) group
<code>[abc]</code>	Range (a or b or c)
<code>[^abc]</code>	Not (a or b or c)
<code>[a-q]</code>	Lower case letter from a to q
<code>[A-Q]</code>	Upper case letter from A to Q
<code>[0-7]</code>	Digit from 0 to 7
<code>\x</code>	Group/subpattern number "x"

Ranges are inclusive.

<https://cheatography.com/davechild/cheat-sheets/regular-expressions/>

Regex Testers : <https://regexr.com/>
<https://regex101.com/>

AWK – THE FIX EVERYTHING OR DIE TRYING

awk is a 'scripting' language made of one-liners that is often used for file manipulation.

It can be used for selecting lines, cutting and pasting files together and even perform arithmetic on file content!

```
$ awk '/,5$/ {print}' patients.txt
```

```
$ awk -F ',' '{print $2,$3}' patients.txt
```

```
$ awk -F ',' '{if ($4 >15) {print}}' patients.txt
```

Why use awk instead a general-purpose programming language?

Because, you can directly invoke it on the command line without needing a script and input/output file handling. It's convenient and efficient!



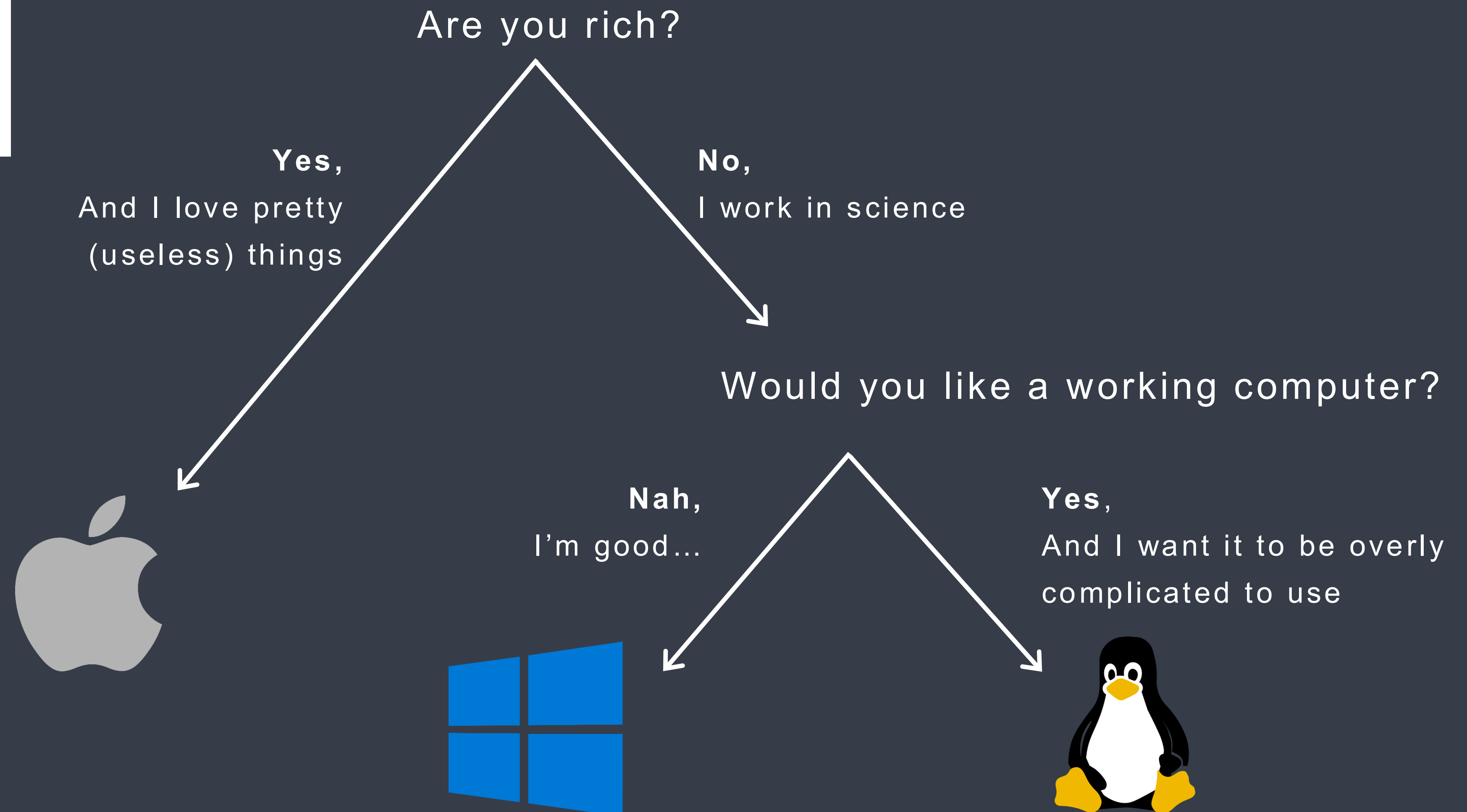
```
Last login: Fri Jun 24 15:05:34 on ttys000
[kgx936@SUN1007442 ~ % echo "Just Bash It"
Just Bash It
[kgx936@SUN1007442 ~ % █
```



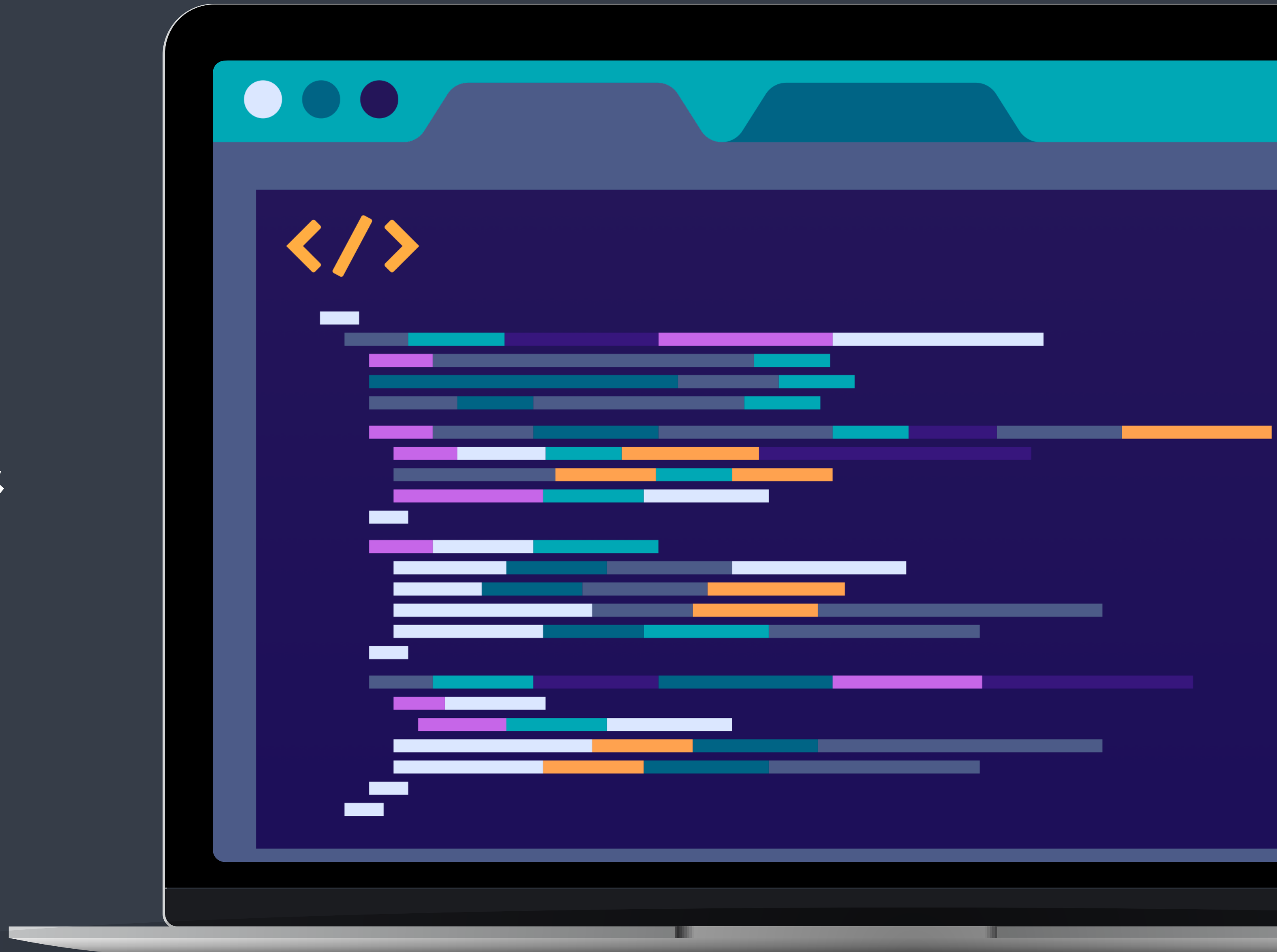
Last exercise of today,
Exercise 6 let's go!

SEE YOU TOMORROW

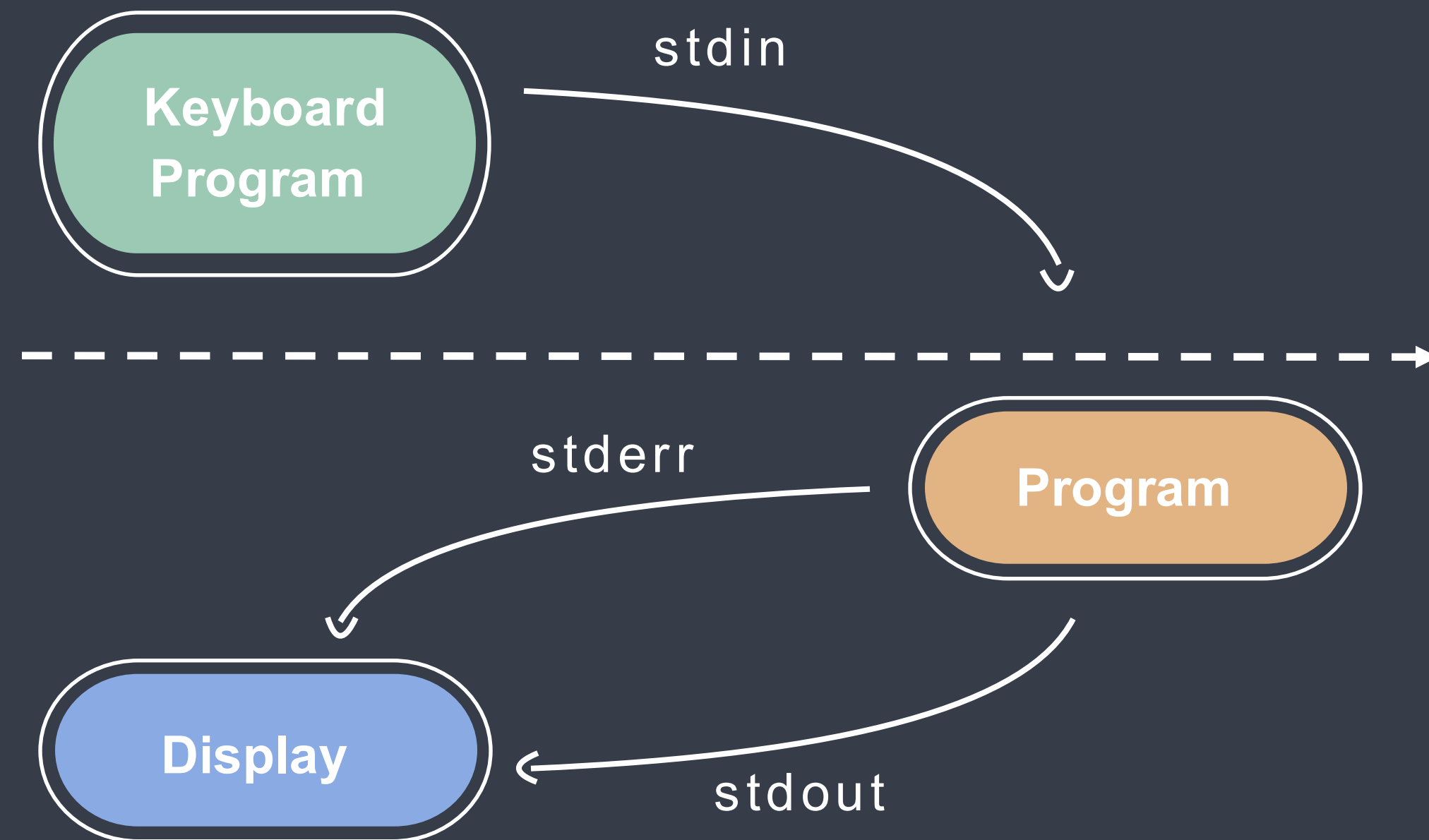
HOW TO PICK AN OPERATING SYSTEM



7. REDIRECTION & PIPES

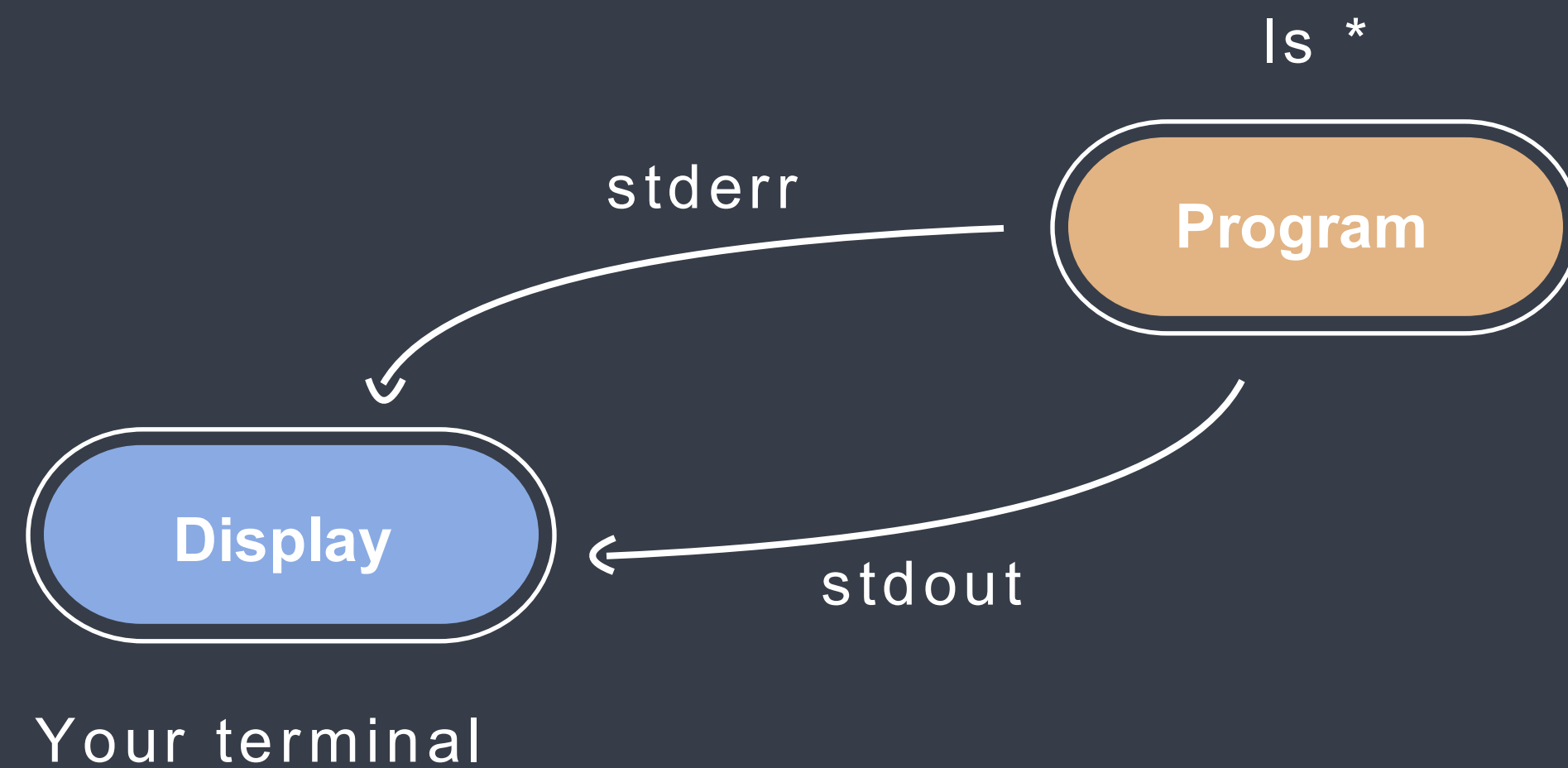


A TALE OF THREE STREAMS



- Data in unix always moves along one of **three (data) streams**:
 - **stdin** (Standard In) – the stream along which input to commands moves
 - **stdout** (Standard out) - the output of commands moves along this stream
 - **stderr** (Standard error) – error messages move via this stream
- Each of the streams can be redirected separately.

A TALE OF THREE STREAMS

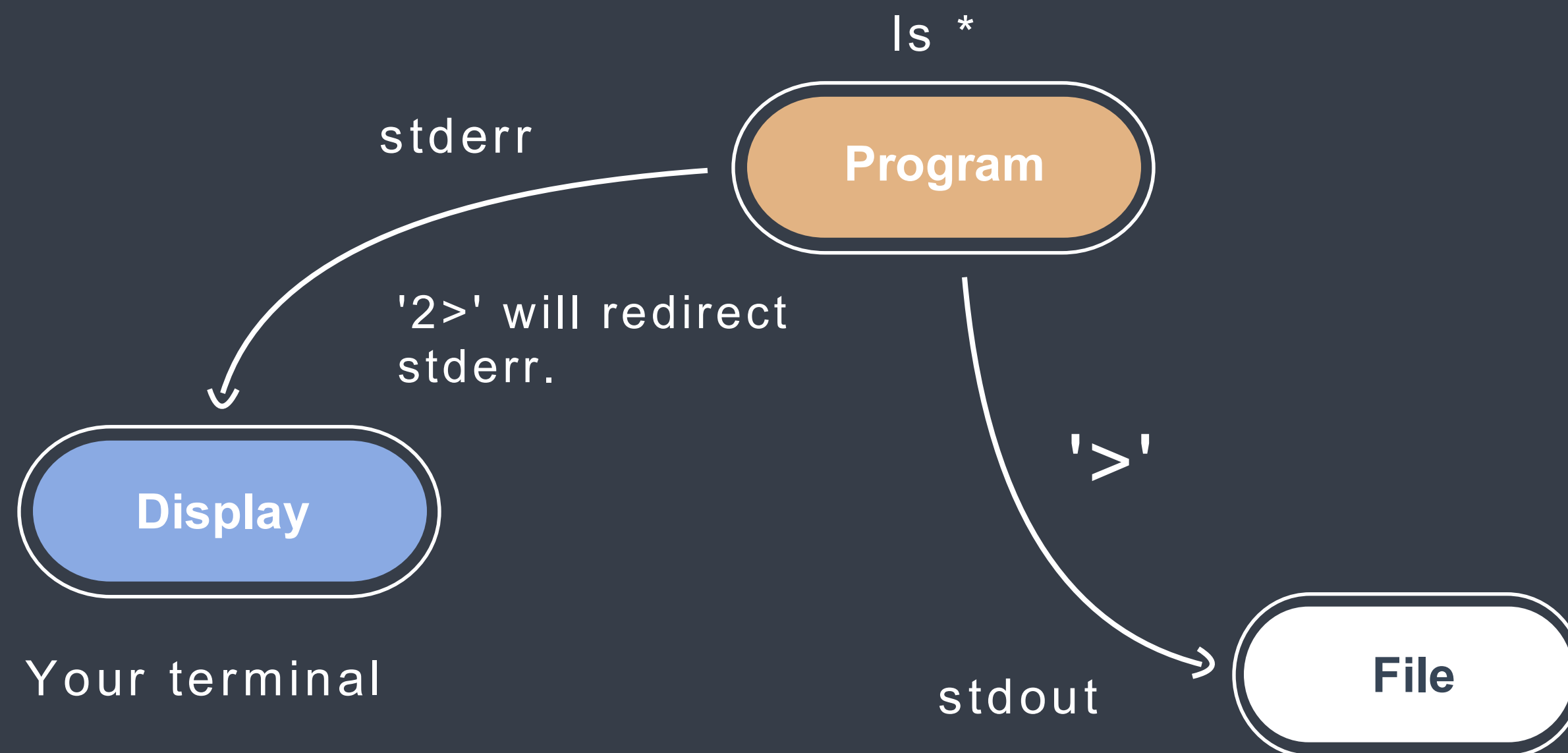


```
BashFigure1.jpeg  
Commandline_Workshop.pptx  
Data  
Examples  
Just_Bash_It_Exercises.md  
Just_Bash_It_Exercises.Rmd  
Just_Bash_It_Presentations.md  
Just_Bash_It_Presentations.Rmd  
LICENSE
```

- **stdout** (standard out) - the output of commands moves along this stream
- **stderr** (standard error) – error messages move via this stream
- By default **stdout** points to the display, your terminal window

REDIRECTING OUTPUT AND ERRORS

But **stdout** and **stderr** can be **redirected** to i.e. a file:



Redirect stdout:

```
$ ls > list_results.txt
```

Redirect stderr:

```
$ ls 2> err.txt
```

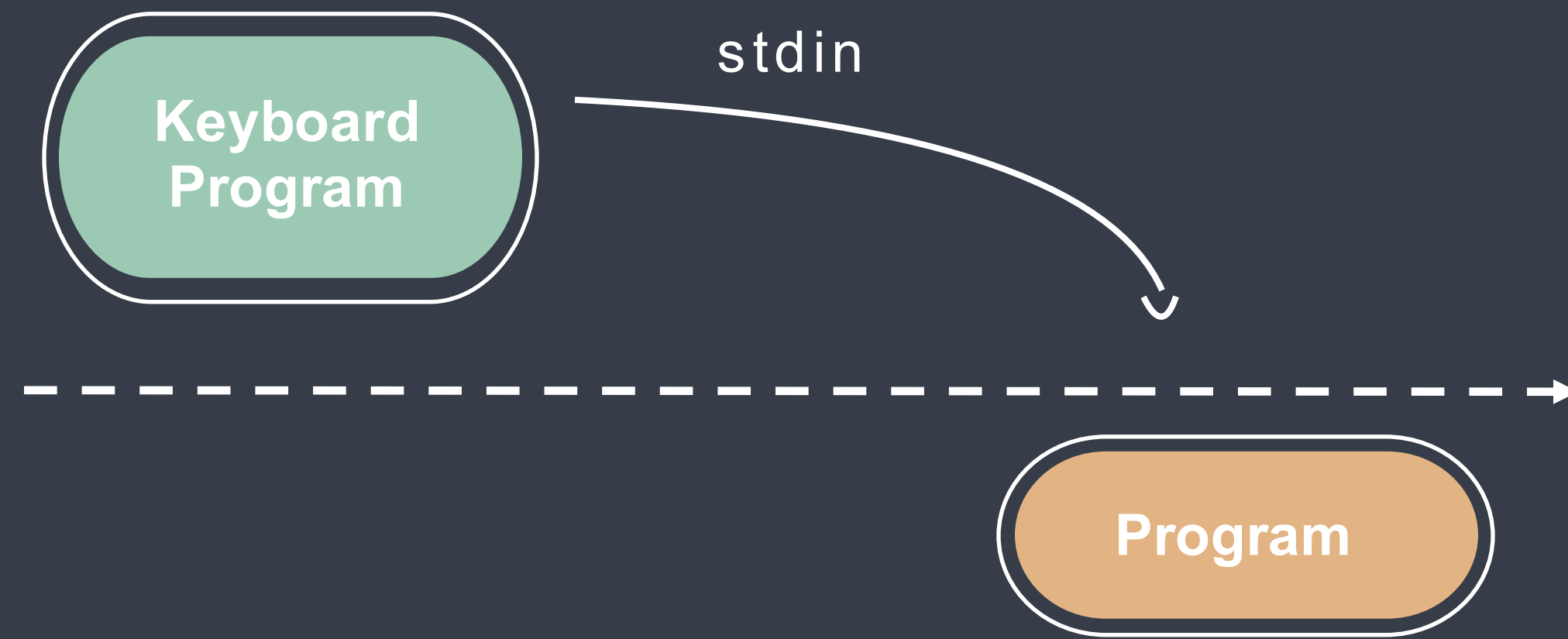
Redirect both into different files:

```
$ ls 2> err.txt > list_results.txt
```

Redirect both into the same file:

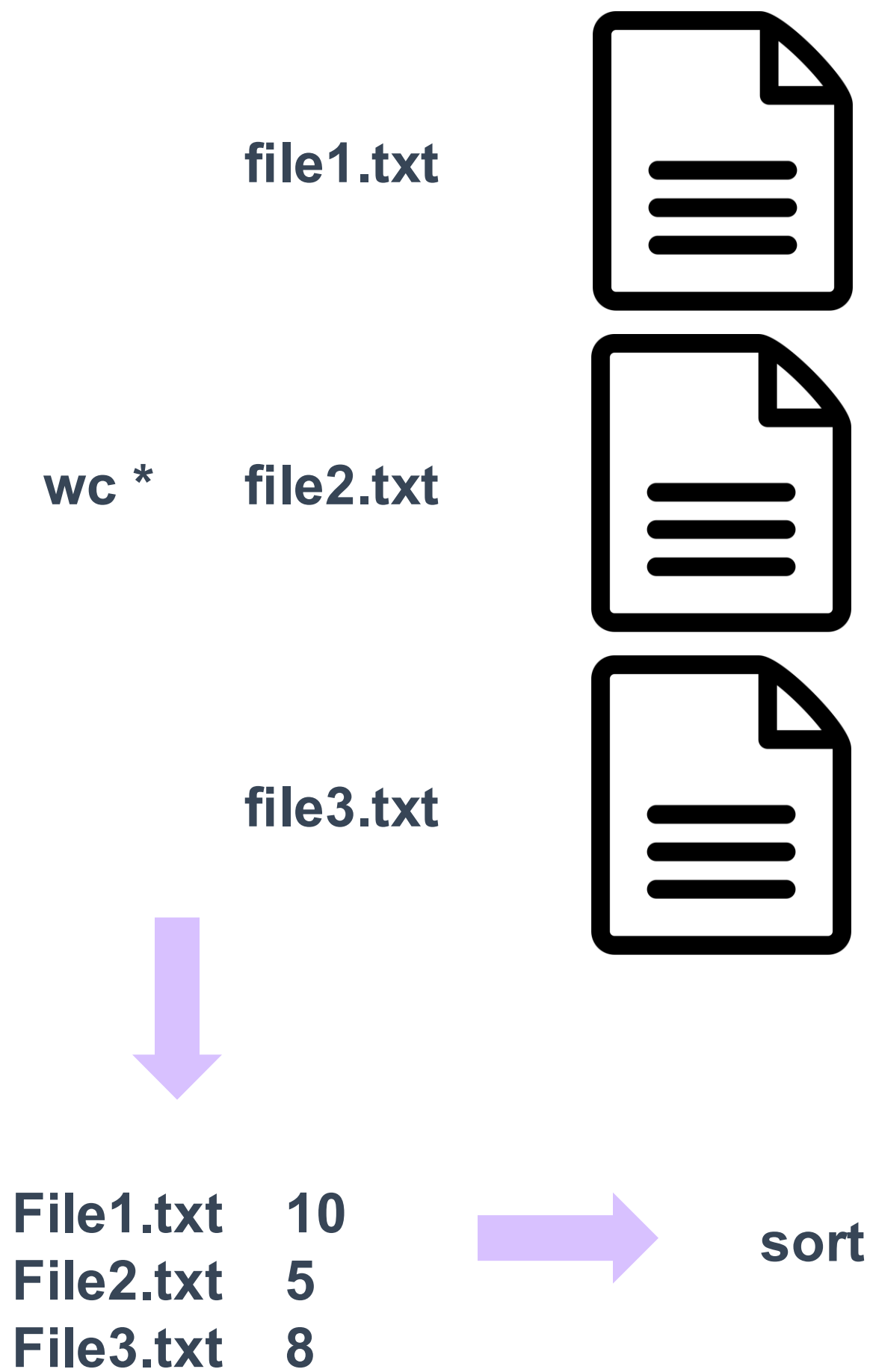
```
$ ls &> both
```

WHAT ABOUT STDIN?



- Many commands require input data to work on, i.e. **wc** works on a file.
- Usually input data comes from the keyboard, i.e. the name of the file to word count.
- But it can also come directly from another program!

CHAINING (PIPING) COMMANDS



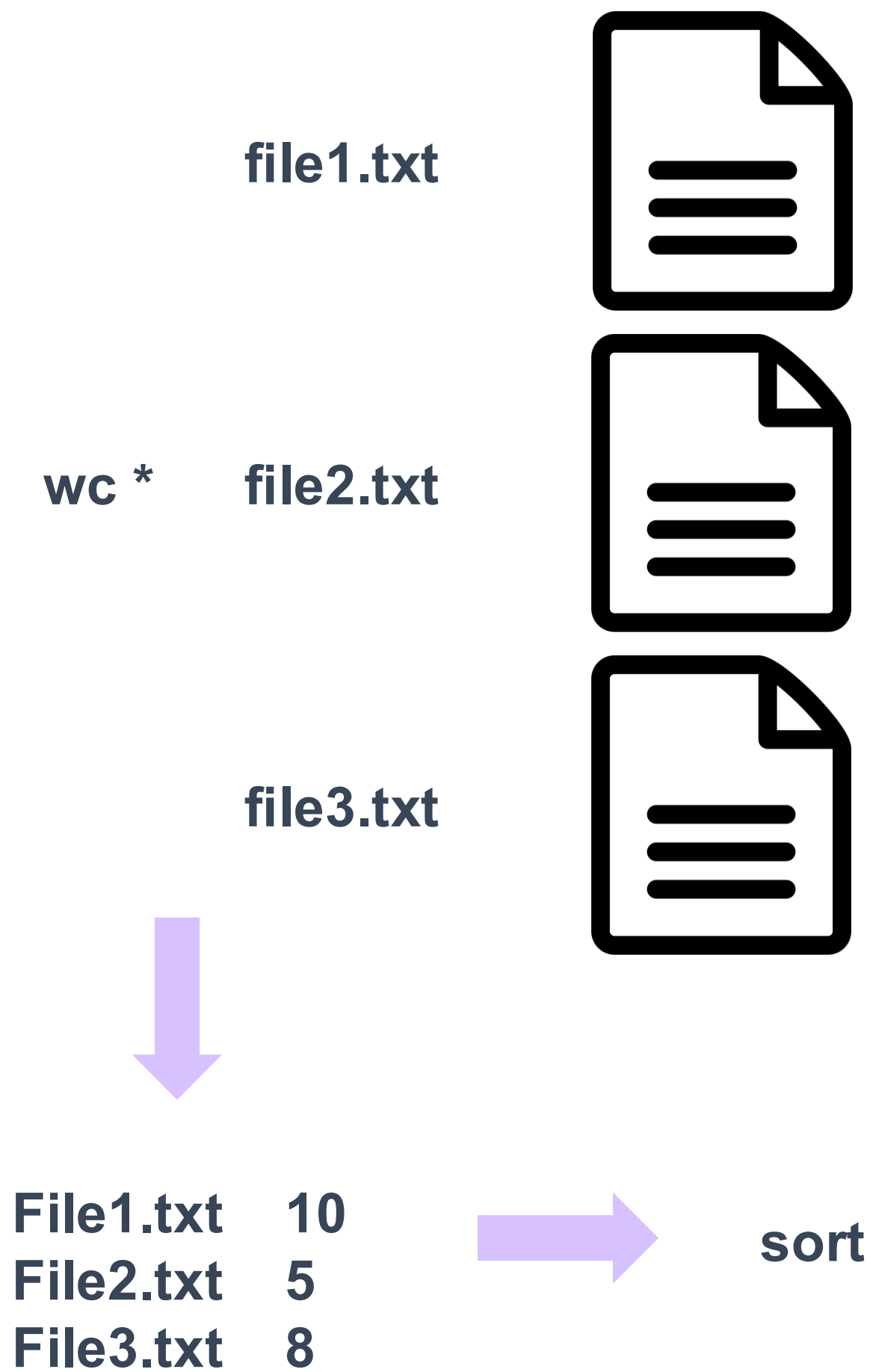
```
$ wc *.txt > file_lengths
```



```
$ sort -n file_lengths
```

But now we have an intermediate file we don't need!

CHAINING (PIPING) COMMANDS

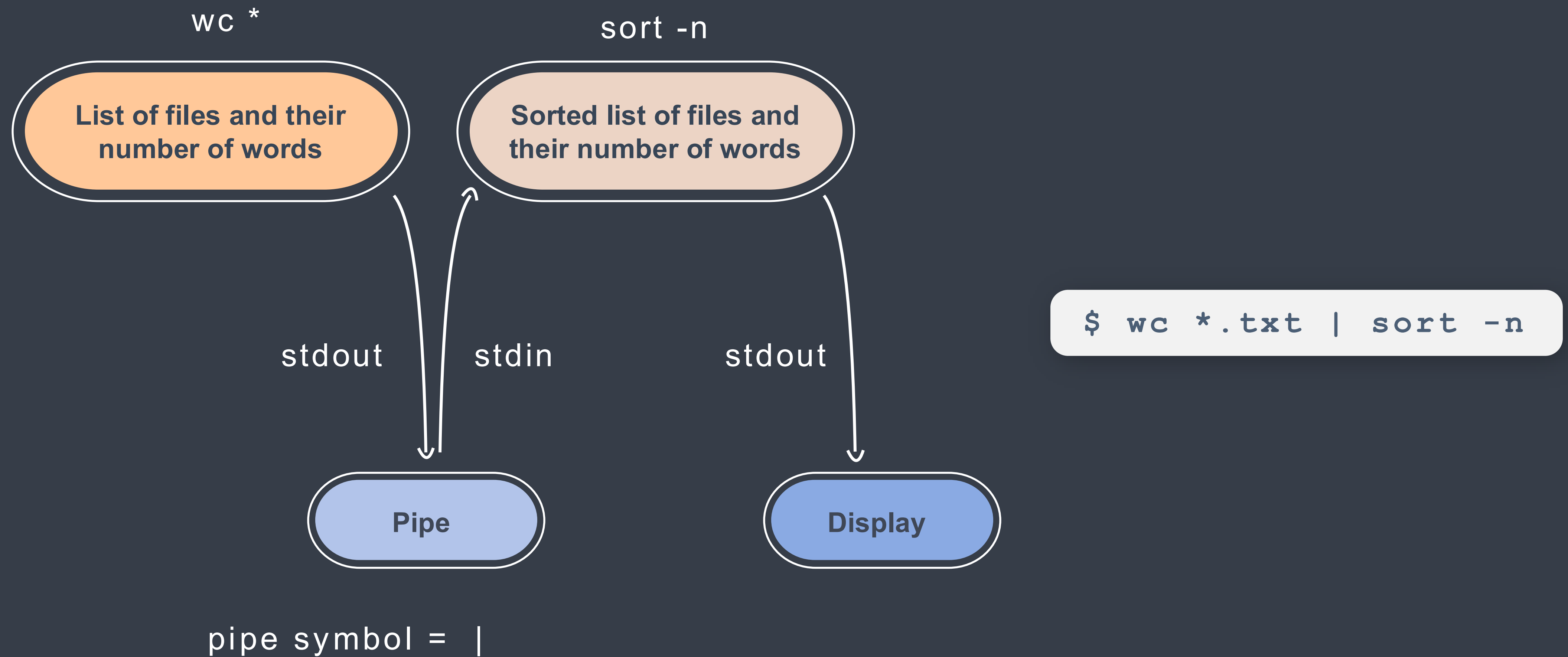


Instead we can use a **pipe** :

```
$ wc *.txt | sort -n
```

pipe symbol = |

CHAINING (PIPING) COMMANDS



CHAINING (PIPING) COMMANDS

Some examples:

- Take the second column and count how often each element occurs:

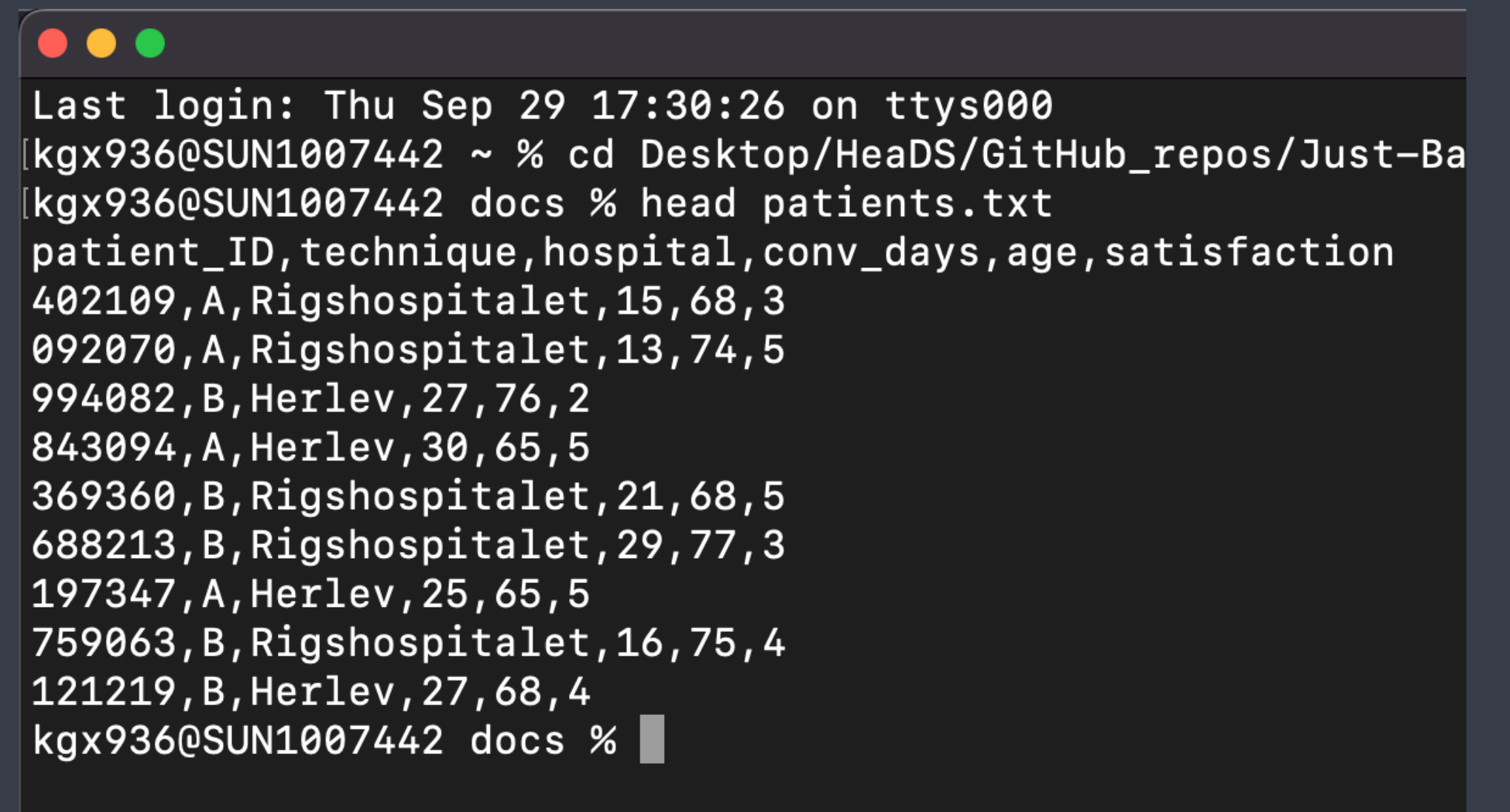
```
$ cut -d ',' -f 2 patients.txt | sort | uniq -c
```

- Get only lines containing 'Herlev' and sort by age:

```
$ grep 'Herlev' patients.txt | sort -t ',' -n -k5
```

- If we don't want the entire line we could also cut out the age column before sorting:

```
$ grep 'Herlev' patients.txt | cut -d ',' -f 5 | sort -n
```

A terminal window with a dark background and light text. It shows the output of a series of commands. The first line is the login banner. The second line shows the user's current directory. The third line shows the command to view the first few lines of 'patients.txt'. The subsequent lines show the output of 'head', which includes a header row and several data rows. The data rows contain patient IDs, techniques, hospitals, conversion days, ages, and satisfaction levels. The terminal window has standard macOS window controls (red, yellow, green buttons) at the top left.

```
Last login: Thu Sep 29 17:30:26 on ttys000
kgx936@SUN1007442 ~ % cd Desktop/HeaDS/GitHub_repos/Just-Ba
kgx936@SUN1007442 docs % head patients.txt
patient_ID,technique,hospital,conv_days,age,satisfaction
402109,A,Rigshospitalet,15,68,3
092070,A,Rigshospitalet,13,74,5
994082,B,Herlev,27,76,2
843094,A,Herlev,30,65,5
369360,B,Rigshospitalet,21,68,5
688213,B,Rigshospitalet,29,77,3
197347,A,Herlev,25,65,5
759063,B,Rigshospitalet,16,75,4
121219,B,Herlev,27,68,4
kgx936@SUN1007442 docs %
```

CHAINING (PIPING) COMMANDS

What about commands that require an input file because of their syntax?

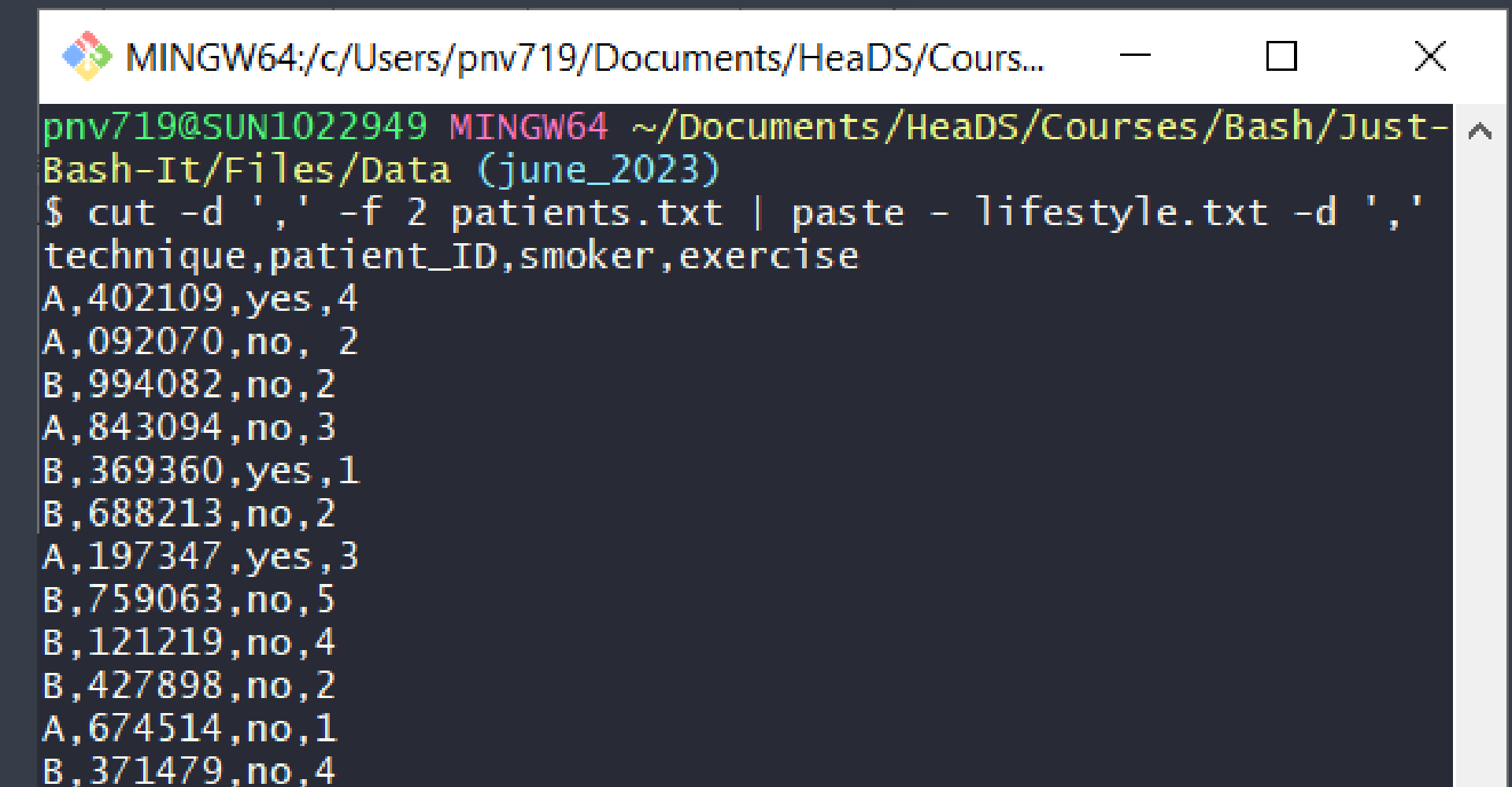
- Take the second column of patients and add it as a column to the lifestyle file:

```
$ cut -d ',' -f 2 patients.txt | paste ?? Lifestyle.txt
```

Solution: we replace the filename with '-'

```
$ cut -d ',' -f 2 patients.txt | paste - Lifestyle.txt
```

Technically '-' means 'read from stdin' which many commands do by default.

A terminal window titled 'MINGW64:/c/Users/pnv719/Documents/HeaDS/Cours...' showing a command and its output. The command is '\$ cut -d ',' -f 2 patients.txt | paste - lifestyle.txt -d ',''. The output is a list of patient data with a new column 'technique' added.

```
MINGW64:/c/Users/pnv719/Documents/HeaDS/Cours...  
pnv719@SUN1022949 MINGW64 ~/Documents/HeaDS/Courses/Bash/Just-Bash-It/Files/Data (june_2023)  
$ cut -d ',' -f 2 patients.txt | paste - lifestyle.txt -d ','  
technique,patient_ID,smoker,exercise  
A,402109,yes,4  
A,092070,no, 2  
B,994082,no,2  
A,843094,no,3  
B,369360,yes,1  
B,688213,no,2  
A,197347,yes,3  
B,759063,no,5  
B,121219,no,4  
B,427898,no,2  
A,674514,no,1  
B,371479,no,4
```

CHEAT SHEET 2

View Files

```
less [file] # view file content
cat [file] # view file content (full)
head / tail -n 10 [file] # view n first/last lines
nano [file] # https://www.nano-editor.org/dist/latest/cheatsheet.html
vim [file] # https://vim.rtorr.com/
```

Compressed Files

```
tar -[f] [file] # .tar files
gzip -[f] [file] # decompress (.tar).gz files
unzip -[f] [file] # decompress .zip files
zless -[f][file] # view .gz file w/o decompression

others: zcat, zmore, gzcat
```

Manipulating Files

```
wc -[f][file] # Count lines, characters, bits
sort -[f][file] # Sort file (by field/column)
uniq -[f][file] # Return unique values
cut -[f][file]: # Extract field/column
paste -[f][files]: # Merge file lines
```

```
sed -[f]'command'[file] # Insertion, deletion, ...
grep -[f]['pattern'][file] # Search for pattern
awk '{pattern}'[file] # Search, replace, extract, ...
find -[f][path]['pattern'] # Search pattern in file name
```

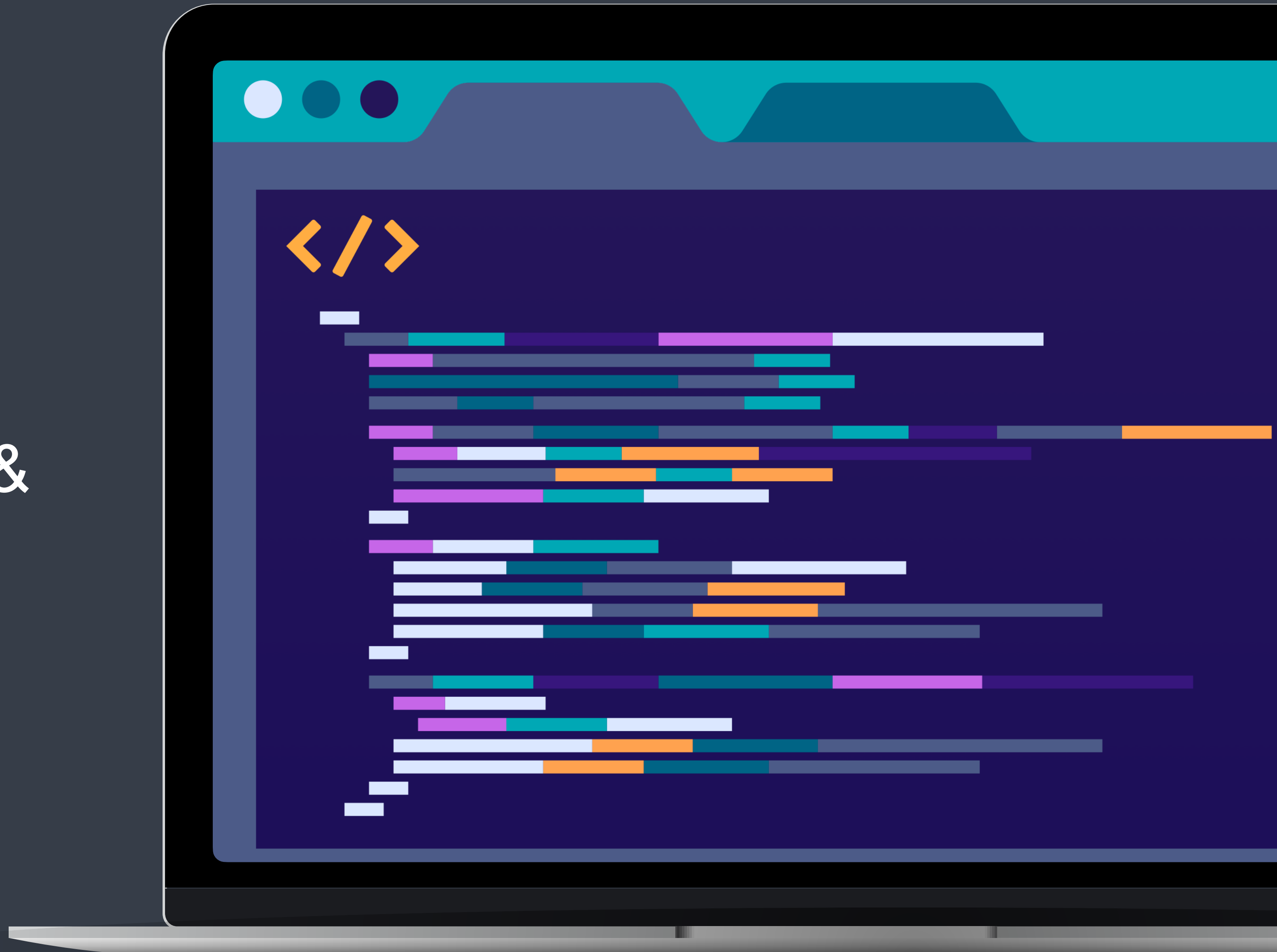


```
Last login: Fri Jun 24 15:05:34 on ttys000
[kgx936@SUN1007442 ~ % echo "Just Bash It"
Just Bash It
[kgx936@SUN1007442 ~ % █
```



Now we are getting ‘fancy’
and chaining some
commands, **Exercise 7!**

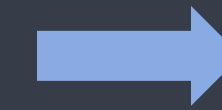
8. SHELL SCRIPTS & LOOPS



WRITING SCRIPTS

Example: I want to create a file with only Herlev patients but also the header.

```
(base) henrike@henrike-ThinkPad-X390:~/Documents/Heads_center_management/courses/Just-Bash-It/Examples/docs$ head -n 1 patients.dat > herlev.dat  
(base) henrike@henrike-ThinkPad-X390:~/Documents/Heads_center_management/courses/Just-Bash-It/Examples/docs$ grep 'Herlev' patients.dat | sort -n -k 5 -t ',' >> herlev.dat
```



```
1 head -n 1 patients.dat > herlev.dat  
2 grep 'Herlev' patients.dat | sort -n -k 5 -t ',' >> herlev.dat
```



```
(base) henrike@henrike-ThinkPad-X390:~/Documents/Heads_center_management/courses/Just-Bash-It/Examples/docs$ bash my_script.sh
```

- Save your commands for later use!
- Re-run anytime, always same result
- Check your script if you don't remember the steps of an analysis
- Back-up your scripts on i.e. github, KU drive, ect.

COMMAND LINE INPUT

- A script is executed with ***bash*** (or simply ***sh***).
- Add **arguments** when you execute your script, these will be passed to it. Useful if you want to specify the file you want to run the script on.
- The **first argument** after the script name will be **\$1** inside the script.
- The **second argument** will be **\$2** and so on.

```
# Comment line script 1  
# Usage: sort_this.sh [file_name]  
  
sort -n $1
```

```
$ bash sort_this.sh patients.txt
```

```
$ bash sort_this.sh my_textfile.txt
```

```
# Comment line script 2  
# Usage: cut_col.sh [file_name]  
[column_number]  
  
cut -d ',' -f $2 $1
```

```
$ bash -x cut_col.sh patients.txt 3
```

LOOP IN BASH



```
#example of a for loop in bash
```

```
for file in *.txt
do
    echo $file
    wc $file
done
```

```
#the general syntax of a for loop is:
```

```
for [iterator] in [generator]
do
    commands
done
```

- Loops allow you to repeat the same action several times, once for each file for example.
- Many bash commands can be run on several files, i.e.

```
$ wc *.txt
```

- However, a loop is useful if you want to execute several commands for each file.

OTHER SCRIPTING

```
$ bash my_script.sh
```

- This is a bash script. It's basically just a text file but we tell the computer to use bash to interpret it by calling it with 'bash my_script.sh'
- We can also run scripts in other languages from the command line, i.e. python:

```
$ python align_reads.py
```

- Scripts in the R language are called with the command 'Rscript':

```
$ Rscript deSEQ_cancer_Data.R
```

- You need to have R and python installed on your computer in order to be able to call them on the command line.

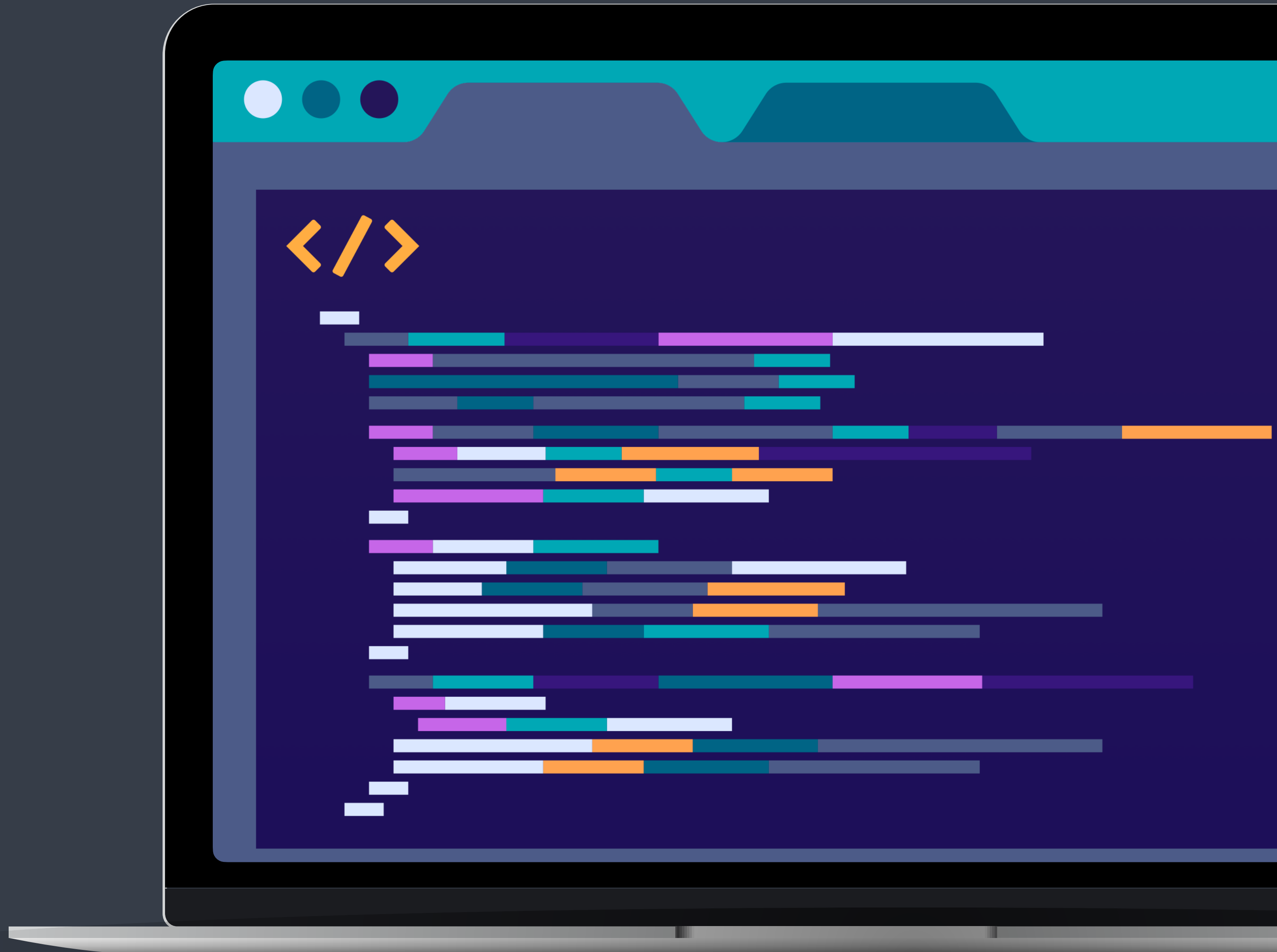


```
Last login: Fri Jun 24 15:05:34 on ttys000
[kgx936@SUN1007442 ~ % echo "Just Bash It"
Just Bash It
[kgx936@SUN1007442 ~ % █
```



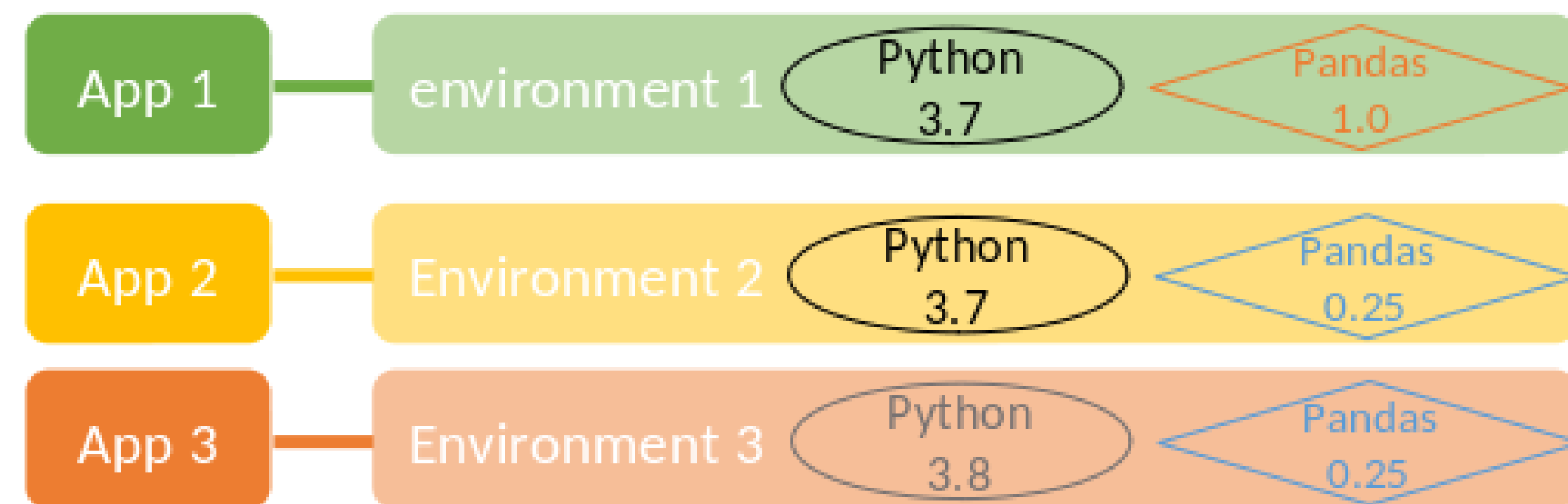
Let's make a script,
Exercise 8!

9. SOFTWARE INSTALLATION UPKEEP & MORE





- **Conda** is an open source package & environment management system.
- Many softwares exist as conda packages, you can use conda to install them and their dependencies.
- **An environment** is a specific combination of packages in a specific version.
- IF you do not update the packages, running an analysis in the same environment will give the same result.
- **Be aware that some channels are not free to use for organizations**



PROS

Comprehensive, many options
Environment management
Well documented

CONS

Can be overwhelming
Some channels are **not** free



So what is free?

- **conda**, the software program for package and environment management
- **packages** in conda-forge and Bioconda channels

What is paid?

- The conda-compatible **packages** in the **default** channel and Anaconda Distribution (if > 200 people)



APT-GET

- **Homebrew** - source software package management system, for OS X, Ubuntu (Linux).
- NOT an environment manager, package manager only.
- Creates separate directories for libs and configurations, adds symlinks to *user/local*.
- Manage all installed softwares via the terminal, install, check, update, remove...

PROS

Easy to use
Good documentation
Commands a few and logical

CONS

Packages, NOT environments

- **apt-get** is a command-line tool for handling packages in Linux (Ubuntu, 'Windows').
- APT = *Advanced Packaging Tool*
- Upgrade and remove of packages along with their dependencies.
- NOT an environment manager, package manager only.

PROS

Works like any linux command
No installation (is default)
Robust to OS updates

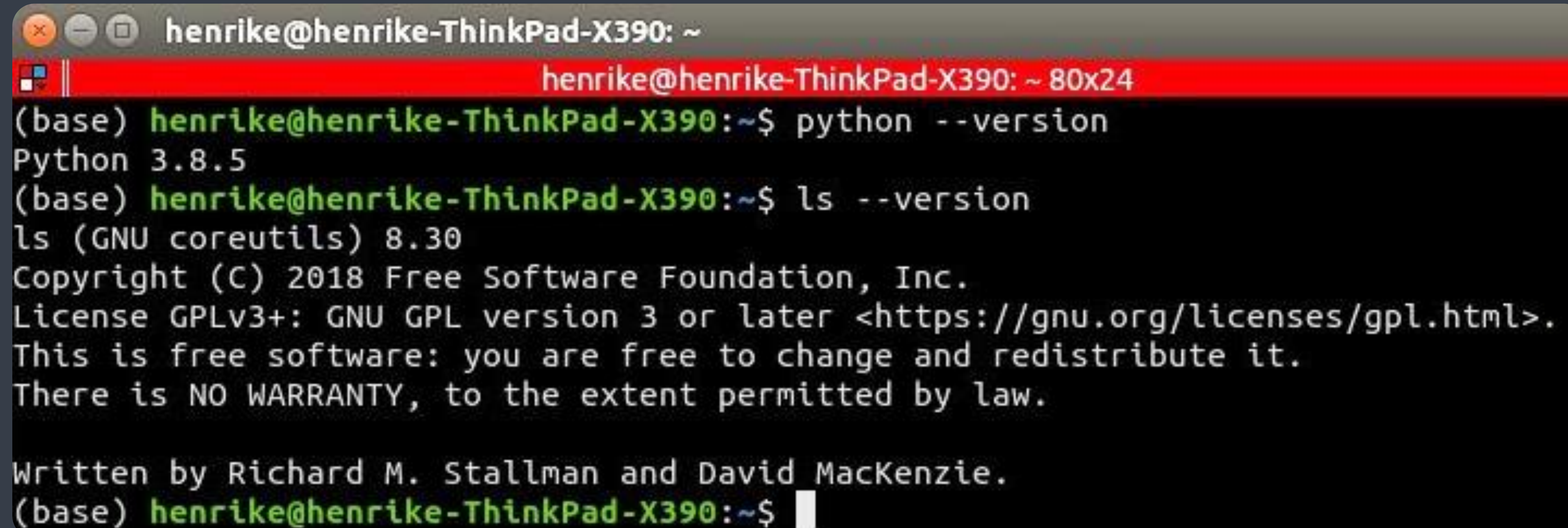
CONS

Packages, NOT environments

SOFTWARE VERSION

You can generally check the version of installed software with:

```
[name of software] --version
```

A terminal window titled 'henrike@henrike-ThinkPad-X390: ~' with a red title bar. The terminal shows two commands and their outputs. The first command is 'python --version' which outputs 'Python 3.8.5'. The second command is 'ls --version' which outputs 'ls (GNU coreutils) 8.30' followed by copyright and license information for GNU GPL version 3 or later, and a statement that the software is free and has no warranty. The prompt '(base) henrike@henrike-ThinkPad-X390:~\$' is visible at the end of each line.

```
henrike@henrike-ThinkPad-X390: ~
henrike@henrike-ThinkPad-X390: ~ 80x24
(base) henrike@henrike-ThinkPad-X390:~$ python --version
Python 3.8.5
(base) henrike@henrike-ThinkPad-X390:~$ ls --version
ls (GNU coreutils) 8.30
Copyright (C) 2018 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>.
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Written by Richard M. Stallman and David MacKenzie.
(base) henrike@henrike-ThinkPad-X390:~$
```

KEEPING THINGS UP TO DATE

Update your software with the same tool you used to install it:

- Installed with **conda**: (update only the current environment)

```
$ conda update [software]
```



- Installed with **homebrew** (updates a software):

```
$ brew upgrade [software]
```



- Installed with **apt-get**:

```
$ apt-get update [software]
```

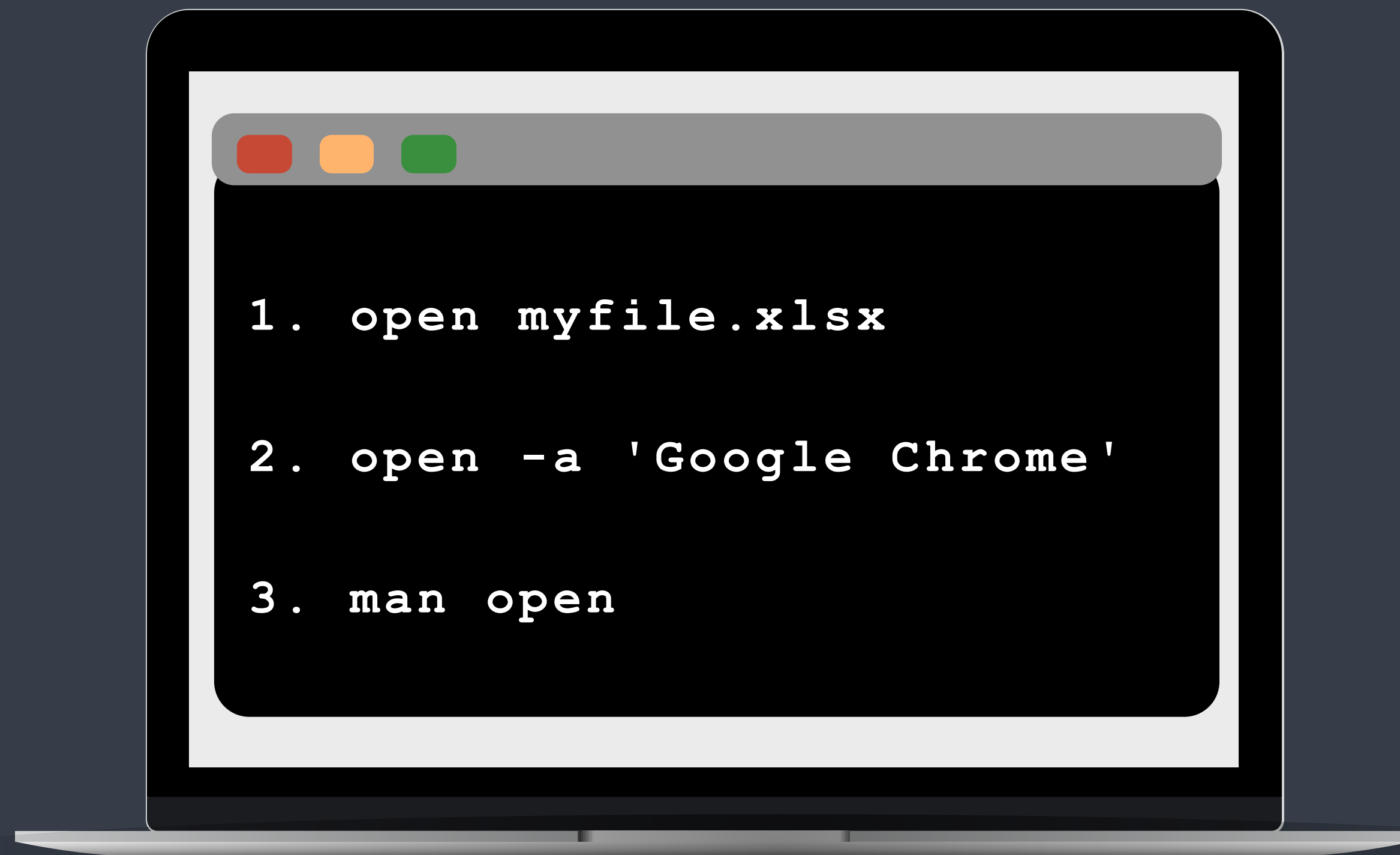


- With **Windows App store**:

*Update via
Windows App store*



OPENING APPS & RUNNING SOFTWARE



1. Some (not all) **files** can be **opened** without specifying an app.
2. Many **apps** can be **launched** with `open -a`.
3. Some **software** are designed to run on both the command line and in a GUI, others are not.

Monitor processes, find them, check memory & 'kill' them when necessary:

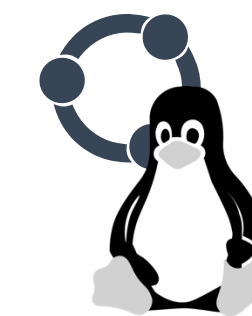
```
$ top (htop)
$ ps aux (processes for all users)

$ ps aux | grep 'Google Chrome'
$ kill [pid]
```

CONFIGURATION FILES

- Config files are used to specify parameters, options, settings and preferences applied to your OS or a software.
- Types of config files; system-wide, program-specific, user-specific.
- Extension will pertain to what is configured.
- Rarely permission to change system-wide files. Software-specific files you can usually edit.
- **N.B** config files are 'hidden', i.e. `ls -la` (or similar) to see them.

UBUNTU - BASH SHELL



/etc

.profile

.bashrc

.bash_profile

/User

.profile

.bashrc

.bash_profile

OS X - Z SHELL (BASH +)



/etc

.zprofile

.zshrc

.zsh_profile

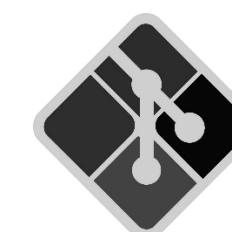
/User

.zprofile

.zshrc

.zsh_profile

WINDOWS - GITNBASH



/User

.bash_profile

CHEAT SHEET 3

Applications & Scripts

top # display running processes in 'real time'

ps (aux) [file] # snapshot of processes

open -a [application] # open an application

kill [pid] # kill/stop a process using pid

bash [file.sh] or sh [file.sh] # Execute/run bash script

[software] --version # check software version

brew install [software] # OS X install software on Mac

apt-get[software] # OS X install software on Ubuntu/Linux

conda # software and environment management system for Windows, Ubuntu & OS X

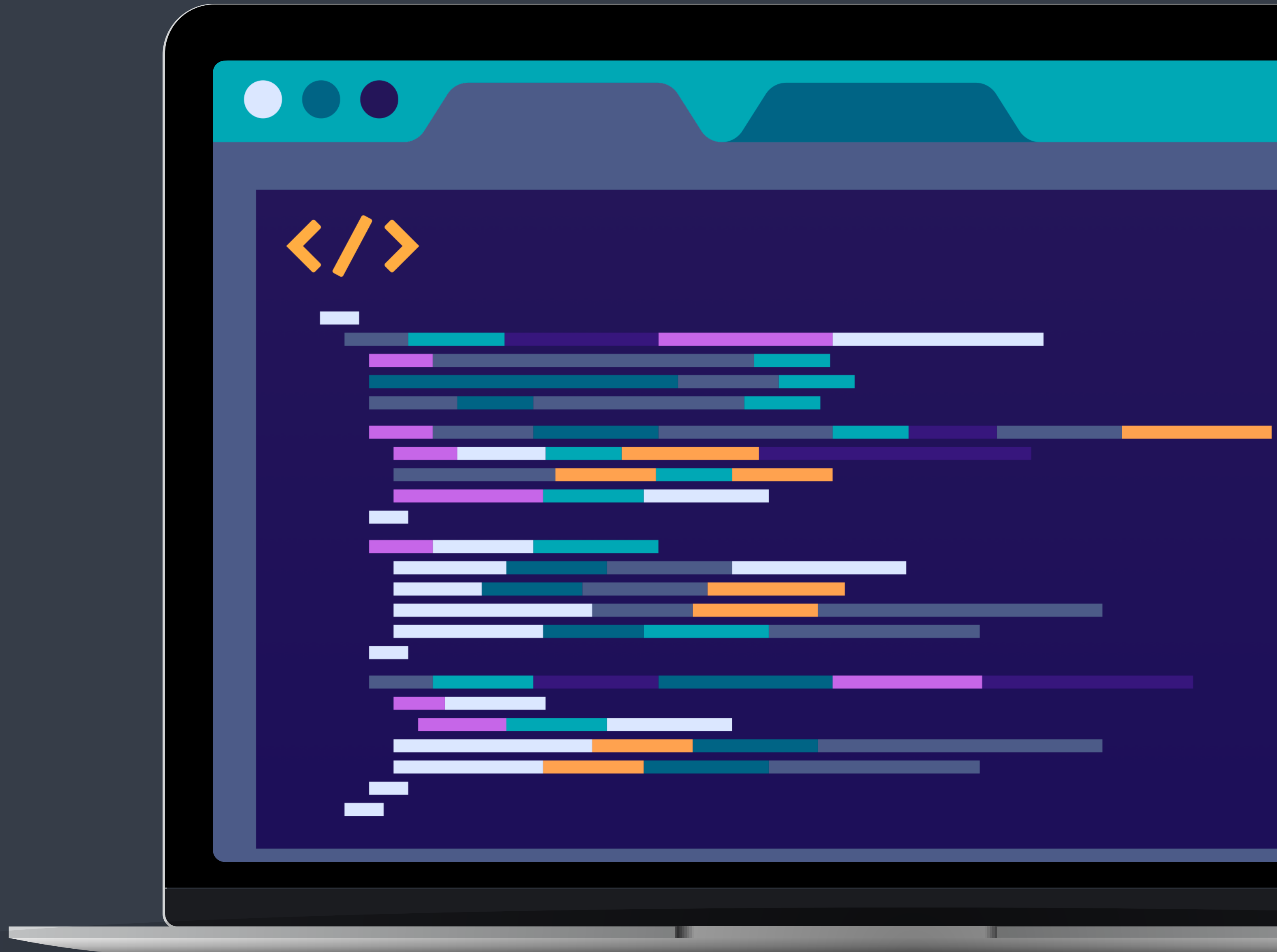


```
Last login: Fri Jun 24 15:05:34 on ttys000
[kgx936@SUN1007442 ~ % echo "Just Bash It"
Just Bash It
[kgx936@SUN1007442 ~ % █
```



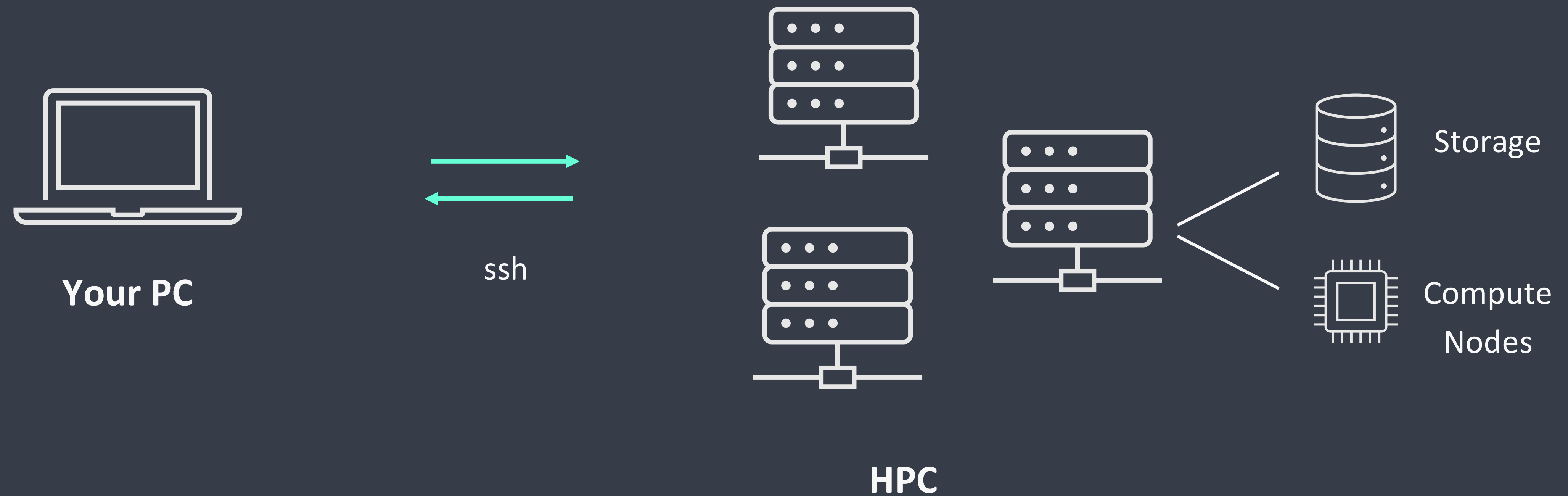
Like a real hacker we'll
install and run a software in
the command line in
Exercise 9!

10. REMOTE COMPUTING ON HPCS



WHAT IS AN HPC

HPC – High Performance Computing system (Server, Supercomputer)



FILE SYSTEMS ON HPCS

Most HPCs have structured file systems that are on different hard drives which mounted to the computing nodes. This means you cannot put things wherever.

Home directory:

- Backed up
- Configuration files
- Small user files

`/home/[user]`



Project/Data/Group directories:

- Backed up
- Large data files, results ect
- Databases

`/projects/RNAseq_a_thal`

`/groups/Jensen_group`

`/data/RH_2025_prj392`

[examples for dir names,
check documentation of
your HPC!]

Scratch directories:

- Not backed up!
- Temporary files

`/scratch`

QUEUING SYSTEMS

On most HPC systems you use a job scheduler to execute programs



No scheduler:

- First come first serve
- All commands compete for the available resources
- No load scheduling
- Interactive
- Fine for small tasks

Scheduler:

- Manages resource allocation
- Jobs can run during the night/ when you are not logged in
- Can ask for interactive job

SLURM

SLURM is both a scheduler and a resource manager.

Submitting a job script to queue:

```
$ sbatch my_job.sh
```

Checking the queue:

```
$ squeue
```

Starting an interactive job:

```
$ srun
```

or

```
$ salloc
```

```
#!/bin/bash
#
#SBATCH --job-name=test
#SBATCH --account=myAmazingProject
#SBATCH --nodes=2
#SBATCH --time=01:00:00
#SBATCH --output=res.txt

paste -d '\t' GenomicRegion.tmp \
gene_names_clean.txt > Annotation_Gene.gff

grep 'tRNA' Annotation_Gene_Len.gff -c
```

An example job script

SOFTWARE ON HPCS

Pre-installed, e.g. in
environment modules

```
$ module load python
```

User-administrated, e.g.
in a conda environment

```
$ conda activate my_env
```

Pre-packaged in containers
like Docker, Singularity

WORKFLOW LANGUAGES

Workflow languages:

- Tools to create reproducible pipelines – more robust bash scripts.

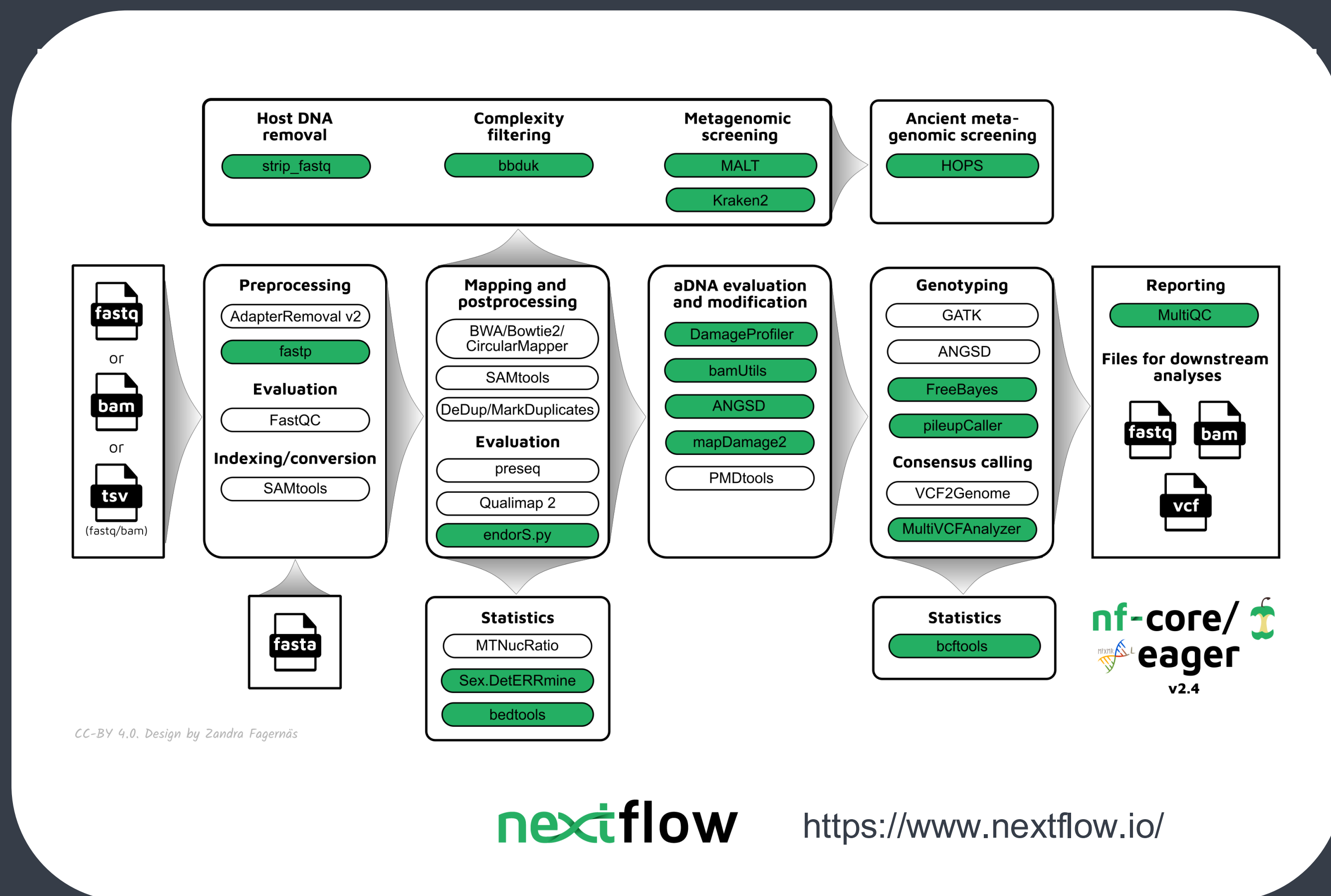
Workflow languages allow you to:

- Control the flow of your pipeline
- Restart at various defined points
- Ensure previous commands have executed without error
- Logging and reporting jobs



snakemake

<https://snakemake.readthedocs.io/en/stable/>





```
Last login: Fri Jun 24 15:05:34 on ttys000
[kgx936@SUN1007442 ~ % echo "Just Bash It"
Just Bash It
[kgx936@SUN1007442 ~ % █
```



Explore the wide world of
the server in **Exercise 10!**

SUDO THE ULTIMATE POWER MOVE – JUST DON'T!



- *sudo* == 'super user do!'
- This command forces an action and it is meant to be used only by system administrators.
- The sudo command has 'elevated privileges'
- Using this command usually requires a confirm in terms of typing a system/computer password.

- Sudo rights are required to alter root administrative system files, i.e. configuration files.
- You may get reported to the admin!

```
Berkas  Sunting  Tampilan  Cari  Terminal  Bantuan
localdisk@kali:~$ sudo ifconfig

We trust you have received the usual lecture from the local System
Administrators. It usually boils down to these three things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

[sudo] password for localdisk: 
```

THANK YOU FOR TODAY

